MYP

Mathematics

2 STUDENT COURSEBOOK

MYP 2 · YEAR 2 · ELEVEN CHAPTERS IN FOUR PARTS

A concept-driven inquiry course in ratio, proportion, algebra, geometry, probability and data. You investigate first, then reason, communicate and apply.

RATIOS · PROPORTIONAL REASONING · RATES · INTEGERS · EXPRESSIONS AND EQUATIONS · PATTERNS AND GENERALISATION · ANGLES AND AREA · SOLIDS AND MEASURES · TRANSFORMATIONS · PROBABILITY · UNIVARIATE DATA

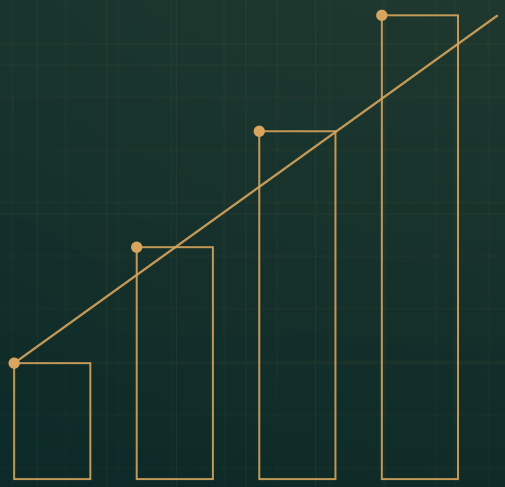
HOW THIS BOOK WORKS

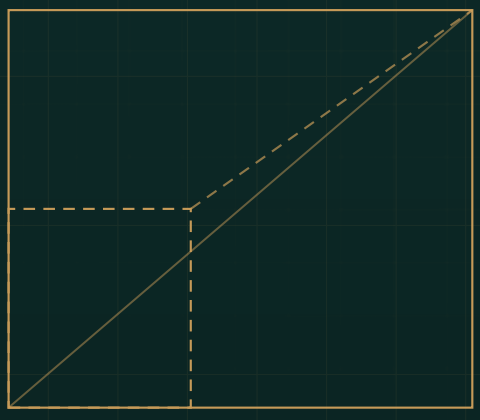
Every chapter opens with a task. You investigate, notice a pattern and state a mathematical conclusion before you meet the formal explanation.

MYP ASSESSMENT CRITERIA

CRITERION A	Knowing and understanding
CRITERION B	Investigating patterns
CRITERION C	Communicating
CRITERION D	Applying mathematics in real-life contexts

GROWTH





PROPORTION

SADIKOU MOUKODDIM

MYP 2 · STUDENT COURSEBOOK

AUTHOR SADIKOU MOUKODDIM

About this edition

This is the second edition of the MYP 2 mathematics coursebook. It was rebuilt from the first-edition MYP Mathematics 2 Student Coursebook, which remains the principal content reference and is preserved unchanged in the project archive.

The edition keeps the first edition's chapter scope and its visual identity, and rebuilds the teaching sequence around inquiry: engagement, inquiry questions, investigation, observation, mathematical conclusion, formal development, worked examples, extensive visual practice, and application, assessment and reflection.

All classroom data sets are simulated and are labelled as such. Real data would require provenance, so no claim in this book rests on unverified statistics.

No publisher, ISBN, print specification or commercial print-readiness claim is made in this file. The book is produced as an A4 review copy: fonts are embedded, pagination is continuous and every page has been checked for overflow.

MYP 2 · STUDENT COURSEBOOK

How to use this book

Each chapter follows the same sequence, so you always know what is coming next. The sequence is deliberate: you investigate before you are told the rule, because a rule you have discovered is a rule you can defend.

The chapter sequence

1. **Engagement.** A short situation that raises the question the chapter will answer.
2. **Inquiry questions.** Factual, conceptual and debatable questions that frame the chapter.
3. **Investigation.** A task you complete before the formal teaching: record evidence in a table and look for a pattern.
4. **Observation and discovery.** What you noticed, stated in your own words and compared with a partner.
5. **Mathematical conclusion.** The rule that the evidence supports, written as a general statement.
6. **Formal development.** The explanation, definitions and notation, with the reasoning made explicit.
7. **Worked examples.** Problem, visual representation, mathematical reasoning, step-by-step solution, final answer and interpretation.
8. **Practice.** Separated into Criterion A, B, C and D practice, with visual representations throughout.
9. **Application, assessment and reflection.** Real-context tasks, criterion-based assessment practice and a review of your progress.

MYP 2 · STUDENT COURSEBOOK

How to use this book

The four MYP mathematics criteria

A · Knowing and understanding

Apply knowledge to familiar and unfamiliar problems, and check results.

B · Investigating patterns

Investigate, describe patterns, generalise and justify — recognising, describing and verifying patterns at Year 2 level rather than producing formal proofs.

C · Communicating

Use correct mathematical language, notation and representations, and reason coherently.

D · Applying mathematics in real-life contexts

Apply mathematics to authentic situations, identify relevant information and comment on the accuracy and sense of the result.

Working with this book

- Practice is separated by criterion so that you can see which skill a task is training. Criterion B practice is additional genuine pattern investigation, not a repeat of the teaching.
- Visuals are part of the mathematics. Where a task supplies a diagram, grid, table, number line or model, use it; where it asks you to sketch, your drawing is part of the answer.
- Answers and student-facing reasoning are collected at the back of the book, with a formula bank, a glossary and references.
- Every chapter ends with project connections: a Service as Action opportunity, a Community Project connection and a Personal Project connection for MYP 5.

MYP 2 · STUDENT COURSEBOOK

MYP mathematics and this coursebook

The IB Middle Years Programme asks students to inquire into concepts through global contexts, to develop approaches to learning (ATL) skills, and to act on what they learn through service as action. In mathematics this means more than practising procedures: students investigate patterns, justify generalisations, communicate with precise language and apply mathematics to situations that matter to them.

MYP mathematics is organised into four branches — number, algebra, geometry and trigonometry, and statistics and probability — and the programme recommends at least 50 hours of teaching per subject each year. This coursebook covers the Year 2 content of those branches across eleven chapters, grouped into four parts.

Assessment in MYP mathematics uses four equally weighted criteria, each with a maximum of 8: A knowing and understanding, B investigating patterns, C communicating, and D applying mathematics in real-life contexts. This book trains each criterion explicitly, and says so on the page where the training happens.

MYP 2 · STUDENT COURSEBOOK

MYP mathematics and this coursebook

Projects, service and action

Service as Action

Each chapter includes a service opportunity with a community need, a mathematical connection, a proposed action, an expected impact and a student reflection. Completing an exercise is not service: service requires action that benefits others and reflection on that action.

Community Project

MYP students completing the programme in Year 3 or Year 4 complete a community project. Each chapter offers a student-led project idea with a research question, the mathematics involved, an action or outcome, beneficiaries and evidence. Follow your school's requirements and supervision.

Personal Project (MYP 5)

Students completing MYP in Year 5 complete a personal project. Each chapter offers an adaptable MYP 5-oriented extension with a personal interest, a learning goal, a product, the mathematics involved, a process and evidence. It is deliberately different from the community project and from service as action.

What this book does not claim

- Programme requirements quoted here follow the official IB MYP curriculum, mathematics and projects pages and the MYP general regulations. Everything else — the chapter sequence, the investigations, the practice sets and the project ideas — is author-designed material built to serve those requirements, and is not an IB publication.
- No ISBN, publisher, imprint or print specification is claimed. The file is an A4 review copy with embedded fonts and continuous pagination, produced for proofing and classroom review rather than for commercial print production.

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Application and assessment · Review · Projects and service

Every chapter follows the same sequence: engagement, inquiry questions, investigation, observation, mathematical conclusion, formal development, worked examples, Criterion A–D practice, application, assessment and reflection, and project connections.

CHAPTER 1 · PART I · NUMBER AND PROPORTION

01

Ratios and equivalent ratios

A ratio compares two quantities. The same ratio can describe a snack pack or a factory order, and equivalent ratios keep that comparison honest.

ENGAGE

A school fair sells boxes of 8 pencils and boxes of 12 pencils. A teacher buys 3 boxes of 8 and a parent buys 2 boxes of 12. Who has more pencils? A ratio answers this in one line.

Framing the inquiry

Key concept: Form
Related concepts: Equivalence · Representation
Global context: Fairness and development
Global context exploration: Inequality, difference and inclusion

Statement of inquiry: The form of a ratio, and the equivalence of its different representations, allow a limited resource to be shared fairly and allow that fairness to be justified and checked.

Questions to guide us

- **Factual:** What does the ratio 3:5 compare, what is its simplest form, and how do you generate ratios equivalent to it?
- **Conceptual:** Why can one ratio describe many different-sized situations, and how does representing it as a unit rate or a ratio table change what you are able to compare?
- **Debatable:** Is a ratio always the fairest way to share a limited resource, and how could you use mathematical evidence to defend your position?

ATL · CRITICAL THINKING. In order for a student to evaluate and improve a mathematical claim, students must test the claim on several cases, find a counterexample if one exists, and revise the statement so it is accurate.

CHAPTER 1 · INVESTIGATION 1.1

Understanding ratio language

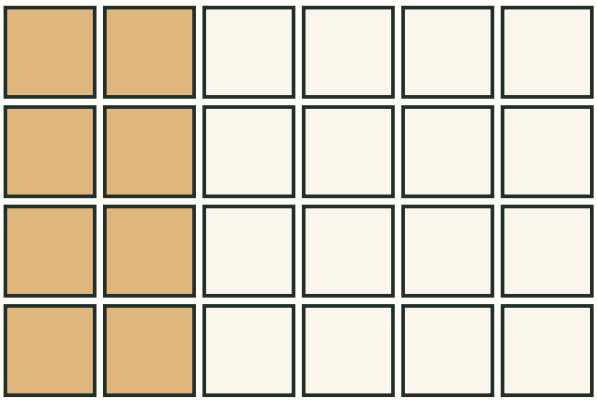
INQUIRY QUESTION

How can one ratio describe a comparison between two quantities, and what does its simplest form tell us about that comparison?

YOUR TASK

A donation of coloured tiles is shared between two classes. Colour a 4 × 6 grid so that the ratio of shaded (class A) to unshaded (class B) tiles is 2:4, then colour a 6 × 9 grid with the same ratio. Record how many tiles each class receives in each grid.

A 4 by 6 grid with 8 shaded tiles



shaded : unshaded = 8 : 16 = 1 : 2

Grid	shaded	unshaded	total
4 × 6	8	16	24
6 × 9	12	24	36
your grid			

CHAPTER 1 · INVESTIGATION 1.1

Understanding ratio language

What do you notice?

- Both grids have shaded:unshaded = 1:2.
- The totals are different but the simplified ratio is the same.
- Doubling the grid doubles both parts of the ratio.

Observation. Multiplying both parts of a ratio by the same number keeps the comparison unchanged.

MATHEMATICAL CONCLUSION

a : b = (ak) : (bk) for any non-zero number k, so one ratio can describe the comparison in any size, and its simplest form is the shortest honest description of that comparison.

Your own words

CHAPTER 1 · 1.1

Understanding ratio language

A ratio compares two quantities of the same kind. The order matters: 3:5 is not the same comparison as 5:3. A ratio does not tell you the total — it tells you how the total is split.

Two different kinds of comparison are easy to confuse. A part-part ratio compares one part with another (red:blue = 3:5). A part-whole ratio compares one part with everything (red:total = 3:8). Writing the wrong one is the most common error in this chapter.

A ratio can be simplified by dividing both parts by the same non-zero number, exactly as with fractions. 12:18 simplifies to 2:3 because both parts divide by 6.

> FORMAL DEVELOPMENT
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> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

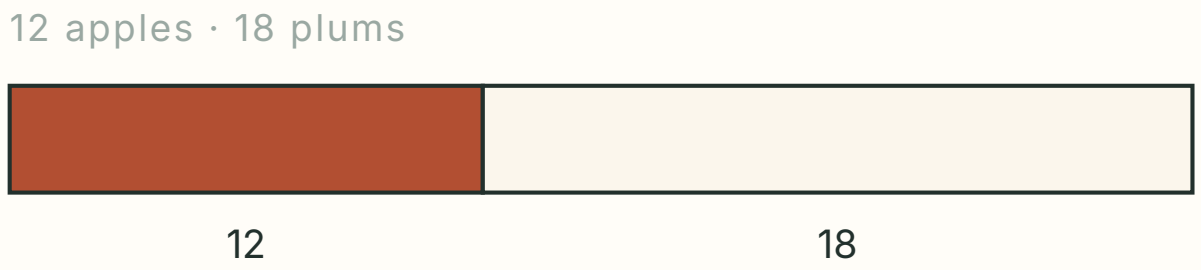
In your own words

CHAPTER 1 · 1.1 · WORKED EXAMPLE 1

Understanding ratio language

WORKED EXAMPLE 1.1.1

Problem. A fruit bowl holds 12 apples and 18 plums. Write the ratio of apples to plums in simplest form, then write the ratio of apples to all the fruit.



Mathematical reasoning. The ratio compares like quantities, so I can divide both parts by their highest common factor without changing the comparison.

1. Apples : plums = 12:18.
2. HCF(12, 18) = 6, so divide both parts by 6.
3. 12:18 = 2:3.
4. Total fruit = 12 + 18 = 30, so apples : total = 12:30 = 2:5.

Final answer. apples : plums = 2:3 and apples : total = 2:5

Interpretation. Simplifying did not change the bowl; it only made the comparison easier to read and to scale.

CHAPTER 1 · 1.1 · WORKED EXAMPLE 2

Understanding ratio language

WORKED EXAMPLE 1.1.2

Problem. A class has 14 boys and 16 girls. A second class has 21 boys and 24 girls. Do the two classes have the same ratio of boys to girls?



Mathematical reasoning. Two ratios are equivalent when one is a multiple of the other. Scaling 14:16 by 1.5 gives 21:24, so the comparison is the same.

1. First class: 14:16. Divide both parts by 2 → 7:8.
2. Second class: 21:24. Divide both parts by 3 → 7:8.
3. Both classes reduce to 7:8, so the ratios are equivalent.

Final answer. Yes — both classes have a boys-to-girls ratio of 7:8

Interpretation. The classes are different sizes, but the balance between boys and girls is identical.

CHAPTER 1 · INVESTIGATION 1.2

Equivalent ratios and simplest form

INQUIRY QUESTION

What stays the same when both parts of a ratio change, and how can you tell whether two ratios describe the same comparison?

YOUR TASK

Take the ratio 3:5. Multiply both parts by 2, 3, 4 and 5 and record the results. Then try to find a different pair of numbers with the same simplest form as 3:5.

multiplier k	3k	5k	ratio
2	6	10	6:10
3	9	15	9:15
4	12	20	12:20
5	15	25	15:25

CHAPTER 1 · INVESTIGATION 1.2

Equivalent ratios and simplest form

What do you notice?

- Every new ratio simplifies back to 3:5.
- The gap between the two parts grows, but the comparison does not change.
- There is no largest equivalent ratio: k can be any non-zero number.

Observation. A single comparison has infinitely many equivalent forms, all multiples of the simplest form.

MATHEMATICAL CONCLUSION

What stays the same is the comparison: equivalent ratios are exactly the ratios that reduce to the same simplest form, whatever the size of the numbers.

Your own words

CHAPTER 1 · 1.2

Equivalent ratios and simplest form

Equivalent ratios are the ratio version of equivalent fractions. To test whether two ratios are equivalent, simplify both to simplest form, or check whether one multiplier turns one into the other.

A unit rate is a ratio scaled so that one part is 1. It is the most useful form for comparing: 8 pencils for 3 dollars is a unit price of 0.375 dollars per pencil.

Simplifying is also a checking tool. If two answers disagree, reduce both to simplest form before deciding which is wrong.

> FORMAL DEVELOPMENT
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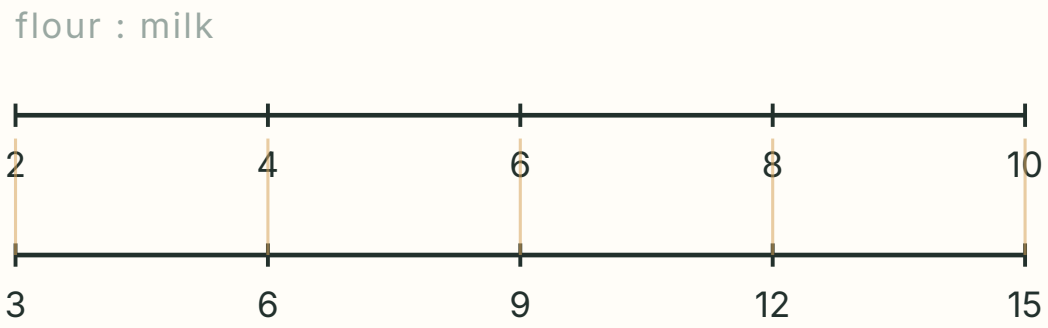
In your own words

CHAPTER 1 · 1.2 · WORKED EXAMPLE 1

Equivalent ratios and simplest form

WORKED EXAMPLE 1.2.1

Problem. A recipe uses 2 cups of flour for every 3 cups of milk. How much flour is needed for 15 cups of milk?



Mathematical reasoning. The ratio is fixed at 2:3, so milk scaled by 5 means flour also scales by 5.

1. flour : milk = 2:3.
2. 15 ÷ 3 = 5, so the milk has been multiplied by 5.
3. Multiply the flour by the same factor: 2 × 5 = 10.

Final answer. 10 cups of flour

Interpretation. The recipe keeps its taste because both parts moved by the same factor.

CHAPTER 1 · 1.2 · WORKED EXAMPLE 2

Equivalent ratios and simplest form

WORKED EXAMPLE 1.2.2

Problem. Show that 12:16 and 27:36 are equivalent.

12:16 = 3:4



Mathematical reasoning. Two ratios are equivalent exactly when their simplest forms are identical.

1. 12:16: HCF is 4, so 12:16 = 3:4.
2. 27:36: HCF is 9, so 27:36 = 3:4.
3. Both simplify to 3:4, so they are equivalent.

Final answer. 12:16 = 27:36 because both equal 3:4

Interpretation. Simplifying first makes the comparison immediate, whatever the sizes.

CHAPTER 1 · INVESTIGATION 1.3

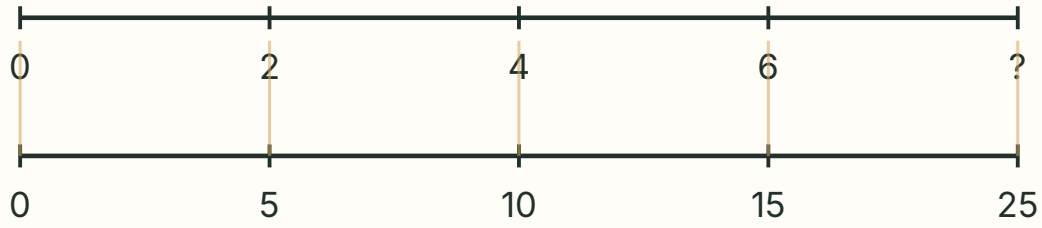
Ratio tables and double number lines

INQUIRY QUESTION

How can a ratio table or a double number line find a missing value in a comparison without guessing?

YOUR TASK

A double number line shows 0, 2, 4, 6 litres of juice above 0, 5, 10, 15 cups. Predict the value above 25 cups, then test your prediction with a table.



litres	2	4	6	8	10
cups	5	10	15	20	25

CHAPTER 1 · INVESTIGATION 1.3

Ratio tables and double number lines

What do you notice?

- Each step of 2 litres matches 5 cups.
- 25 ÷ 5 = 5 steps, so 5 × 2 = 10 litres.
- The line and the table give the same answer.

Observation. A double number line and a ratio table are the same relationship in two notations.

MATHEMATICAL CONCLUSION

A missing value is found by locating the multiplier in one row and applying it to the other, which is why the table and the number line never need trial and error.

Your own words

CHAPTER 1 · 1.3

Ratio tables and double number lines

A ratio table lists matching pairs of quantities. Reading down the table shows one quantity scaling; reading across shows the fixed comparison. A double number line does the same job on a line, which makes missing values visible.

Tables are the natural home for ratio reasoning because they keep the relationship visible while you scale. To find a missing value, first find the multiplier between two known values in the same column.

The same structure appears later as a table of values for a proportional relationship, and as a coordinate graph. Recognising it now saves time in chapters 2 and 6.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.3 · WORKED EXAMPLE 1

Ratio tables and double number lines

WORKED EXAMPLE 1.3.1

Problem. A shade of green uses 3 parts blue to 5 parts yellow. Complete the table and find how much blue is needed for 40 litres of yellow.

blue	3	6	9	12	?
yellow	5	10	15	20	40

Ratio table for the green shade

Mathematical reasoning. The yellow column has been multiplied by 8 to reach 40, so the blue column must be multiplied by 8.

1. Multiplier: $40 \div 5 = 8$.
2. Blue needed: $3 \times 8 = 24$.
3. Check: 24:40 simplifies to 3:5.

Final answer. 24 litres of blue

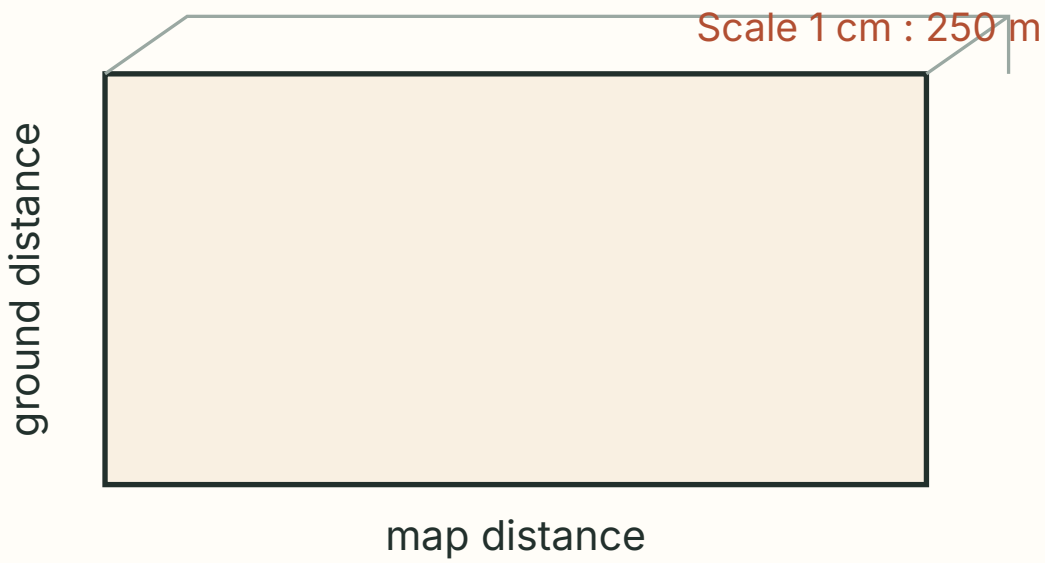
Interpretation. The table makes the single multiplier obvious, so the answer needs no trial and error.

CHAPTER 1 · 1.3 · WORKED EXAMPLE 2

Ratio tables and double number lines

WORKED EXAMPLE 1.3.2

Problem. A map scale is 1 cm to 250 m. How many centimetres on the map show 2 km?



Mathematical reasoning. Convert to the same unit first, then use the ratio table to scale.

1. 2 km = 2 000 m.
2. 2 000 ÷ 250 = 8.
3. The map distance is 8 cm.

Final answer. 8 cm

Interpretation. A scale is a ratio table with only one row: every ground length is 250 times its map length.

CHAPTER 1 · INVESTIGATION 1.4

Sharing and comparing with ratios

INQUIRY QUESTION

When is a sharing rule fair, and how can you check that a sharing really matches the rule that was agreed?

YOUR TASK

A school receives a donation of 120 books to share between two year groups in the ratio 5:3. Try sharing it as 60 and 60, then as 75 and 45, then as 5 parts and 3 parts. Which sharing matches the ratio, and how do you know?

sharing	group 1	group 2	simplifies to
equal	60	60	1:1
5 parts and 3 parts	75	45	5:3

CHAPTER 1 · INVESTIGATION 1.4

Sharing and comparing with ratios

What do you notice?

- 60:60 simplifies to 1:1, not 5:3.
- 120 ÷ 8 = 15, so one part is 15 books.
- 5 × 15 = 75 and 3 × 15 = 45, which adds back to 120.

> **Observation.** A sharing is correct only when the parts add to the total and simplify to the given ratio.

MATHEMATICAL CONCLUSION

A sharing is fair when the rule is stated first and the result is checked: find the value of one part, then verify both the total and the simplest form.

Your own words

CHAPTER 1 · 1.4

Sharing and comparing with ratios

Sharing an amount in a ratio has three steps: add the parts, divide the amount by the total to find the value of one part, then multiply by each part. Skipping the first step is the classic error.

Comparing ratios is easier when both are written with the same second part, or when both are converted to unit rates. Whichever you choose, say which you used.

A ratio also supports reasoning about fairness: if two people contribute in the ratio 3:2, a fair split of the reward is not automatically 3:2 unless the agreed rule says so. State the rule before you calculate.

> FORMAL DEVELOPMENT
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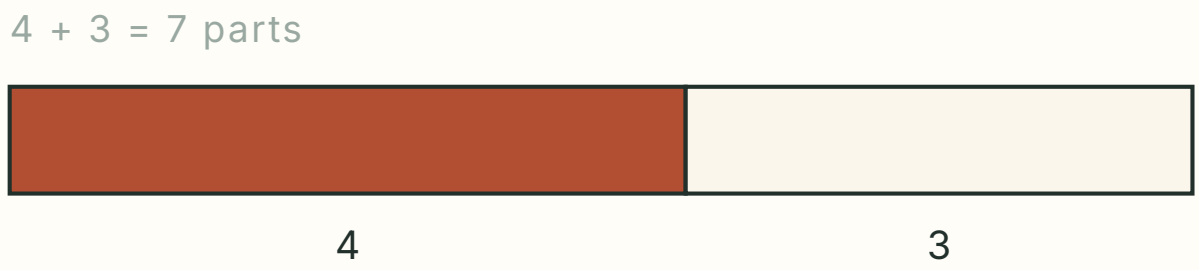
In your own words

CHAPTER 1 · 1.4 · WORKED EXAMPLE 1

Sharing and comparing with ratios

WORKED EXAMPLE 1.4.1

Problem. Share 56 sweets between two classes in the ratio 4:3.



Mathematical reasoning. The value of one part is the total divided by the number of parts.

1. Total parts: 4 + 3 = 7.
2. One part: 56 ÷ 7 = 8 sweets.
3. First class: 4 × 8 = 32. Second class: 3 × 8 = 24.
4. Check: 32 + 24 = 56.

Final answer. 32 sweets and 24 sweets

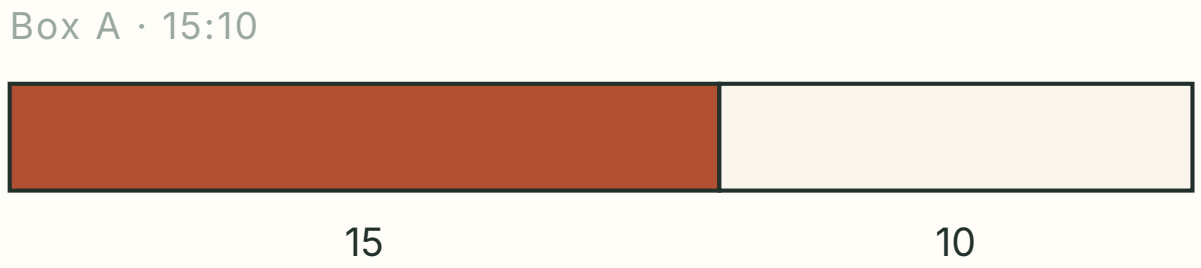
Interpretation. The check is part of the answer: the parts must add back to the total.

CHAPTER 1 · 1.4 · WORKED EXAMPLE 2

Sharing and comparing with ratios

WORKED EXAMPLE 1.4.2

Problem. Box A holds 15 red and 10 blue counters. Box B holds 21 red and 14 blue. Which box has the greater proportion of red counters?



Mathematical reasoning. Compare the same part in both ratios: red to total, or red to blue.

1. Box A: 15:10 = 3:2, so red is 3 of 5 parts = 60%.
2. Box B: 21:14 = 3:2, so red is also 60%.
3. The proportions are equal.

Final answer. Neither — both boxes have the same proportion of red counters

Interpretation. The totals are different, so only the simplified comparison answers the question.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

1.A1 FLUENCY

Write the ratio 8:12 in simplest form.

1.A2 FLUENCY

A donation of 6 red and 9 blue notebooks is shared between two year groups. Write red : blue : total in simplest form.

1.A3 FLUENCY

A school receives 45 exercise books and 60 pencils to share in one ratio. Write the ratio of books to pencils in simplest form.

1.A4 REASONING

Class A shares 6 apples and 9 oranges; class B shares 10 apples and 15 oranges. Are the two sharings equivalent? Justify.

1.A5 FLUENCY

A trail mix has 30 nuts and 20 raisins. Write nuts : total in simplest form.

1.A6 REASONING

Explain the difference between a part-part and a part-whole ratio using 5:7, and say which one describes 5 seats out of 12 on a student council.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

1.A7 FLUENCY

Two classes contribute 0.4 and 0.6 of their tuck-shop profit to a shared fund. Write the ratio of their contributions in simplest form.

1.A8 FLUENCY

A choir has 18 girls and 12 boys. Write boys : everyone.

1.A9 FLUENCY

Find the missing value: 7:9 = 21:?

1.A10 APPLICATION

Two teams score in the ratio 4:5. The first scores 32 points. How many does the second score?

1.A11 APPLICATION

A recipe needs 5 parts flour to 2 parts fat. With 400 g of fat, how much flour?

1.A12 REASONING

A donation is shared 3:4 between two year groups. Is 3:4 the same comparison as 4:3? Explain using the two groups.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

1.B1 FLUENCY

A donation is shared 5:8 between two year groups. Generate three equivalent sharings and record the multiplier you used.

1.B2 REASONING

Investigate: a donation shared 6:10 is halved before it is divided. Does multiplying both parts by 0.5 give a valid sharing, and is it still fair?

1.B3 REASONING

Find the odd one out: 9:12, 12:16, 15:20, 14:18, and say which sharing is not equivalent to the others.

1.B4 REASONING

Investigate the pattern 2:3, 4:6, 6:9, 8:12. Write the next two terms and a rule.

1.B5 REASONING

A pattern of squares has shaded : unshaded = 1:3 in figure 1. Predict the counts in figure 5.

1.B6 REASONING

Explain why 2:3 and 200:300 describe the same comparison but very different totals, using two donations of different sizes.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

1.B7 REASONING

Test the claim: if a:b = c:d then a + b = c + d. Give a counterexample using two donations.

1.B8 REASONING

Investigate: can two different simplest forms describe the same situation?

1.B9 FLUENCY

Find k so that 4:7 = 4k:35, and explain what k means when a sharing is scaled up.

1.B10 REASONING

Describe the relationship between the two rows of a ratio table.

1.B11 FLUENCY

Predict the missing pair in the sharing table 3:5, 6:10, 9:15, ?, 15:25 and justify the rule you used.

CRITERION C Communicating

Criterion C practice · Communicating

1.C1 FLUENCY

Complete: 5:8 = ? : 40, and explain which sharing this represents when 40 items are divided between two groups.

1.C2 REASONING

Explain in words how to use a double number line to solve 3:7 = 18:?, and say why the multiplier is the same on both lines.

1.C3 APPLICATION

A double number line shows 0, 4, 8 minutes above 0, 3, 6 pages of a shared reading record. How many pages in 20 minutes?

1.C4 FLUENCY

Write a ratio table for 7:9 up to four rows, and explain what each row represents when a donation is shared.

1.C5 REASONING

Which representation — table, double number line or bar model — best shows a missing value? Justify.

1.C6 APPLICATION

A relief map uses 1 cm : 5 km. Show 15 km as a table row and explain what the scale means.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

1.C7 REASONING

Convert the ratio 3:5 into a unit rate and explain what it means for one part of a sharing.

1.C8 FLUENCY

A co-operative sells shares at a constant price. Complete the table: cost 2, ?, 8, 10 dollars for 1, 3, ?, 5 shares.

1.C9 REASONING

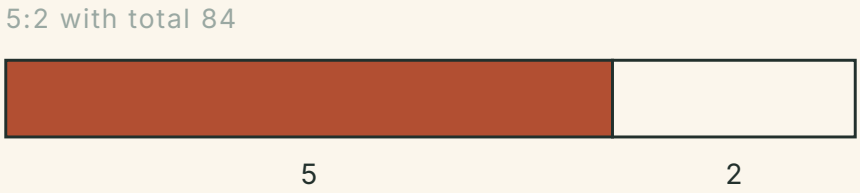
Explain why a ratio table and a coordinate graph can show the same relationship.

1.C10 APPLICATION

A water delivery line shows 0, 5, 10 litres above 0, 2, 4 bottles of a shared supply. How many litres fill 9 bottles?

1.C11 FLUENCY

Draw a bar model for 5:2 and label the value of one part when the total is 84.



1.C12 FLUENCY

A recipe scales from 4 servings to 10. Use a table to find the multiplier.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

1.D1 FLUENCY

Share 120 books between two year groups in the ratio 5:3.

1.D2 APPLICATION

A charity splits 2 400 dollars in the ratio 3:5. How much is the larger share?

1.D3 APPLICATION

Two classes collect bottles in the ratio 7:5. The smaller class collects 120. How many altogether?

1.D4 REASONING

A fair-share rule gives 60% of a 90-dollar profit to the maker and 40% to the seller. Express this as a ratio and find each share.

1.D5 REASONING

A school day is shared between five subjects in the ratio 2:2:1:1:1. How many minutes each in a 360-minute day?

1.D6 REASONING

Compare 15 girls out of 25 in class A with 18 girls out of 30 in class B. Which class has the greater proportion of girls, and what does this say about equal representation?

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

1.D7 APPLICATION

A nurse mixes 3 parts medicine to 7 parts water. How much medicine in 500 ml of mixture?

1.D8 REASONING

Three friends contribute 4, 6 and 10 hours. Share a 200-dollar prize in the ratio of hours and comment on fairness.

1.D9 APPLICATION

A paint mix is 1 part white to 4 parts blue. How much white for 2 litres of paint?

1.D10 REASONING

Two shops sell the same juice: 2 litres for 3.40 dollars and 3 litres for 4.80 dollars. Which is better value?

1.D11 CHALLENGE

A class votes yes:no = 5:3. If 12 more students vote yes the ratio becomes 3:1. How many voted?

1.D12 REASONING

Explain, with a calculation, why sharing in a ratio is not always the same as sharing equally.

CHAPTER 1 · APPLICATION AND ASSESSMENT

Applying chapter 1

These tasks use ratio language, equivalent ratios, ratio tables and sharing. Choose the representation that makes the comparison easiest to explain.

A1 · The community garden harvest

A garden produces 84 kg of vegetables. Three families worked in the ratio 3:4:5 of hours. Share the harvest in the ratio of hours, then explain whether this sharing is fair.

A2 · The dilution problem

A cleaner is sold as a concentrate. 250 ml of concentrate makes 2 litres of working solution. How much concentrate is needed for a 15-litre batch, and what is the ratio of concentrate to water in simplest form?

Criteria A and D · show the table you used and check that the shares add to 84 kg.

CHAPTER 1 · REVIEW

What you should now be able to do

Key ideas

- A ratio compares quantities of the same kind, and the order matters.
- Part-part and part-whole ratios answer different questions.
- Equivalent ratios are multiples of the same simplest form.
- A unit rate scales a ratio so that one part is 1, which makes comparison easy.
- To share an amount in a ratio, find the value of one part, then check the total.

Vocabulary

ratio · part-part · part-whole · simplest form · equivalent ratio · unit rate · ratio table · double number line · share

CHAPTER 1 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

1.R1 Simplify 36:48.

1.R2 Share 90 in the ratio 4:5.

1.R3 Complete 6:7 = ? : 42.

1.R4 Which is better value: 5 for 4 dollars or 8 for 6 dollars?

1.R5 A table shows 3, 6, 9 litres above 4, 8, 12 cups. How many cups for 15 litres?

1.R6 Explain why 2:5 and 5:2 are different comparisons.

CHAPTER 1 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> The form of a ratio, and the equivalence of its different representations, allow a limited resource to be shared fairly and allow that fairness to be justified and checked.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of equivalence and representation appear in your work?
- How is the global context of fairness and development, and its exploration of inequality, difference and inclusion, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of ratio reasoning for deciding what is fair, and what would you still like to ask?

CHAPTER 1 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Ratio reasoning is used wherever a limited resource is shared. Use it to act, not only to calculate.

Service as Action

COMMUNITY NEED	Some families in the school community cannot afford new school supplies.
MATHEMATICAL CONNECTION	Ratios and unit rates to compare donations and share resources fairly.
PROPOSED ACTION	Collect and sort donated supplies, then share them between two year groups in a ratio based on need.
EXPECTED IMPACT	Every learner in the chosen year groups starts the term with the same basic kit.
STUDENT REFLECTION	Record the ratio you used, why it was fair, and what you would change next time.

Community Project

PROJECT IDEA	A supplies-sharing audit for the school.
NEED	Donated supplies are currently handed out first come, first served.
RESEARCH QUESTION	What sharing rule gives the fairest distribution of the donated supplies?
MATHEMATICS	Ratios, equivalent ratios, unit rates and percentage comparison.
ACTION OR OUTCOME	A written sharing rule and a distribution plan for the next donation drive.
BENEFICIARIES	Students in the year groups receiving supplies.
EVIDENCE	The audit tables, the rule and a photograph of the sorted supplies.

Personal Project connection · MYP 5

PERSONAL INTEREST	Cooking, budgeting or sport team selection.
LEARNING GOAL	To use ratio reasoning to make a decision you can justify.
PRODUCT OR OUTCOME	A scaled recipe, a budget plan or a squad rotation table.
MATHEMATICS	Equivalent ratios, unit rates and proportional scaling.
PROCESS	Collect the numbers, build a ratio table, then compare options by unit rate.
EVIDENCE	The table, the decision and a short justification.

CHAPTER 2 · PART I · NUMBER AND PROPORTION

02

Proportional reasoning

Proportional reasoning moves from ratios to relationships: two quantities linked by a constant multiplier, drawn as a straight line through the origin.

ENGAGE

A recipe for 4 people uses 300 g of rice. How much rice for 6 people, and how much for 10? One relationship answers both questions.

Framing the inquiry

Key concept: Relationships
Related concepts: Equivalence · Representation · Logic
Global context: Globalisation and sustainability
Global context exploration: Markets, commodities and commercialisation

Statement of inquiry: A proportional relationship models two quantities that change together; the equivalence of its representations as a table, a unit rate and a straight line through the origin rests on the logic of one constant multiplier, and allows costs and scales to be compared, justified and predicted.

Questions to guide us

- **Factual:** What is a proportional relationship, what is its constant of proportionality, and how do you recognise one in a table, an equation and a graph?
- **Conceptual:** Why does every proportional relationship graph as a straight line through the origin, and what does the slope tell you that a table alone does not?
- **Debatable:** When two offers are compared per unit, is the cheaper unit price always the fairer choice for a buyer with very little money?

ATL · TRANSFER. In order for a student to apply skills and knowledge in unfamiliar situations, students must select a structure that fits the new situation, apply it independently, and test the result for sense.

CHAPTER 2 · INVESTIGATION 2.1

Recognising proportional relationships

INQUIRY QUESTION

How can you tell whether two quantities really change together in a proportional relationship, and what single number describes it?

YOUR TASK

For each table, divide the second row by the first and record the quotients. Decide whether the relationship is proportional.

x	1	2	3	4
A	5	10	15	20
B	5	9	13	17
C	4	8	12	16

CHAPTER 2 · INVESTIGATION 2.1

Recognising proportional relationships

What do you notice?

- A gives quotients 5, 5, 5, 5 — proportional.
- B gives quotients 5, 4.5, 4.33, 4.25 — not proportional.
- C gives quotients 4, 4, 4, 4 — proportional.

Observation. A relationship is proportional exactly when every quotient is the same.

MATHEMATICAL CONCLUSION

A relationship is proportional exactly when every quotient is the same, and that one quotient is the constant of proportionality k in $y = kx$.

Your own words

CHAPTER 2 · 2.1

Recognising proportional relationships

Two quantities are in a proportional relationship when one is always a constant multiple of the other. That constant is the constant of proportionality, often written k, and $y = kx$.

A quick test: divide each pair of matching values. If every quotient is the same, the relationship is proportional. If even one quotient differs, it is not.

Proportional relationships are a special case of linear relationships. Every proportional relationship is linear, but a linear relationship with a non-zero starting value is not proportional.

The unit rate is the constant of proportionality when the second quantity is 1. Comparing unit rates is the most reliable way to decide which of two offers is better value.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.1 · WORKED EXAMPLE 1

Recognising proportional relationships

WORKED EXAMPLE 2.1.1

Problem. A taxi charges 3.50 dollars per kilometre with no fixed fee. Is the cost proportional to distance? What is k?

distance (km)	1	2	5	10
cost ($)	3.50	7.00	17.50	35.00

Cost grows with distance

Mathematical reasoning. There is no fixed starting fee, so cost = k × distance with k = 3.50.

1. Divide cost by distance for each row: 3.50, 3.50, 3.50, 3.50.
2. Every quotient is the same, so the relationship is proportional.
3. k = 3.50, so cost = 3.50 × distance.

Final answer. Yes, proportional with k = 3.50 dollars per kilometre

Interpretation. Because k is constant, doubling the distance exactly doubles the cost.

CHAPTER 2 · 2.1 · WORKED EXAMPLE 2

Recognising proportional relationships

WORKED EXAMPLE 2.1.2

Problem. A plumber charges a 40-dollar call-out fee plus 25 dollars per hour. Is the charge proportional to hours worked?

hours	1	2	3	4
charge ($)	65	90	115	140

Charge with a fixed fee

Mathematical reasoning. A proportional relationship must have zero charge at zero hours. Here one hour already costs 65 dollars.

1. At 0 hours the charge is still 40 dollars, not 0.
2. The quotients are 65, 45, 38.3, 35 — not constant.
3. So the relationship is linear but not proportional.

Final answer. No — there is a fixed starting value of 40 dollars

Interpretation. A graph of this relationship is a straight line that does not pass through the origin.

CHAPTER 2 · INVESTIGATION 2.2

Unit rates and comparison

INQUIRY QUESTION

How does converting two different offers to the same unit rate turn an unfair comparison into a fair one?

YOUR TASK

Find three prices in a shop catalogue per 100 g, then convert each to price per kilogram and rank them.

product	price per 100 g	price per kg	rank
product 1			
product 2			
product 3			

CHAPTER 2 · INVESTIGATION 2.2

Unit rates and comparison

What do you notice?

- Multiplying by 10 converts per 100 g to per kg.
- Ranking by the converted rate is the same as ranking by unit price.
- The cheapest per kg is not always the cheapest packet.

Observation. Converting to a common unit rate makes different pack sizes comparable.

MATHEMATICAL CONCLUSION

Converting to a common unit rate puts both offers on the same basis, and that unit rate is the constant of proportionality, so it is unique to the situation.

Your own words

CHAPTER 2 · 2.2

Unit rates and comparison

A unit rate scales a ratio so the second quantity is 1: dollars per item, kilometres per hour, grams per dollar. Unit rates make unlike quantities comparable.

To compare two offers, convert both to the same unit rate. Whichever is smaller per unit is better value when you are paying; whichever is larger is better when you are earning.

Unit rates also let you convert units cleanly: 60 km/h is 1 km per minute, because 60 ÷ 60 = 1.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.2 · WORKED EXAMPLE 1

Unit rates and comparison

WORKED EXAMPLE 2.2.1

Problem. Shop A sells 5 kg of rice for 12 dollars. Shop B sells 8 kg for 18 dollars. Which is better value?



Mathematical reasoning. Convert both to cost per kilogram, then compare.

1. Shop A: 12 ÷ 5 = 2.40 dollars per kg.
2. Shop B: 18 ÷ 8 = 2.25 dollars per kg.
3. 2.25 < 2.40, so Shop B is better value.

Final answer. Shop B at 2.25 dollars per kg

Interpretation. Unit rates turn two different offers into two comparable numbers.

CHAPTER 2 · 2.2 · WORKED EXAMPLE 2

Unit rates and comparison

WORKED EXAMPLE 2.2.2

Problem. A cyclist travels 45 km in 3 hours. Find the speed in km/h and the distance covered in 20 minutes.



Mathematical reasoning. Speed is the constant of proportionality from time to distance.

1. Speed: 45 ÷ 3 = 15 km/h.
2. 20 minutes = 1/3 hour.
3. Distance: 15 × 1/3 = 5 km.

Final answer. 15 km/h and 5 km in 20 minutes

Interpretation. The unit rate is the bridge between any two matching values.

CHAPTER 2 · INVESTIGATION 2.3

Graphs of proportional relationships

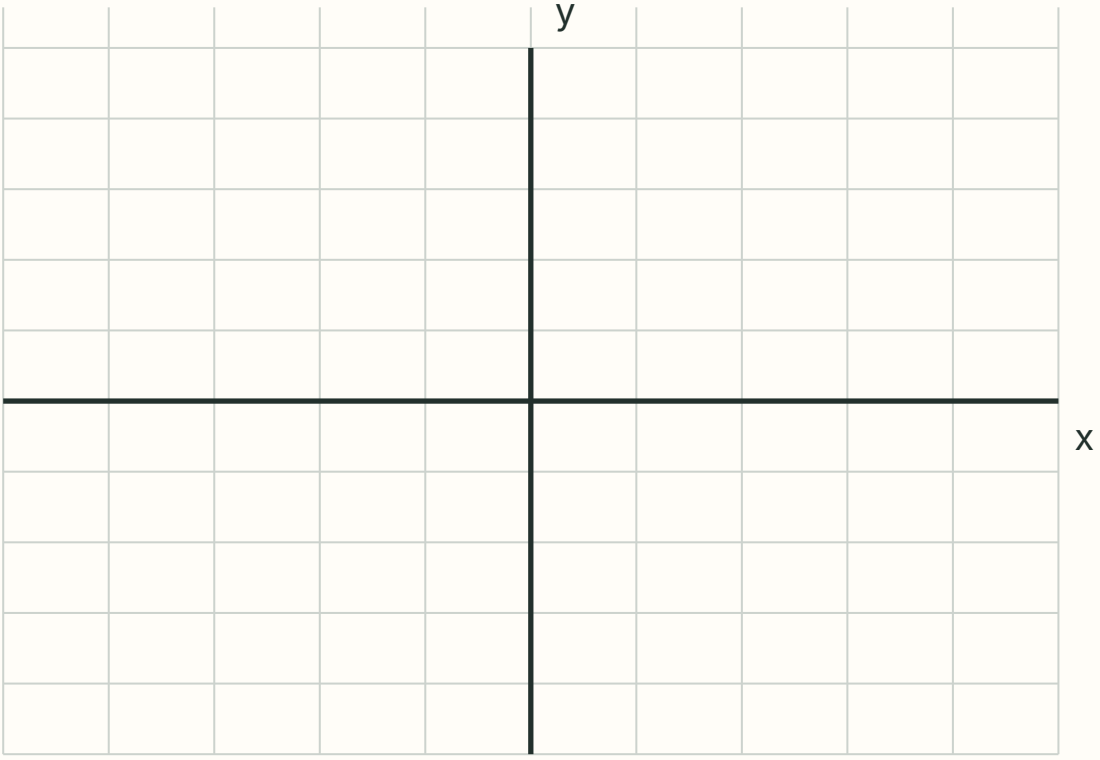
INQUIRY QUESTION

What does the graph of a proportional relationship show that a table of values cannot show?

YOUR TASK

Plot the points from a ratio table for 2:5 and join them. Then plot a table for a relationship that adds 2 each time. Compare the two graphs.

A proportional line and an additive line



x	1	2	3	4
proportional	2.5	5	7.5	10
additive	2	4	6	8

CHAPTER 2 · INVESTIGATION 2.3

Graphs of proportional relationships

What do you notice?

- The proportional line passes through the origin; the additive line does not.
- Both lines are straight, so both are linear.
- Only the proportional line has a constant quotient.

> **Observation.** Straight line does not mean proportional — passing through the origin is the extra condition.

MATHEMATICAL CONCLUSION

The graph shows the rate of growth as steepness: proportional relationships graph as straight lines through the origin whose slope is the constant k.

Your own words

CHAPTER 2 · 2.3

Graphs of proportional relationships

A proportional relationship graphs as a straight line through the origin. The slope of the line is the constant of proportionality.

Reading a value off the graph is a form of interpolation; reading beyond the data is extrapolation. Both are valid when the relationship genuinely stays proportional, and both need a note about that assumption.

The graph also shows what the table hides: the steepness of the line tells you at a glance which relationship grows faster.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.3 · WORKED EXAMPLE 1

Graphs of proportional relationships

WORKED EXAMPLE 2.3.1

Problem. A machine fills 60 bottles per minute. Draw the graph of bottles against time and read off the number filled in 7 minutes.

time (min)	1	2	5	10
bottles	60	120	300	600



Mathematical reasoning. Bottles = 60 × time, so the graph is a line through the origin with slope 60.

1. Plot the points (1, 60), (2, 120), (5, 300) and (10, 600).
2. Join them with a straight line through (0, 0).
3. At 7 minutes the line reaches 7 × 60 = 420.

Final answer. 420 bottles in 7 minutes

Interpretation. Reading from the line is legitimate only because the rate stays at 60 per minute.

CHAPTER 2 · 2.3 · WORKED EXAMPLE 2

Graphs of proportional relationships

WORKED EXAMPLE 2.3.2

Problem. Two printers print at 12 and 18 pages per minute. Sketch both lines and state which is steeper and why.

time (min)	1	2	3
printer P	12	24	36
printer Q	18	36	54

Two proportional relationships

Mathematical reasoning. Both lines pass through the origin; the steeper line has the larger constant.

1. P: pages = 12 × time. Q: pages = 18 × time.
2. At 3 minutes P gives 36 and Q gives 54.
3. Q's line is steeper because 18 > 12.

Final answer. Printer Q, with a constant of 18 pages per minute

Interpretation. On a graph, the constant of proportionality is the steepness of the line.

CHAPTER 2 · INVESTIGATION 2.4

Scaling and real-world models

INQUIRY QUESTION

When a shape or a quantity is scaled, what changes and what must stay the same for the model to remain faithful?

YOUR TASK

Enlarge a 3 cm by 2 cm rectangle by factors 2, 3 and 4. Record length, area and volume (if the rectangle becomes a prism of fixed height 1 cm) at each step.

factor	length	area	volume
1	6 cm	6 cm²	6 cm³
2	12 cm	24 cm²	24 cm³
3	18 cm	54 cm²	54 cm³
4	24 cm	96 cm²	96 cm³

CHAPTER 2 · INVESTIGATION 2.4

Scaling and real-world models

What do you notice?

- Lengths multiply by k, areas by k^2 and volumes by k^3.
- A fixed-height prism makes area and volume numerically equal, which is why the distinction is easy to miss.
- Changing the height too would separate area from volume.

Observation. Different measures scale by different powers of the scale factor.

MATHEMATICAL CONCLUSION

Under a scale factor k, lengths scale by k, areas by k^2 and volumes by k^3, so the shape stays the same while each measure changes by its own power of k.

Your own words

CHAPTER 2 · 2.4

Scaling and real-world models

Scaling multiplies every length by the same factor. Areas scale by the square of that factor and volumes by its cube — a fact that surprises most learners the first time.

A scale drawing keeps angles unchanged and multiplies every length by the scale factor. A scale model is a three-dimensional version of the same idea.

Percentage change is a proportional relationship with k = the percentage written as a decimal. A 15% increase multiplies by 1.15.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

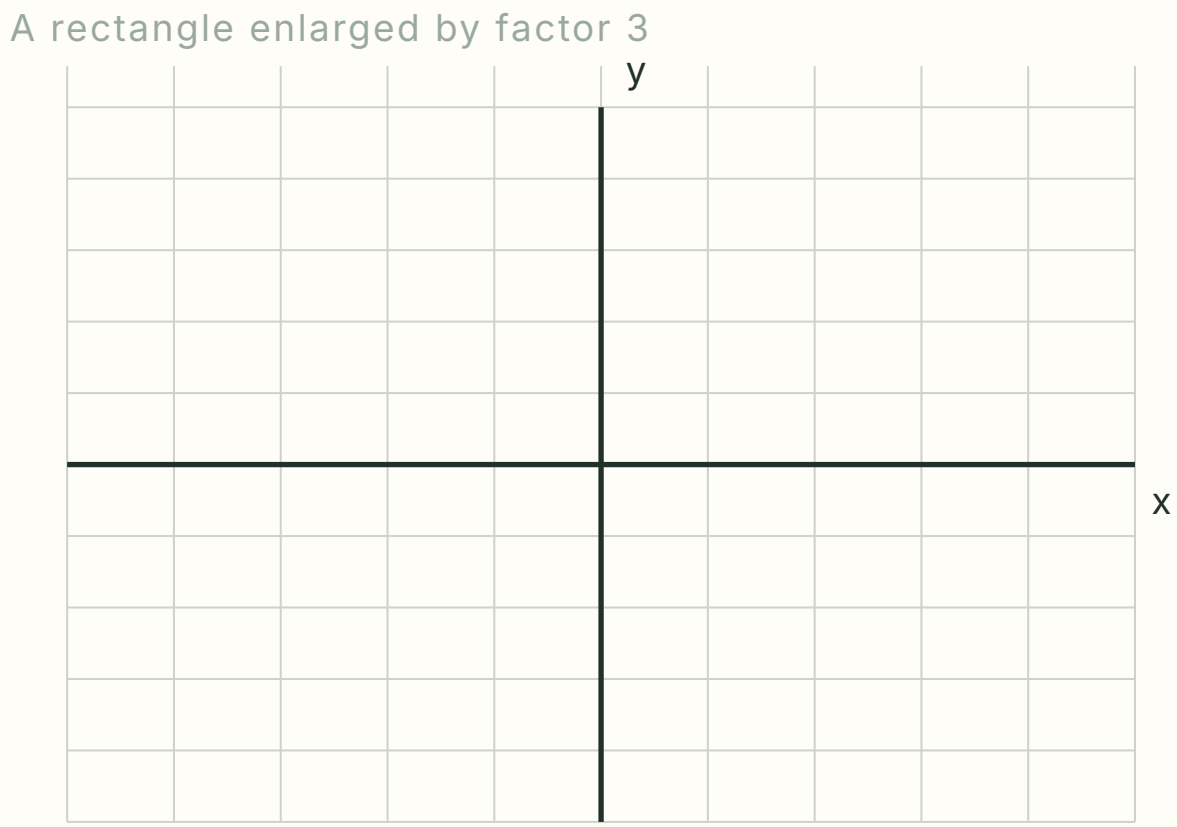
In your own words

CHAPTER 2 · 2.4 · WORKED EXAMPLE 1

Scaling and real-world models

WORKED EXAMPLE 2.4.1

Problem. A rectangle is 6 cm by 4 cm. It is enlarged by scale factor 3. Find the new dimensions and the new area.



Mathematical reasoning. Lengths scale by 3, so the area scales by $3^2 = 9$.

1. New dimensions: 18 cm by 12 cm.
2. New area: $18 \times 12 = 216$ cm^2.
3. Original area: 24 cm^2, and $24 \times 9 = 216$ cm^2.

Final answer. 18 cm by 12 cm; area 216 cm² (nine times the original)

Interpretation. The area grows much faster than the lengths, which is why scale models are not simple copies.

CHAPTER 2 · 2.4 · WORKED EXAMPLE 2

Scaling and real-world models

WORKED EXAMPLE 2.4.2

Problem. A price of 80 dollars increases by 15%. Find the new price.



Mathematical reasoning. A 15% increase multiplies by 1 + 0.15 = 1.15.

1. Multiplier: 1.15.
2. 80 × 1.15 = 92.

Final answer. 92 dollars

Interpretation. Percentage change is proportional reasoning with a known constant.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

2.A1 FLUENCY
A table gives x = 2, 4, 6 and y = 7, 14, 21. Is the relationship proportional? What is k?

2.A2 FLUENCY
A market stall sells rice at 4 dollars per kilogram. Write the equation linking cost c and mass x, and find the cost of 9 kg.

2.A3 REASONING
Is y = 2x + 3 proportional? Justify.

2.A4 FLUENCY
A rope costs 6 dollars per metre. Write the equation linking cost c and length l.

2.A5 APPLICATION
A tap fills a tank at 8 litres per minute. How much in 12 minutes?

2.A6 REASONING
Explain what the constant of proportionality tells you when a wholesaler prices goods by the kilogram.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

2.A7 APPLICATION

A car uses 0.07 litres per km. How much fuel for 350 km?

2.A8 REASONING

Test whether the price list (3 kg, 12 dollars), (5 kg, 20 dollars), (7 kg, 25 dollars) is proportional, and say what a buyer would conclude.

2.A9 APPLICATION

A market office prints 15 flyers per minute for traders. How many flyers in 4.5 minutes?

2.A10 REASONING

A linear relationship has $y = 5$ when $x = 0$. Can it be proportional?

2.A11 REASONING

A plant grows 2 cm each week. Write the relationship and find the height after 8 weeks from 10 cm.

2.A12 FLUENCY

Find k when $y = 2.5x$, and say what k means if y is the cost in dollars and x the mass in kilograms.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

2.B1 FLUENCY

A delivery van covers 240 km in 4 hours. Find the unit rate and say what it tells a trader about delivery cost per kilometre.

2.B2 APPLICATION

Which is better value: 3 kg for 7.50 dollars or 5 kg for 12 dollars?

2.B3 FLUENCY

Convert 90 km/h to km per minute and explain which unit is more useful for a short delivery across a market.

2.B4 APPLICATION

Investigate: a worker earns 120 dollars for 8 hours. Find the hourly rate and the pay for 15 hours.

2.B5 REASONING

Explain how a unit rate helps compare two different-sized packs.

2.B6 FLUENCY

Find the unit rate for 8 apples costing 6 dollars, and say what it means.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

2.B7 APPLICATION

A car travels 150 km on 12 litres. Find km per litre.

2.B8 REASONING

Investigate the pattern: speed 30, 60, 90, 120 km/h against time 4, 2, 1.33, 1 h. What is constant?

2.B9 APPLICATION

A baker's recipe for 4 loaves uses 500 g of flour. How much flour for 7 loaves, and what is the unit rate per loaf?

2.B10 REASONING

Explain why a unit rate is also the constant of proportionality.

2.B11 FLUENCY

Convert 0.5 dollars per 100 ml to dollars per litre.

CRITERION C Communicating

Criterion C practice · Communicating

2.C1 FLUENCY

Sketch the graph of cost against mass for rice at 3 dollars per kg, and state its slope.

2.C2 REASONING

Explain why a graph of price against quantity must pass through the origin, using the idea that no goods cost nothing.

2.C3 REASONING

Two market lines pass through the origin with slopes 2 and 5 dollars per kg. Which grows faster, and how do you know from the graph?

2.C4 FLUENCY

Read from the line $y = 4x$ the value of y when $x = 6$.

2.C5 REASONING

Explain the difference between interpolation and extrapolation on a proportional graph.

2.C6 APPLICATION

A graph of delivery cost against distance is a line through the origin passing (2, 90). Find the cost per kilometre.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

2.C7 FLUENCY

Draw the table of values for y = 2.5x from x = 0 to 4.

2.C8 REASONING

Explain why a straight line of price against quantity that does not pass through the origin cannot represent a unit price.

2.C9 APPLICATION

A graph shows the volume of milk delivered against time with slope 0.4 litres per minute. How much after 12 minutes?

2.C10 REASONING

Explain how the slope of a proportional line relates to the ratio table of price against quantity.

2.C11 FLUENCY

Sketch the cost lines for 0.50 dollars and 2 dollars per item on the same axes and say which is steeper.

2.C12 APPLICATION

A line through the origin passes through (5, 3). Find the equation and say what it could represent in a market.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

2.D1 APPLICATION

A trader's site plan uses the scale 1:200. A stall measures 4.5 cm on the plan. Find its real length in metres.

2.D2 APPLICATION

A scale model of a market stall is 12 cm long for a real stall of 4.8 m. Find the scale factor.

2.D3 APPLICATION

A 6 cm cube carton is enlarged by factor 4 for a display. Find the new volume and the ratio of new to old volume.

2.D4 APPLICATION

A salary of 2 400 dollars rises by 8%. Find the new salary.

2.D5 REASONING

A price falls by 20% from 75 dollars. Find the new price and explain the multiplier.

2.D6 REASONING

A square market awning has its side doubled. By what factors do its perimeter and its area grow?

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

2.D7 APPLICATION
A market poster 8 cm by 6 cm is enlarged so the width becomes 20 cm. Find the new height.

2.D8 REASONING
A shirt costs 35 dollars plus 15% tax. Find the total and explain the two steps.

2.D9 REASONING
A tank holds 500 litres and leaks 8 litres per hour. Is the volume proportional to time?

2.D10 APPLICATION
Two similar market stalls have floor areas 20 m^2 and 80 m^2. Find the scale factor between them.

2.D11 APPLICATION
A trader's sales rise from 15 to 18 units per day. Find the percentage increase.

2.D12 REASONING
Explain, with an example, why doubling the side of a square stall does not double its floor area.

CHAPTER 2 · APPLICATION AND ASSESSMENT

Applying chapter 2

Use tables, unit rates and graphs. In every task, state the constant that links the two quantities.

B1 · The school trip budget

A trip costs 18 dollars per student plus a 150-dollar bus. Write the total cost for n students, decide whether the cost is proportional to n, and find the cost for 45 students.

B2 · The scale model

A model of the school is built at scale 1:200. The real building is 18 m tall. Find the model height, and find by what factor the model's surface area differs from the real one.

Criteria B and D · justify your decision with the right unit and state the constant.

CHAPTER 2 · REVIEW

What you should now be able to do

Key ideas

- A proportional relationship is $y = kx$ with a constant quotient.
- The unit rate is k when the second quantity is 1.
- Every proportional relationship is linear, but not every linear relationship is proportional.
- A proportional graph is a straight line through the origin with slope k.
- Under scale factor k, lengths scale by k, areas by k^2 and volumes by k^3.
- Percentage change is proportional reasoning with multiplier $1 \pm$ the rate.

Vocabulary

proportional · constant of proportionality · unit rate · origin test · slope · interpolate · extrapolate · scale factor · scale drawing · similar

CHAPTER 2 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

2.R1 Is y = 7x proportional? What is k?

2.R2 A tap fills 3.5 litres per minute. Volume after 18 minutes?

2.R3 Which is better value: 2 kg for 5 dollars or 3 kg for 7 dollars?

2.R4 A 5 cm cube is enlarged by factor 3. Find the volume ratio.

2.R5 A price of 60 dollars rises by 12%. Find the new price.

2.R6 Explain why y = x + 4 is not proportional.

CHAPTER 2 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

A proportional relationship models two quantities that change together; the equivalence of its representations as a table, a unit rate and a straight line through the origin rests on the logic of one constant multiplier, and allows costs and scales to be compared, justified and predicted.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of equivalence, representation and logic appear in your work?
- How is the global context of globalisation and sustainability, and its exploration of markets, commodities and commercialisation, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a proportional model, and what would you still like to ask?

CHAPTER 2 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Proportional reasoning is the mathematics of scaling: recipes, maps, models and currency. Use it on a real project.

Service as Action

COMMUNITY NEED	Local small producers struggle to price goods fairly across different pack sizes.
MATHEMATICAL CONNECTION	Unit rates and proportional reasoning to compare and set prices.
PROPOSED ACTION	Build a price-per-unit chart for three local products and share it with the producers.
EXPECTED IMPACT	Producers can price competitively and buyers can compare fairly.
STUDENT REFLECTION	Explain how unit rates changed the pricing decisions.

Community Project

PROJECT IDEA	A school fair pricing guide.
NEED	Stalls price similar items in different-sized portions, so buyers cannot compare.
RESEARCH QUESTION	Which stall offers the best value per unit, and how should items be priced?
MATHEMATICS	Unit rates, proportional relationships and percentage comparison.
ACTION OR OUTCOME	A price-per-unit poster and a recommendation for next year's fair.
BENEFICIARIES	Fair visitors and stall holders.
EVIDENCE	The comparison table, the poster and photographs of the stalls.

Personal Project connection · MYP 5

PERSONAL INTEREST	Fitness, cooking or travel planning.
LEARNING GOAL	To plan a schedule or budget using proportional reasoning.
PRODUCT OR OUTCOME	A training plan, a scaled menu or a travel budget.
MATHEMATICS	Unit rates, proportional graphs and percentage change.
PROCESS	Collect data, build a table or graph, then use the constant to predict.
EVIDENCE	The table or graph and the predictions made from it.

CHAPTER 3 · PART I · NUMBER AND PROPORTION

03

Rates and conversions

A rate compares quantities of different kinds. Converting units, reading speed and working with density are all rate problems in disguise.

ENGAGE

A lorry travels 240 km in 3 hours. A cyclist travels 45 km in 2 hours. Which is faster, and by how much?

Framing the inquiry

Key concept: Logic
Related concepts: Representation · Equivalence
Global context: Scientific and technical innovation
Global context exploration: Precision, accuracy and measurement

Statement of inquiry: A rate compares quantities of different kinds, and the logic of converting units precisely relies on the equivalence of different representations of the same quantity, which allows speeds, densities and pressures to be measured, compared and trusted.

Questions to guide us

- **Factual:** What two units does a rate compare, and how do you convert between compound units such as km/h and m/s?
- **Conceptual:** Why does converting to the smaller unit before dividing protect the accuracy of a rate, and how does a unit rate make unlike quantities comparable?
- **Debatable:** Is an average speed ever an honest description of a journey, and what evidence would you need to defend your answer?

ATL · COMMUNICATION. In order for a student to communicate information and ideas effectively, students must organise a structure for a purpose, use appropriate notation, and check that a reader can follow the argument.

CHAPTER 3 · INVESTIGATION 3.1

Understanding rates

INQUIRY QUESTION

How can one number compare two journeys of different lengths and durations?

YOUR TASK

Walk or wheel a marked distance three times at different paces. Record distance and time, then compute the average speed for each trial and for all three together.

trial	distance (m)	time (s)	speed (m/s)
1			
2			
3			
all three			

CHAPTER 3 · INVESTIGATION 3.1

Understanding rates

What do you notice?

- Each speed is distance ÷ time for that trial.
- The combined speed is total distance ÷ total time.
- The combined speed is not the mean of the three trial speeds.

> **Observation.** Average speed is a single rate computed from totals.

MATHEMATICAL CONCLUSION

One number — total distance ÷ total time — compares journeys of any length and duration, which is why average speed is defined from totals.

Your own words

CHAPTER 3 · 3.1

Understanding rates

A rate compares quantities of different kinds, so it always carries two units: kilometres per hour, dollars per kilogram, pages per minute. A ratio compares quantities of the same kind.

Speed, price per unit and flow are the three rates learners meet most often. Each is a constant of proportionality between its two quantities.

Average speed is total distance divided by total time, not the average of the speeds. Confusing the two is the most common error in rate problems.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 3 · 3.1 · WORKED EXAMPLE 1

Understanding rates

WORKED EXAMPLE 3.1.1

Problem. A cyclist covers 45 km in 2 hours. Find the average speed.



Mathematical reasoning. Average speed is total distance divided by total time.

1. Distance = 45 km and time = 2 h.
2. 45 ÷ 2 = 22.5.

Final answer. 22.5 km/h

Interpretation. The units km/h come directly from distance divided by time.

CHAPTER 3 · 3.1 · WORKED EXAMPLE 2

Understanding rates

WORKED EXAMPLE 3.1.2

Problem. A journey has two parts: 60 km at 30 km/h, then 60 km at 60 km/h. Find the average speed for the whole journey.

part	distance	time
1	60 km	2 h
2	60 km	1 h
total	120 km	3 h

Total distance ÷ total time

Mathematical reasoning. The average speed uses the total distance and the total time, not the mean of the two speeds.

1. Time for part 1: 60 ÷ 30 = 2 h. Time for part 2: 60 ÷ 60 = 1 h.
2. Total distance 120 km; total time 3 h.
3. 120 ÷ 3 = 40 km/h.

Final answer. 40 km/h

Interpretation. The mean of 30 and 60 would be 45 km/h, which is wrong — the slow part takes longer.

CHAPTER 3 · INVESTIGATION 3.2

Converting units

INQUIRY QUESTION

How can the same rate be written in different units without changing its value?

YOUR TASK

Convert 5 km/h into m/s two ways: first dividing by 3.6, then converting km to m and hours to s separately. Compare the two answers.

method	working	answer
÷ 3.6	5 ÷ 3.6	1.388… m/s
unit by unit	5 000 ÷ 3 600	1.388… m/s

CHAPTER 3 · INVESTIGATION 3.2

Converting units

What do you notice?

- Both methods give the same value.
- 5 000 ÷ 3 600 = 50 ÷ 36 = 25 ÷ 18, a recurring decimal.
- Leaving the answer as a fraction 25/18 m/s is exact.

Observation. Different conversion routes give the same rate.

MATHEMATICAL CONCLUSION

Converting units is proportional reasoning with a fixed constant, so the same rate can be written in different units without changing its value.

Your own words

CHAPTER 3 · 3.2

Converting units

Conversion is proportional reasoning with a known constant. 1 km = 1 000 m means every kilometre is 1 000 metres, so distance in metres = 1 000 × distance in kilometres.

To convert a compound rate, convert one unit at a time and keep the fraction structure. 60 km/h becomes 60 000 m/h, then 1 000 m/min, then about 16.7 m/s.

Choosing the order of conversion matters for accuracy: converting to the smaller unit first avoids repeating decimals.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 3 · 3.2 · WORKED EXAMPLE 1

Converting units

WORKED EXAMPLE 3.2.1

Problem. Convert 72 km/h into m/s.



Mathematical reasoning. Convert kilometres to metres and hours to seconds separately, keeping the fraction.

1. 72 km/h = 72 000 m/h.
2. 1 h = 3 600 s, so 72 000 ÷ 3 600 = 20.

Final answer. 20 m/s

Interpretation. Dividing by 3.6 converts km/h to m/s exactly, because 1 000 ÷ 3 600 = 1/3.6.

CHAPTER 3 · 3.2 · WORKED EXAMPLE 2

Converting units

WORKED EXAMPLE 3.2.2

Problem. A bottle holds 1.5 litres. How many 250 ml glasses does it fill?

litres	ml	glasses
1.5	1500	6

Litres to millilitres

Mathematical reasoning. Convert to the same unit, then divide by the glass size.

1. 1.5 litres = 1 500 ml.
2. 1 500 ÷ 250 = 6.

Final answer. 6 glasses

Interpretation. Converting to the smaller unit first makes the division exact.

CHAPTER 3 · INVESTIGATION 3.3

Density and pressure

INQUIRY QUESTION

How can a rate compare objects of different sizes and shapes fairly?

YOUR TASK

Weigh three objects of similar size but different material. Estimate each volume by displacement in a measuring cylinder, then compute density and rank them.

object	mass (g)	volume (cm³)	density (g/cm³)
1			
2			
3			

CHAPTER 3 · INVESTIGATION 3.3

Density and pressure

What do you notice?

- Objects of similar size can have very different masses.
- Density separates them because the volume is nearly the same.
- The densest object sinks first in water (density 1 g/cm^3).

Observation. Density is mass per unit volume, so it compares objects of different sizes fairly.

MATHEMATICAL CONCLUSION

Density and pressure are rates, so dividing by the size — volume or area — compares objects of different sizes fairly: density = mass ÷ volume and pressure = force ÷ area.

Your own words

CHAPTER 3 · 3.3

Density and pressure

Density is mass per unit volume: density = mass ÷ volume. Two objects with the same volume can have very different masses, which is what density measures.

Pressure is force per unit area: pressure = force ÷ area. A smaller area with the same force gives a larger pressure, which is why a knife is sharp.

Both are rates, so both use the same reasoning: fix the denominator to 1 and compare.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 3 · 3.3 · WORKED EXAMPLE 1

Density and pressure

WORKED EXAMPLE 3.3.1

Problem. A block has mass 480 g and volume 240 cm^3. Find its density and identify the material if aluminium is 2.7 g/cm^3 and wood is 0.6 g/cm^3.

mass	volume	density
480 g	240 cm^3	?

Mass ÷ volume

Mathematical reasoning. Density is mass per unit volume, so divide.

1. 480 ÷ 240 = 2 g/cm^3.
2. 2 is not 2.7 or 0.6, so the material is neither of the two listed.

Final answer. 2 g/cm^3 — not aluminium and not the wood given

Interpretation. Density identifies materials because each substance has its own value.

CHAPTER 3 · 3.3 · WORKED EXAMPLE 2

Density and pressure

WORKED EXAMPLE 3.3.2

Problem. A force of 600 N acts on an area of 0.5 m^2. Find the pressure, and find the pressure if the area is halved.

force	area	pressure
600 N	0.5 m^2	?

Force ÷ area

Mathematical reasoning. Pressure is force per unit area, so halving the area doubles the pressure.

1. 600 ÷ 0.5 = 1 200 Pa.
2. Halving the area to 0.25 m^2 gives 600 ÷ 0.25 = 2 400 Pa.

Final answer. 1 200 Pa, and 2 400 Pa when the area is halved

Interpretation. Pressure is inversely proportional to area when the force is fixed.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

3.A1 APPLICATION
A train travels 300 km in 4 hours. Find its average speed.

3.A2 REASONING
Define a rate and give one example with its units.

3.A3 FLUENCY
A machine makes 240 bottles in 8 minutes. Find the rate per minute.

3.A4 FLUENCY
A tap fills 45 litres in 9 minutes. Find the flow rate.

3.A5 REASONING
Explain why average speed is not usually the mean of two speeds.

3.A6 APPLICATION
A cyclist rides 12 km in 40 minutes. Find the speed in km/h.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

3.A7 FLUENCY

A worker earns 96 dollars in 6 hours. Find the hourly rate.

3.A8 APPLICATION

A delivery van uses 8 litres of fuel per 100 km. How much fuel for a 350 km route, and how accurate must the measurement be?

3.A9 APPLICATION

A label printer prints 18 labels per minute. How many labels in 15 minutes?

3.A10 REASONING

Explain the difference between a ratio and a rate.

3.A11 CHALLENGE

Two taps fill a tank in 6 and 12 minutes. Together, in how many minutes?

3.A12 APPLICATION

A journey of 150 km takes 2.5 hours. Find the speed.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

3.B1 FLUENCY

Convert 5.4 km into metres.

3.B2 FLUENCY

Convert 250 cm^2 into m^2.

3.B3 FLUENCY

Convert 3.5 hours into minutes.

3.B4 FLUENCY

Investigate: convert 90 km/h to m/s by dividing by 3.6.

3.B5 FLUENCY

Convert 2.5 m^3 into litres.

3.B6 REASONING

Explain why converting area and volume uses powers of ten, not ten itself.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

3.B7 APPLICATION

Convert 45 miles per hour to km/h using 1 mile = 1.6 km.

3.B8 FLUENCY

Investigate: how many 20-cent coins make 15.40 dollars, and what does this tell you about the precision of the coin count?

3.B9 REASONING

Convert 0.75 kg per litre to g per cm^3.

3.B10 REASONING

Explain why you should convert to the smaller unit before dividing.

3.B11 FLUENCY

Convert 1.2 tonnes to kg.

CRITERION C Communicating

Criterion C practice · Communicating

3.C1 FLUENCY

A block has mass 900 g and volume 300 cm^3. Find its density.

3.C2 REASONING

Explain what density tells you that mass alone does not.

3.C3 FLUENCY

A force of 200 N acts on 4 m^2. Find the pressure.

3.C4 REASONING

Explain why a sharp knife cuts better than a blunt one.

3.C5 REASONING

Two objects have the same mass but different volumes. Which is denser?

3.C6 APPLICATION

A liquid has density 1.2 g/cm^3. Find the mass of 250 cm^3.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

3.C7 REASONING
Explain the relationship between pressure and area when force is fixed.

3.C8 APPLICATION
A metal cube of side 4 cm has mass 512 g. Find its density.

3.C9 REASONING
Convert 1 000 kg/m^3 into g/cm^3.

3.C10 REASONING
Explain how density can identify an unknown material.

3.C11 APPLICATION
A pressure of 300 Pa acts on 0.2 m^2. Find the force.

3.C12 REASONING
Explain why a heavy lorry uses wide tyres.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

3.D1 APPLICATION

A car travels 120 km at 60 km/h then 120 km at 40 km/h. Find the average speed.

3.D2 APPLICATION

A fuel tank holds 60 litres. The car uses 7.5 litres per 100 km. Find the range.

3.D3 APPLICATION

A delivery van carries 1.2 tonnes. The load area is 3 m^2. Find the pressure in Pa if the weight force is 12 000 N.

3.D4 APPLICATION

Two cyclists start together. One rides at 18 km/h, the other at 15 km/h. How far apart after 2 hours?

3.D5 APPLICATION

A recipe for 6 people uses 750 g of rice. How much for 25 people?

3.D6 APPLICATION

A ship's hull displaces 4 000 m^3 of seawater of density 1 025 kg/m^3. Find the mass displaced.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

3.D7 APPLICATION

A runner completes 10 km in 42 minutes. Find the speed in m/s.

3.D8 APPLICATION

A tap fills 0.4 m^3 per hour. Find the flow in litres per minute.

3.D9 REASONING

Explain, with a calculation, why two runners with the same average speed can have very different races.

3.D10 APPLICATION

A machine prints 12 labels per minute. How long for 300 labels?

3.D11 APPLICATION

Convert 5 gallons to litres using 1 gallon = 4.5 litres.

3.D12 APPLICATION

A block of wood is 10 cm by 5 cm by 4 cm and has mass 100 g. Find its density.

CHAPTER 3 · APPLICATION AND ASSESSMENT

Applying chapter 3

Convert first, calculate second. State the units of every answer.

C1 · The school run

A bus travels 18 km at 36 km/h, stops for 5 minutes, then travels 12 km at 24 km/h. Find the total time and the average speed for the whole run.

C2 · The packing problem

A box of mass 12 kg rests on a base of 0.3 m^2. Find the pressure in Pa, then find the base area needed to halve the pressure.

Criteria B and D · state the formula before substituting.

CHAPTER 3 · REVIEW

What you should now be able to do

Key ideas

- A rate compares different kinds of quantity and carries two units.
- Average speed = total distance ÷ total time.
- Converting units is proportional reasoning with a known constant.
- Areas and volumes convert with powers of ten, not ten itself.
- Density = mass ÷ volume and pressure = force ÷ area.

Vocabulary

rate · unit · average speed · flow rate · conversion · density · pressure · compound unit

CHAPTER 3 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

3.R1 A car travels 210 km in 3 hours. Find the speed.

3.R2 Convert 2.4 km/h into m/s.

3.R3 A mass of 600 g has volume 150 cm^3. Find the density.

3.R4 A force of 450 N acts on 1.5 m^2. Find the pressure.

3.R5 Explain why area conversions use 100 and 10 000 rather than 10.

3.R6 A tap fills 90 litres in 12 minutes. Find the flow rate.

CHAPTER 3 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> A rate compares quantities of different kinds, and the logic of converting units precisely relies on the equivalence of different representations of the same quantity, which allows speeds, densities and pressures to be measured, compared and trusted.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of logic and the related concepts of representation and equivalence appear in your work?
- How is the global context of scientific and technical innovation, and its exploration of precision, accuracy and measurement, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a single rate as a description, and what would you still like to ask?

CHAPTER 3 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Rates govern travel, trade and consumption. Measure a rate in your community and report it with correct units.

Service as Action

COMMUNITY NEED	Water is collected and carried by hand in parts of the community.
MATHEMATICAL CONNECTION	Flow rates and unit conversion to size containers and plan collection.
PROPOSED ACTION	Measure a tap's flow rate and use it to design a water collection schedule for a household.
EXPECTED IMPACT	Less time spent collecting water and a reliable daily supply.
STUDENT REFLECTION	Record the measurements and explain how the rate informed the plan.

Community Project

PROJECT IDEA	A school traffic and speed survey.
NEED	Drivers speed near the school gate at drop-off and pick-up times.
RESEARCH QUESTION	What is the average speed of traffic outside the school, and how does it vary by time of day?
MATHEMATICS	Rates, unit conversion and average speed.
ACTION OR OUTCOME	A speed report with a recommendation for a traffic-calming measure.
BENEFICIARIES	Students, families and staff.
EVIDENCE	The measurement table, the report and photographs of the site.

Personal Project connection · MYP 5

PERSONAL INTEREST	Cycling, running or gaming.
LEARNING GOAL	To measure and improve a personal performance rate.
PRODUCT OR OUTCOME	A training log with speeds and a target pace.
MATHEMATICS	Rates, unit conversion and averages.
PROCESS	Measure distance and time, compute the rate, then set a target.
EVIDENCE	The log, the rate calculations and the target.

CHAPTER 4 · PART I · NUMBER AND PROPORTION

04

Integers and their operations

Integers extend the number line in both directions. Once direction is clear, the rules for operating with negative numbers follow from the model.

ENGAGE

The temperature at dawn is −4 °C. By noon it has risen 9 degrees, and by midnight it falls 11 degrees. What is the temperature at midnight?

Framing the inquiry

Key concept: Logic
Related concepts: Representation · Patterns
Global context: Orientation in space and time
Global context exploration: Natural and human landscapes and resources

Statement of inquiry: Signed numbers are a representation of position relative to a chosen reference point, and the logic of direction on the number line, together with the patterns in the sign rules, allows changes in temperature, depth and balance to be described, ordered and predicted.

Questions to guide us

- **Factual:** What is an integer, how are integers ordered on the number line, and what does the sign of a number tell you?
- **Conceptual:** Why does subtracting a negative number increase the value, and how does the number line explain each of the sign rules?
- **Debatable:** Is a negative balance or a negative temperature a real quantity, or only a convention about where we place zero?

ATL · REFLECTION. In order for a student to evaluate their own learning and progress, students must consider the process of learning, identify strengths and weaknesses, and set a plan for improvement.

CHAPTER 4 · INVESTIGATION 4.1

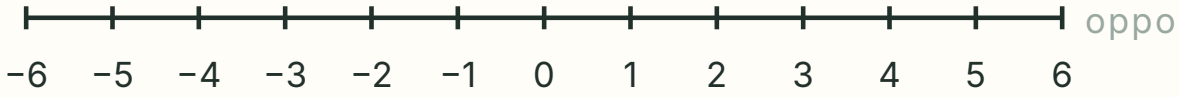
The integer number line

INQUIRY QUESTION

How can a single line represent numbers both above and below a reference point, and how does it decide which of two integers is greater?

YOUR TASK

Mark the integers from −6 to +6 on a number line. Then find pairs of numbers that are the same distance from zero, and record their sum.



pair	distances from 0	sum
−3 and 3	3 and 3	0
−6 and 6	6 and 6	0
your pair		

CHAPTER 4 · INVESTIGATION 4.1

The integer number line

What do you notice?

- Each pair sits symmetrically about zero.
- Every such pair sums to zero.
- The only number that is its own opposite is 0.

Observation. Opposites are symmetric about zero and sum to zero.

MATHEMATICAL CONCLUSION

One line with a chosen reference point represents numbers in both directions: opposites are symmetric about zero and sum to zero ($a + (-a) = 0$), and the further right a number lies the greater it is.

Your own words

CHAPTER 4 · 4.1

The integer number line

Integers are the whole numbers together with their negatives: ... −3, −2, −1, 0, 1, 2, 3 ... The number line gives them an order: a number further right is always greater.

Zero is the reference point. A positive number lies to the right of zero, a negative number to the left. The sign gives the direction and the digit gives the distance from zero.

Opposites are pairs like +5 and −5. They sit the same distance from zero on opposite sides, and they add to zero.

Inequality signs can be read directly from the line: $-7 < -2$ because −7 lies further left.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

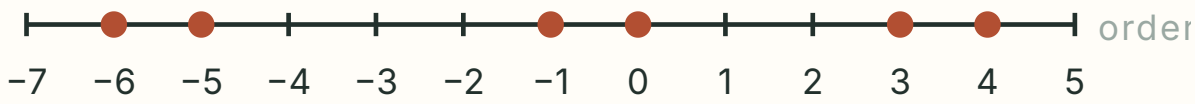
In your own words

CHAPTER 4 · 4.1 · WORKED EXAMPLE 1

The integer number line

WORKED EXAMPLE 4.1.1

Problem. Order the integers −5, 3, −1, 0, 4, −6 from smallest to greatest.



Mathematical reasoning. On the number line, the smallest value is furthest left.

1. Place all six values on the line.
2. Read from left to right: −6, −5, −1, 0, 3, 4.

Final answer. −6, −5, −1, 0, 3, 4

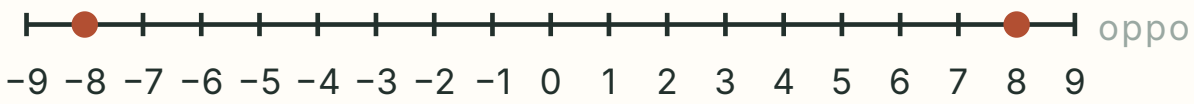
Interpretation. The line turns an abstract ordering into a left-to-right reading.

CHAPTER 4 · 4.1 · WORKED EXAMPLE 2

The integer number line

WORKED EXAMPLE 4.1.2

Problem. Find the opposite of −8 and state what the two numbers add to.



Mathematical reasoning. An opposite is the same distance from zero on the other side.

1. The distance from −8 to 0 is 8 units.
2. The point 8 units to the right of 0 is +8.
3. −8 + 8 = 0.

Final answer. The opposite of −8 is +8, and they sum to 0

Interpretation. Opposites cancel because they represent equal and opposite directions.

CHAPTER 4 · INVESTIGATION 4.2

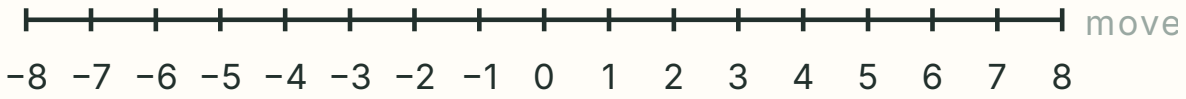
Adding and subtracting integers

INQUIRY QUESTION

How can the number line explain what happens when a negative number is added or subtracted?

YOUR TASK

Start at 0 on a number line and apply the moves +4, −6, +3, −2. Record the position after each move, then find a different order that gives the same final position.



move	position
start	0
+4	4
−6	−2
+3	1
−2	−1

CHAPTER 4 · INVESTIGATION 4.2

Adding and subtracting integers

What do you notice?

- The final position is −1 whatever the order.
- Only the total move matters, not the order of the moves.
- Reversing the order of addition always gives the same result.

Observation. Integer addition is commutative and associative.

MATHEMATICAL CONCLUSION

Addition is a move along the line, so subtracting a negative moves right, and the total of a set of moves does not depend on the order in which they are made.

Your own words

CHAPTER 4 · 4.2

Adding and subtracting integers

Addition is a move along the line: adding a positive moves right, adding a negative moves left. Subtraction is the same move in the opposite direction.

The two rules follow from this model. Adding a negative is the same as subtracting the positive: $5 + (-3) = 5 - 3 = 2$. Subtracting a negative is the same as adding the positive: $5 - (-3) = 5 + 3 = 8$.

A written calculation can always be checked on a number line. If the move and the answer disagree, the mistake is in the arithmetic, not in the model.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

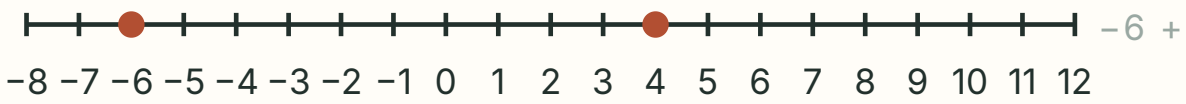
In your own words

CHAPTER 4 · 4.2 · WORKED EXAMPLE 1

Adding and subtracting integers

WORKED EXAMPLE 4.2.1

Problem. Evaluate −6 + 10 and explain the move on the number line.



Mathematical reasoning. Start at −6 and move 10 units to the right.

1. Start at −6.
2. Move 10 units right: −5, −4, −3, −2, −1, 0, 1, 2, 3, 4.
3. The landing point is 4.

Final answer. −6 + 10 = 4

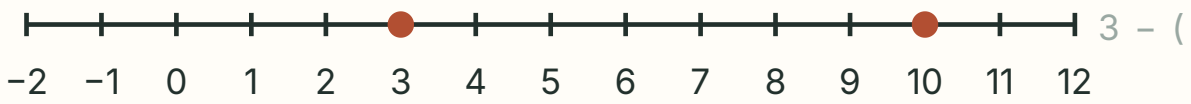
Interpretation. A positive addition always moves right, whatever the starting point.

CHAPTER 4 · 4.2 · WORKED EXAMPLE 2

Adding and subtracting integers

WORKED EXAMPLE 4.2.2

Problem. Evaluate 3 − (−7) and explain why the value increases.



Mathematical reasoning. Subtracting a negative is adding the opposite, so the move is to the right.

1. 3 − (−7) = 3 + 7.
2. Start at 3 and move 7 units right.
3. The landing point is 10.

Final answer. 3 − (−7) = 10

Interpretation. Removing a debt increases what you have, which is why the value grows.

CHAPTER 4 · INVESTIGATION 4.3

Multiplying and dividing integers

INQUIRY QUESTION

Where do the sign rules for multiplication and division come from, and why is a negative times a negative positive?

YOUR TASK

Complete the multiplication table for $(-2) \times n$ for $n = -4$ to 4 and describe the pattern.

n	-4	-3	-2	-1	0	1	2	3	4
$(-2) \times n$	8	6	4	2	0	-2	-4	-6	-8

CHAPTER 4 · INVESTIGATION 4.3

Multiplying and dividing integers

What do you notice?

- Each step to the right reduces the product by 2.
- The sign changes when n changes sign.
- The product is zero only when n is zero.

Observation. The sign of a product is fixed by the signs of the factors.

MATHEMATICAL CONCLUSION

The sign rules come from continuing a pattern rather than from a separate law: same signs give a positive product or quotient, different signs give a negative one.

Your own words

CHAPTER 4 · 4.3

Multiplying and dividing integers

Multiplication is repeated addition, so $3 \times (-4)$ means three moves of -4: -12. The sign rule follows: a positive times a negative is negative.

The other cases complete the pattern. Negative times positive is negative, negative times negative is positive. The pattern in the table of products is the reason for the rule, not a separate law.

Division uses the same sign rules as multiplication, because division is multiplication by the reciprocal.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 4 · 4.3 · WORKED EXAMPLE 1

Multiplying and dividing integers

WORKED EXAMPLE 4.3.1

Problem. Complete the pattern: $4 \times 3 = 12$, $3 \times 3 = 9$, $2 \times 3 = 6$, $1 \times 3 = 3$, $0 \times 3 = 0$, $-1 \times 3 = ?$

multiplier	product
3	9
2	6
1	3
0	0
−1	?

A product pattern

Mathematical reasoning. Each step reduces the product by 3, so the next term must be −3.

1. 12, 9, 6, 3, 0, then −3.
2. So $(-1) \times 3 = -3$.

Final answer. $(-1) \times 3 = -3$

Interpretation. The sign rules come from continuing a pattern, not from an arbitrary law.

CHAPTER 4 · 4.3 · WORKED EXAMPLE 2

Multiplying and dividing integers

WORKED EXAMPLE 4.3.2

Problem. Evaluate (−5) × (−4) and explain the result.

a	b	a × b
−5	−4	?

Negative times negative

Mathematical reasoning. Continuing the pattern of (−5) × 4, (−5) × 3, … shows the products rising by 5 each step.

1. (−5) × 4 = −20, (−5) × 3 = −15, (−5) × 2 = −10, (−5) × 1 = −5, (−5) × 0 = 0.
2. The next step is (−5) × (−1) = +5, and so (−5) × (−4) = +20.

Final answer. (−5) × (−4) = 20

Interpretation. A negative times a negative is positive because the pattern demands it.

CHAPTER 4 · INVESTIGATION 4.4

Integers in context

INQUIRY QUESTION

How can signed numbers describe a change in a real quantity such as temperature, depth or balance?

YOUR TASK

Record the temperature at the same time on five consecutive days, including at least one below zero. Find the range and the mean, and explain what the negative value means.

day	temperature	change from previous da
1		
2		
3		
4		
5		

CHAPTER 4 · INVESTIGATION 4.4

Integers in context

What do you notice?

- The change between days is found by subtraction.
- A negative change means the temperature fell.
- The range is the largest value minus the smallest.

Observation. Signed numbers record both direction and size of change.

MATHEMATICAL CONCLUSION

A change is the difference of two signed values measured from the same reference point, so its sign shows the direction of the change.

Your own words

CHAPTER 4 · 4.4

Integers in context

Negative numbers model anything measured against a reference: temperature, height above sea level, bank balance, golf scores. The context decides what zero means.

Choosing a sensible reference point makes a problem easier. Sea level is zero for height; the starting balance is zero for a bank account.

When several changes happen in sequence, add them as signed numbers. The order does not matter, only the total change.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

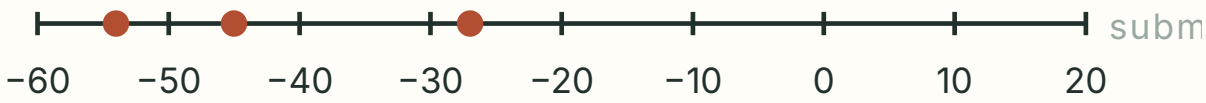
In your own words

CHAPTER 4 · 4.4 · WORKED EXAMPLE 1

Integers in context

WORKED EXAMPLE 4.4.1

Problem. A submarine descends 45 m, rises 18 m, then descends 27 m. Find its final depth relative to the surface.



Mathematical reasoning. Treat each move as a signed number and add them.

1. Start at 0.
2. Descend 45 m: −45. Rise 18 m: −45 + 18 = −27. Descend 27 m: −27 − 27 = −54.
3. The final position is −54.

Final answer. 54 m below the surface

Interpretation. Sea level is the zero reference, and depth is a negative height.

CHAPTER 4 · 4.4 · WORKED EXAMPLE 2

Integers in context

WORKED EXAMPLE 4.4.2

Problem. An account holds 120 dollars. A payment of 145 dollars is made. Find the balance and describe what it means.

transaction	balance
start	120
payment	−25

A negative balance

Mathematical reasoning. Subtract the payment from the balance.

1. $120 - 145 = -25$.

Final answer. −25 dollars, an overdraft of 25 dollars

Interpretation. A negative balance is meaningful only because zero is the starting amount.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

4.A1 FLUENCY

A survey records five points on a hillside relative to a marker at 0 m: −3, 0, 5, −7, 2 metres. Place them on a number line and order them from lowest to highest.

4.A2 FLUENCY

Write the opposite of −12 and explain what the two numbers describe about two points either side of a reference level.

4.A3 FLUENCY

Two boreholes reach −5 m and −9 m below the reference level. Which is greater, and which is deeper? Justify with the number line.

4.A4 FLUENCY

A shelter is 4 m below the reference level and a marker is 4 m above it. Write the sum of their heights as an integer.

4.A5 REASONING

Explain what zero represents in a temperature scale.

4.A6 FLUENCY

Write an integer for '8 m below sea level'.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

4.A7 FLUENCY

Order the heights −2, 4, −6, 1, −1 metres of five landmarks from greatest to smallest.

4.A8 REASONING

Find the opposite of the opposite of −7, and explain what two reflections in the reference level do to a height.

4.A9 REASONING

Explain why every integer has exactly one opposite, using the number line as a mirror in zero.

4.A10 REASONING

Is 0 a height above or below the reference level? Justify your answer.

4.A11 APPLICATION

A lift moves from floor −2 to floor 5. How many floors does it pass?

4.A12 FLUENCY

Two wells are 6 m apart in depth and one of them is at −1 m. Write the two possible depths.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

4.B1 FLUENCY

A hiker starts 7 m above a marker and descends 10 m. Write the calculation for the final height and evaluate it.

4.B2 FLUENCY

A submarine at −8 m rises 3 m. Write this as a subtraction of a negative and evaluate it.

4.B3 FLUENCY

The temperature at a mountain station is −12 °C and rises 5 °C. Write the calculation and the new temperature.

4.B4 REASONING

Investigate: does the order of addition change the result for −4 + 9 and 9 + (−4)?

4.B5 FLUENCY

A drone at 6 m altitude descends 11 m. Draw the move on a number line and give the final altitude.

4.B6 REASONING

Investigate the pattern −3, −1, 1, 3. Write the next two terms and a rule.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

4.B7 REASONING

Explain why subtracting a negative increases the value, using a forecast that removes an expected fall in temperature.

4.B8 FLUENCY

A mine lowers its cage by 5 m in each of three loads. Write the total change as a repeated addition and evaluate it.

4.B9 APPLICATION

A temperature of −6 °C rises by 4 °C. Find the new temperature.

4.B10 REASONING

Explain how a number line shows that a temperature of 3 °C falling by 8 °C gives −5 °C.

4.B11 FLUENCY

Evaluate −2 − 7 and describe the move in terms of depth below a reference level.

CRITERION C Communicating

Criterion C practice · Communicating

4.C1 REASONING

Explain in your own words why (−4) × 6 is negative, using the pattern in (−4) × 3, (−4) × 2, (−4) × 1, (−4) × 0.

4.C2 APPLICATION

A lift descends 4 m every minute for 6 minutes. Write the calculation as a multiplication, draw a number line for the first three minutes, and state the final depth.

4.C3 REASONING

Write a rule in words that works for any two integers with different signs, and test it on 24 ÷ (−6).

4.C4 REASONING

Explain why (−2) × (−3) is positive, using a pattern in the products.

4.C5 REASONING

Two students disagree about (−7) × (−1): one says −7, the other says 7. Decide who is right and write the reasoning you would show them.

4.C6 REASONING

Explain why 0 × (−9) = 0, using the idea of repeated moves of −9 taken no times.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

4.C7 REASONING

Explain the sign rule for division and show how it follows from the rule for multiplication.

4.C8 APPLICATION

A scuba diver ascends in 5 equal stages covering −30 m in total. Write the division that gives one stage, explain the sign, and evaluate it.

4.C9 APPLICATION

Explain what $(-2)^3$ means as repeated multiplication, evaluate it, and say what the sign tells you about the direction of the change.

4.C10 REASONING

Explain why $(-2)^4$ is positive but $(-2)^3$ is negative, using the idea of pairs of negative factors.

4.C11 REASONING

Write a general rule for the sign of a power of a negative number, and use it to predict $(-1) \times (-1) \times (-1)$.

4.C12 APPLICATION

A debt of 12 dollars is shared equally among 4 people. Write the division, evaluate it, and explain what the sign of the answer means.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

4.D1 APPLICATION
A bank balance of 250 dollars has payments of 180 and 120 dollars. Find the balance.

4.D2 APPLICATION
A diver at −20 m descends 15 m then rises 22 m. Find the final depth.

4.D3 APPLICATION
The temperature falls from 3 °C to −8 °C. Find the change.

4.D4 REASONING
A golf score is 4 above par on day 1 and 3 below par on day 2. Find the total.

4.D5 APPLICATION
A company gains 12 000 dollars then loses 15 000 dollars. Find the net change.

4.D6 REASONING
Explain, with a calculation, what a negative height above sea level means.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

4.D7 APPLICATION

A lift starts at floor −3, rises 8 floors, then falls 5. Find the final floor.

4.D8 APPLICATION

A temperature of −5 °C changes by −7 °C. Find the new temperature.

4.D9 REASONING

A debt of 40 dollars is reduced by a payment of 55 dollars. Find the balance.

4.D10 REASONING

Explain why the order of a series of deposits and withdrawals does not change the final balance.

4.D11 APPLICATION

A mountain is 1 200 m above sea level and a trench is 800 m below. Find the difference in height.

4.D12 APPLICATION

A construction site records its daily changes in material stock, in tonnes, as +3, −1, +4, −2. Find the net change over the four days.

CHAPTER 4 · APPLICATION AND ASSESSMENT

Applying chapter 4

Draw a number line for every task. Direction is the whole argument.

D1 · The cold week

Temperatures for five days are −3, 2, −5, 1 and −4 °C. Find the mean temperature, the range, and the total change from day 1 to day 5.

D2 · The bank statement

An account starts at 180 dollars. Transactions are −95, +40, −120 and +60 dollars. Find the final balance and explain what a negative balance would mean here.

Criteria C and D - show the signed additions and check the total.

CHAPTER 4 · REVIEW

What you should now be able to do

Key ideas

- Integers extend the number line in both directions from zero.
- A number further right is greater.
- Adding a negative moves left; subtracting a negative moves right.
- Same signs give a positive product or quotient; different signs give a negative one.
- Zero is a reference point, and the context decides what it means.

Vocabulary

integer · opposite · number line · reference point · signed number · ascending · descending · commutative

CHAPTER 4 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

4.R1 Order −4, 7, −1, 0, 3.

4.R2 Evaluate −9 + 4.

4.R3 Evaluate 5 − (−8).

4.R4 Evaluate (−6) × (−3).

4.R5 Evaluate (−20) ÷ 4.

4.R6 A lift goes from −4 to 6. How many floors?

CHAPTER 4 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> Signed numbers are a representation of position relative to a chosen reference point, and the logic of direction on the number line, together with the patterns in the sign rules, allows changes in temperature, depth and balance to be described, ordered and predicted.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of logic and the related concepts of representation and patterns appear in your work?
- How is the global context of orientation in space and time, and its exploration of natural and human landscapes and resources, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a number line as a model, and what would you still like to ask?

CHAPTER 4 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Negative numbers appear wherever a quantity can fall below a reference point. Find one in your community.

Service as Action

COMMUNITY NEED	Households in the community struggle to track small debts and repayments.
MATHEMATICAL CONNECTION	Integers to model balances that can fall below zero.
PROPOSED ACTION	Design a simple ledger template using signed numbers and teach it to a group of traders.
EXPECTED IMPACT	Clearer records and fewer disputed balances.
STUDENT REFLECTION	Explain how signed numbers made the records clearer.

Community Project

PROJECT IDEA	A community temperature and energy audit.
NEED	Cooling and heating costs are hard to predict without temperature data.
RESEARCH QUESTION	How does the daily temperature range relate to energy use in the community?
MATHEMATICS	Integers, signed differences and averages.
ACTION OR OUTCOME	A temperature log and a recommendation for energy saving.
BENEFICIARIES	Households in the community.
EVIDENCE	The log, the calculations and the recommendation.

Personal Project connection · MYP 5

PERSONAL INTEREST	Sport, gaming or personal finance.
LEARNING GOAL	To use signed numbers to track progress against a target.
PRODUCT OR OUTCOME	A score log or savings tracker with positive and negative entries.
MATHEMATICS	Integers, addition of signed numbers and range.
PROCESS	Record entries, compute the running balance, and interpret the sign.
EVIDENCE	The log and a written interpretation of the final value.

CHAPTER 5 · PART II · ALGEBRA AND PATTERN

05

Algebraic expressions and equations

Letters stand for numbers. Building, simplifying and solving expressions turns a single calculation into a rule that works for every value.

ENGAGE

A taxi costs 3 dollars plus 2 dollars per kilometre. Instead of calculating each fare separately, can one formula give every fare?

Framing the inquiry

Key concept: Logic
Related concepts: Patterns · Representation
Global context: Scientific and technical innovation
Global context exploration: Products, processes and solutions

Statement of inquiry: A variable represents an unknown or changing quantity, and the logic of equivalence between different representations of the same relationship, together with the patterns in its structure, allows one expression or equation to model many situations, so that processes can be designed, solved and checked.

Questions to guide us

- **Factual:** What does a variable represent, what are like terms, and how do you build an expression from a written description?
- **Conceptual:** Why can 3x and 5x be collected but 3x and 5 cannot, and how does expanding show that two expressions are equivalent?
- **Debatable:** Is algebra a new kind of mathematics or a shorthand for arithmetic, and what would count as evidence for your view?

ATL · CREATIVE THINKING. In order for a student to generate novel ideas and consider new perspectives, students must use brainstorming and visual diagrams to develop ideas, and make guesses and ask 'what if' questions.

CHAPTER 5 · INVESTIGATION 5.1

Building expressions

INQUIRY QUESTION

How can a letter stand for a number so that one expression describes every possible case?

YOUR TASK

Build expressions for the number of tiles in each of three patterns of squares, then test your expression on a new figure.

Figure 1, 2 and 3 of the pattern

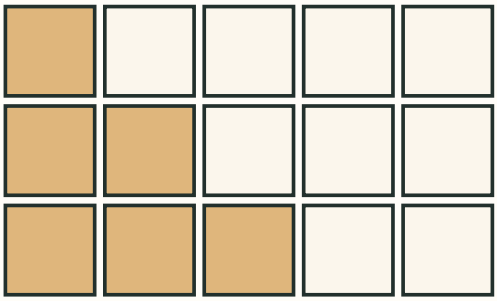


figure 1: 1 tile · figure 2: 3 · figure 3: 6

figure	tiles counted	expression	test figure 5
1			
2			
3			

CHAPTER 5 · INVESTIGATION 5.1

Building expressions

What do you notice?

- Counting works for small figures but not for large ones.
- An expression works for every figure.
- Two different expressions can be equivalent, e.g. 2n + 2 and 2(n + 1).

Observation. A variable replaces an unknown count so one expression covers all figures.

MATHEMATICAL CONCLUSION

A variable lets one expression act as a rule for every case: substituting a value applies the rule.

Your own words

CHAPTER 5 · 5.1

Building expressions

A variable is a letter that stands for a number that can change. A constant is a fixed number. In 3x + 5, x is the variable and 5 is the constant.

An expression is built from a situation: 'three more than twice a number' becomes 2x + 3. The order of operations decides the structure, so read the words carefully.

Substitution replaces the variable with a value and evaluates. In 3x + 5 with x = 4, the value is 17.

Algebra is a shorthand: 2(x + 3) means 'double the total', which is different from 2x + 3, which means 'double, then add three'.

> FORMAL DEVELOPMENT
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.1 · WORKED EXAMPLE 1

Building expressions

WORKED EXAMPLE 5.1.1

Problem. A garden is x metres long and 5 metres wider than it is long. Write expressions for the width and the perimeter.

quantity	expression
length	x
width	x + 5
perimeter	4x + 10

From words to expressions

Mathematical reasoning. Perimeter is twice the sum of length and width, so I substitute each expression.

1. Width = x + 5 metres.
2. Perimeter = 2(length + width) = 2(x + x + 5).
3. = 2(2x + 5) = 4x + 10 metres.

Final answer. width = x + 5 m and perimeter = 4x + 10 m

Interpretation. One expression now describes every garden of this shape, whatever x is.

CHAPTER 5 · 5.1 · WORKED EXAMPLE 2

Building expressions

WORKED EXAMPLE 5.1.2

Problem. Evaluate 4a − 3b when a = 5 and b = 2.

a	b	4a − 3b
5	2	?

Substituting values

Mathematical reasoning. Replace each letter with its value, then follow the order of operations.

1. 4a = 4 × 5 = 20.
2. 3b = 3 × 2 = 6.
3. 20 − 6 = 14.

Final answer. 14

Interpretation. Substitution turns an expression into an ordinary calculation.

CHAPTER 5 · INVESTIGATION 5.2

Simplifying and expanding

INQUIRY QUESTION

What makes two expressions the same, and how can you show it without testing every number?

YOUR TASK

Take the expression $2(x + 3)$. Expand it, then test both forms with $x = 1$, 2 and 5. Record whether the values agree.

x	2(x + 3)	2x + 6
1	8	8
2	10	10
5	16	16

CHAPTER 5 · INVESTIGATION 5.2

Simplifying and expanding

What do you notice?

- Both forms give the same value for every x tested.
- Expanding changes the form, not the value.
- Testing with a value is a reliable check on any simplification.

Observation. Equivalent expressions have the same value for every substitution.

MATHEMATICAL CONCLUSION

Two expressions are equivalent when they agree for every value of the variable, which is why expanding and collecting change the form but never the value.

Your own words

CHAPTER 5 · 5.2

Simplifying and expanding

Like terms have the same variables raised to the same powers. $3x$ and $5x$ are like terms; $3x$ and $5x^2$ are not. Only like terms can be collected.

Simplifying collects like terms: $3x + 5x = 8x$. The coefficients add; the variable part is unchanged.

Expanding multiplies each term inside a bracket by the term outside: $3(x + 4) = 3x + 12$. Expanding and collecting are opposite processes, and both are checked by substituting a value.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.2 · WORKED EXAMPLE 1

Simplifying and expanding

WORKED EXAMPLE 5.2.1

Problem. Simplify $5x + 3 - 2x + 7$.

term	type
$5x$	x term
3	constant
$-2x$	x term
7	constant

Sorting the terms of $5x + 3 - 2x + 7$

Mathematical reasoning. Group the x terms and the constant terms separately.

1. x terms: $5x - 2x = 3x$.
2. Constants: $3 + 7 = 10$.
3. $3x + 10$.

Final answer. $3x + 10$

Interpretation. Only like terms combine, so the variable part never changes.

CHAPTER 5 · 5.2 · WORKED EXAMPLE 2

Simplifying and expanding

WORKED EXAMPLE 5.2.2

Problem. Expand and simplify $4(x + 3) - 2(x - 1)$.

step	expression
expand	$4x + 12 - 2x + 2$
collect	$2x + 14$

Expand then collect

Mathematical reasoning. Multiply each bracket by the term outside, watching the signs, then collect.

1. $4(x + 3) = 4x + 12$.
2. $-2(x - 1) = -2x + 2$.
3. Collect: $4x - 2x + 12 + 2 = 2x + 14$.

Final answer. $2x + 14$

Interpretation. The minus sign before a bracket changes the sign of every term inside.

CHAPTER 5 · INVESTIGATION 5.3

Solving equations

INQUIRY QUESTION

How can an equation be solved by keeping it true at every step?

YOUR TASK

Solve $4x + 3 = 19$ by trial, then by the balance method. Compare the two routes.

method	working	answer
trial	$x = 3$ gives 15; $x = 4$ gives 19	$x = 4$
balance	$4x = 16$, $x = 4$	$x = 4$

CHAPTER 5 · INVESTIGATION 5.3

Solving equations

What do you notice?

- Trial works but is slow and misses non-integer answers.
- The balance method finds every answer exactly.
- Both routes give x = 4, and both pass the check.

Observation. The balance method is a systematic version of trial and improvement.

MATHEMATICAL CONCLUSION

An equation is solved by keeping it true at every step: applying the same operation to both sides preserves the solution.

Your own words

CHAPTER 5 · 5.3

Solving equations

An equation states that two expressions are equal. Solving it means finding the value that makes the statement true.

The balance method keeps the equation true by doing the same thing to both sides: add, subtract, multiply or divide by the same amount.

Always check a solution by substituting it back into the original equation. A solution that fails the check is a signal that a sign or a step went wrong.

> FORMAL DEVELOPMENT
>
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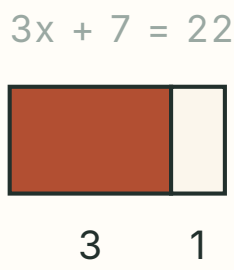
In your own words

CHAPTER 5 · 5.3 · WORKED EXAMPLE 1

Solving equations

WORKED EXAMPLE 5.3.1

Problem. Solve 3x + 7 = 22.



Mathematical reasoning. Undo the +7 by subtracting 7 from both sides, then divide by 3.

1. 3x + 7 − 7 = 22 − 7, so 3x = 15.
2. 3x ÷ 3 = 15 ÷ 3, so x = 5.
3. Check: 3(5) + 7 = 22.

Final answer. x = 5

Interpretation. The balance method works because the equation stays true at every step.

CHAPTER 5 · 5.3 · WORKED EXAMPLE 2

Solving equations

WORKED EXAMPLE 5.3.2

Problem. Solve $2(x - 4) = 10$.



Mathematical reasoning. Expand first, or divide both sides by 2. Either route is valid.

1. Divide both sides by 2: $x - 4 = 5$.
2. Add 4 to both sides: $x = 9$.
3. Check: $2(9 - 4) = 10$.

Final answer. $x = 9$

Interpretation. Choosing the shorter route reduces the chance of an arithmetic slip.

CHAPTER 5 · INVESTIGATION 5.4

Equations in context

INQUIRY QUESTION

How can an equation model a real situation, and how do you decide whether its solution is meaningful?

YOUR TASK

Write a word problem for $3x + 5 = 20$, solve it, then write a second problem whose solution is not a whole number and discuss what that means.

problem	equation	solution	meaningful?
your problem	3x + 5 = 20	x = 5	yes
second problem	2x + 1 = 4	x = 1.5	depends on contex

CHAPTER 5 · INVESTIGATION 5.4

Equations in context

What do you notice?

- The same equation can describe many situations.
- A fractional answer is meaningful for money but not for people.
- Interpreting the answer is part of solving the problem.

Observation. The context decides whether a solution is acceptable.

MATHEMATICAL CONCLUSION

Solve, then check that the answer makes sense in the original situation, because the context decides which solutions are acceptable.

Your own words

CHAPTER 5 · 5.4

Equations in context

Word problems become equations in three steps: define the variable with its unit, write the relationship as an equation, solve and interpret the answer in the original context.

The interpretation step matters: a negative length or a fractional number of people tells you the answer is not meaningful in the situation, even if the algebra is correct.

Inequalities describe ranges rather than single values. They are written with the same balance method, and the direction of the inequality sign is preserved when both sides are multiplied by a positive number.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

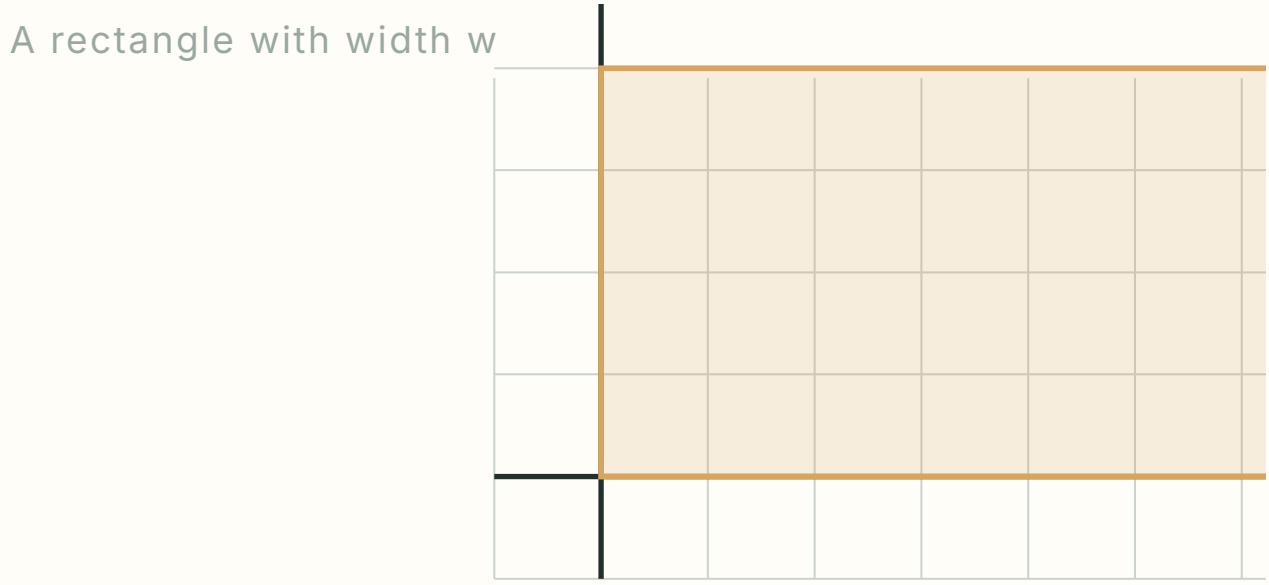
In your own words

CHAPTER 5 · 5.4 · WORKED EXAMPLE 1

Equations in context

WORKED EXAMPLE 5.4.1

Problem. A rectangle has perimeter 36 cm and is twice as long as it is wide. Find its dimensions.



Mathematical reasoning. Write the perimeter in terms of w, set it equal to 36 and solve.

1. Let the width be w, so the length is 2w.
2. Perimeter = 2(w + 2w) = 6w.
3. 6w = 36, so w = 6 and the length is 12.
4. Check: 2(6 + 12) = 36.

Final answer. 6 cm by 12 cm

Interpretation. The answer is interpreted as dimensions, not as a bare number.

CHAPTER 5 · 5.4 · WORKED EXAMPLE 2

Equations in context

WORKED EXAMPLE 5.4.2

Problem. A phone plan costs 12 dollars a month plus 0.05 dollars per minute. How many minutes can be used for a bill of at most 25 dollars?

minutes	cost
100	17
200	22
260	25

Cost against minutes

Mathematical reasoning. Write the cost as an expression, then write the limit as an inequality.

1. Cost = 12 + 0.05m.
2. 12 + 0.05m ≤ 25.
3. 0.05m ≤ 13, so m ≤ 260.
4. At most 260 minutes.

Final answer. at most 260 minutes

Interpretation. An inequality gives a range of allowed values, not a single answer.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

5.A1 FLUENCY

A machine's output is 7 units more than the number of cycles n it completes. Write an expression for the output.

5.A2 FLUENCY

A process costs 3 dollars per cycle plus a 2-dollar start-up fee. Evaluate the cost for 4 cycles.

5.A3 REASONING

In the formula C = 5y − 8 for the cost of a prototype, identify the variable and the constant and say what each represents.

5.A4 FLUENCY

A coating design uses twice the batch number decreased by 5 grams. Write an expression for the mass.

5.A5 FLUENCY

A finish is applied in 4 layers, each adding 2 mm to a base of a mm. Evaluate 4(a + 2) when a = 3 and say what it models.

5.A6 REASONING

Explain the difference between 2x + 3 and 2(x + 3) when x is the number of parts in a kit.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

5.A7 FLUENCY

A book costs p dollars and a pen costs 2 dollars less. Write the pen's cost.

5.A8 FLUENCY

A tank holds 10 litres and loses 3k litres when a valve is open for k hours. Evaluate the remaining volume for k = 2.

5.A9 FLUENCY

Write an expression for the cost of n pens at 3 dollars each plus a 5-dollar bag.

5.A10 REASONING

Explain what a variable is, using the number of parts n in a product design as your example.

5.A11 FLUENCY

A square panel of side a has area a^2 and its frame adds 2a. Evaluate $a^2 + 2a$ for a = 3 and say what each term counts.

5.A12 FLUENCY

Write an expression for 'the product of 4 and a number, increased by 1'.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

5.B1 FLUENCY

A kit contains 7 bolts and 2 nuts per unit. Simplify the total number of fixings for x units.

5.B2 FLUENCY

One carton design uses 5a cm^2 of board plus 3 cm^2 of tape; a second uses 2a cm^2 plus 8 cm^2. Simplify the total for both.

5.B3 FLUENCY

A solution is made by tripling a mixture of x ml of acid and 5 ml of water. Expand to show the amounts.

5.B4 FLUENCY

Two processes add 2(x + 4) and 3(x − 1) grams of reagent. Expand and simplify the total.

5.B5 REASONING

Investigate: are 2(x + 3) and 2x + 3 equivalent? Test with x = 2.

5.B6 FLUENCY

A process takes 4m minutes; a shortcut saves 6m minutes but adds 2 minutes of setup. Simplify the net time.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

5.B7 REASONING

Investigate the pattern of areas: 2n, 2(n + 1), 2n + 2. Are they the same?

5.B8 FLUENCY

Expand −(x − 4) and explain what the outer minus sign does to each term of a design specification.

5.B9 FLUENCY

Simplify 3(x + 2) − (x + 2) and explain why the two parts can be collected.

5.B10 REASONING

Explain why 3x and 5 cannot be collected when you are counting parts and fixed costs.

5.B11 FLUENCY

Expand 5(2x − 3) and say what each term counts in the design.

CRITERION C Communicating

Criterion C practice · Communicating

5.C1 FLUENCY

A beam must be 15 cm long and 9 cm is already cut. Write the equation for the remaining length x, solve it, and explain each step in words.

5.C2 FLUENCY

Four identical supports carry 28 kg. Write the equation, solve it, and explain why dividing both sides keeps the equation true.

5.C3 FLUENCY

A device costs 5 dollars per part plus a 2-dollar fixed charge, and the budget is 17 dollars. Write the equation, solve it, and explain the two steps.

5.C4 REASONING

Explain why you do the same thing to both sides of an equation, using a balance as a representation.

5.C5 FLUENCY

Solve $5(x - 2) = 20$ and explain the order of the two operations you used.

5.C6 FLUENCY

Solve $3x - 4 = 2x + 5$ by first explaining what it means for two designs to cost the same.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

5.C7 REASONING

Explain how to check a solution, and check x = 7 in 4x = 28.

5.C8 FLUENCY

A process runs x/3 hours plus 1 hour of setup and totals 5 hours. Write the equation, solve it, and explain the meaning of x.

5.C9 FLUENCY

Solve 7 − x = 3 and explain the mistake in the answer x = −4.

5.C10 REASONING

Explain the difference between an expression and an equation, using a design example.

5.C11 FLUENCY

Solve 2(x + 1) = 3(x − 1) and show the check by substitution.

5.C12 FLUENCY

Solve 6x − 9 = 3x + 12 and explain in words what the solution means for the two designs being compared.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

5.D1 APPLICATION

A rectangle has length 3 cm more than its width and perimeter 26 cm. Find the dimensions.

5.D2 APPLICATION

Three identical shelves hold 72 books in total, with one more book on each higher shelf. Find the three numbers.

5.D3 APPLICATION

A plumber charges 40 dollars plus 30 dollars per hour. The bill is 190 dollars. How many hours?

5.D4 APPLICATION

A factory output is doubled and then 6 defective units are removed, leaving 18 good units. Find the output.

5.D5 APPLICATION

A taxi costs 3 dollars plus 2 dollars per km. A ride costs 21 dollars. Find the distance.

5.D6 REASONING

Solve the inequality $3x + 2 \le 20$ and interpret the answer for the number of parts a machine can process.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

5.D7 APPLICATION

A father is three times as old as his son. Their ages total 48. Find both ages.

5.D8 APPLICATION

A rectangle's length is twice its width. Its area is 50 cm^2. Find the width.

5.D9 APPLICATION

A phone plan is 15 dollars plus 0.10 dollars per minute. The bill is at most 30 dollars. Find the maximum minutes.

5.D10 REASONING

Explain, with an example, why an answer can be correct algebraically but meaningless in context.

5.D11 APPLICATION

Two production batches differ by 7 units and total 35 units. Find the batch sizes.

5.D12 APPLICATION

A 2-litre tank is filled at 0.25 litres per minute by a pump. After how many minutes is it full, and how would you check the answer?

CHAPTER 5 · APPLICATION AND ASSESSMENT

Applying chapter 5

Build the expression first, then substitute. Say what each letter means.

E1 · The garden design

A rectangular garden has length 4 m more than its width and a perimeter of 48 m. Find the dimensions and the area, and explain each step of your method.

E2 · The fair budget

A class pays 80 dollars for a stall plus 3 dollars per item sold. They want to keep costs under 200 dollars. How many items can they sell, and what is the cost at that limit?

Criteria B and D · set up the variable, solve the inequality, then interpret the answer.

CHAPTER 5 · REVIEW

What you should now be able to do

Key ideas

- A variable stands for a number that can change.
- Only like terms can be collected.
- Expanding multiplies each term inside a bracket by the term outside.
- The balance method keeps an equation true by doing the same to both sides.
- Always substitute a solution back into the original equation.
- An inequality gives a range of values, and the context decides which are meaningful.

Vocabulary

variable · constant · expression · like terms · simplify · expand · equation · solution · inequality · balance method

CHAPTER 5 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

5.R1 Simplify $6x + 4 - 2x$.

5.R2 Expand $3(2x - 4)$.

5.R3 Solve $5x - 3 = 22$.

5.R4 Write an expression for 'five less than three times a number'.

5.R5 Solve $2(x + 3) = x + 10$.

5.R6 Solve $4x + 1 \leq 17$ and interpret it as a number of tickets.

CHAPTER 5 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

A variable represents an unknown or changing quantity, and the logic of equivalence between different representations of the same relationship, together with the patterns in its structure, allows one expression or equation to model many situations, so that processes can be designed, solved and checked.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of logic and the related concepts of patterns and representation appear in your work?
- How is the global context of scientific and technical innovation, and its exploration of products, processes and solutions, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a mathematical model of a real situation, and what would you still like to ask?

CHAPTER 5 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Algebra models any situation where a rule repeats. Model one in your community.

Service as Action

COMMUNITY NEED	A community group needs to plan a small event within a fixed budget.
MATHEMATICAL CONNECTION	Expressions and equations to model costs and find limits.
PROPOSED ACTION	Build a cost model for the event and present the maximum affordable options.
EXPECTED IMPACT	The event runs within budget with no surprises.
STUDENT REFLECTION	Explain how the equation made the budget decisions transparent.

Community Project

PROJECT IDEA	A school fair cost model.
NEED	Stalls are priced by guesswork, so some run at a loss.
RESEARCH QUESTION	What fixed and variable costs must each stall cover to break even?
MATHEMATICS	Expressions, equations and inequalities.
ACTION OR OUTCOME	A break-even model and a pricing guide for stall holders.
BENEFICIARIES	Stall holders and the school.
EVIDENCE	The cost model, the guide and the final event accounts.

Personal Project connection · MYP 5

PERSONAL INTEREST	Gaming, crafting or personal budgeting.
LEARNING GOAL	To model a personal cost with an equation.
PRODUCT OR OUTCOME	A budget model for a project or purchase.
MATHEMATICS	Expressions, equations and inequalities.
PROCESS	Define the variable, build the expression, solve and interpret.
EVIDENCE	The model, the solution and the decision it supported.

CHAPTER 6 · PART II · ALGEBRA AND PATTERN

06

Numerical patterns and generalisation

A pattern is a sequence; a generalisation is a rule. This chapter turns counting patterns into formulas and graphs.

ENGAGE

A staircase of blocks has 1 block in the first step, 3 in the second, 6 in the third. How many blocks are in the 20th step?

Framing the inquiry

Key concept: Patterns
Related concepts: Relationships · Logic
Global context: Personal and cultural expression
Global context exploration: Artistry, craft, creation and beauty

Statement of inquiry: A numerical pattern describes relationships between position and value, and the logic of generalisation allows one rule to describe and predict every term, which is how patterns in design and nature are created, extended and reproduced.

Questions to guide us

- **Factual:** What is a numerical sequence, what is the nth term, and how do you describe a pattern in words and in symbols?
- **Conceptual:** How do you know a rule is correct for every term and not only for the terms you checked, and what do the differences tell you about the rule?
- **Debatable:** Is finding a pattern the same as proving it, and what would you accept as proof?

ATL · CRITICAL THINKING. In order for a student to practise observing carefully, students must recognise patterns and relationships, and test a generalisation on cases not yet seen.

CHAPTER 6 · INVESTIGATION 6.1

Describing sequences

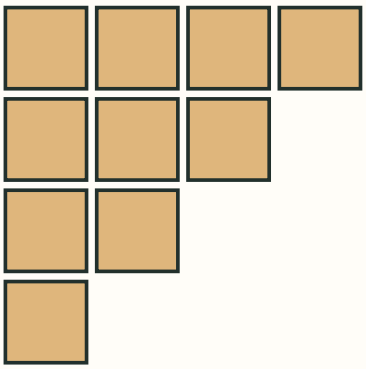
INQUIRY QUESTION

How can a list of numbers be described so that any term can be found without listing them all?

YOUR TASK

Build a staircase of blocks with 1, 3, 6, 10 blocks. Record the number of blocks and find a rule for the nth staircase.

Staircases of 1, 3, 6 and 10 blocks



n	1	2	3	4	5
blocks	1	3	6	10	15

CHAPTER 6 · INVESTIGATION 6.1

Describing sequences

What do you notice?

- The differences are 2, 3, 4, 5 — not constant.
- The second differences are all 1.
- Two copies of the staircase make an n by (n + 1) rectangle.

Observation. Triangular numbers grow quadratically because their second differences are constant.

MATHEMATICAL CONCLUSION

A list can be described by its differences: constant differences give a linear rule and constant second differences a quadratic one, so the nth triangular number is n(n + 1)/2.

Your own words

CHAPTER 6 · 6.1

Describing sequences

A sequence is an ordered list of numbers. A term is one entry; its position is usually called n. The nth term rule gives the value at position n.

To find a rule, look at the differences between consecutive terms. Constant differences mean a linear rule; constant second differences mean a quadratic rule.

A rule is a generalisation: it claims to work for every position. Testing one new term is evidence, and checking the structure is proof.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 6 · 6.1 · WORKED EXAMPLE 1

Describing sequences

WORKED EXAMPLE 6.1.1

Problem. Find the next two terms of 4, 9, 14, 19, … and write a rule for the nth term.

n	1	2	3	4	5	6
term	4	9	14	19	24	29

Differences of 5

Mathematical reasoning. The differences are all 5, so the rule is linear with slope 5.

1. Differences: 5, 5, 5, 5.
2. A linear rule has the form an + b. With difference 5, a = 5.
3. At n = 1 the term is 4, so 5(1) + b = 4 and b = −1.
4. Rule: 5n − 1. Check at n = 4: 5(4) − 1 = 19.

Final answer. 24, 29 and the nth term is 5n − 1

Interpretation. The rule predicts any term without listing the ones before it.

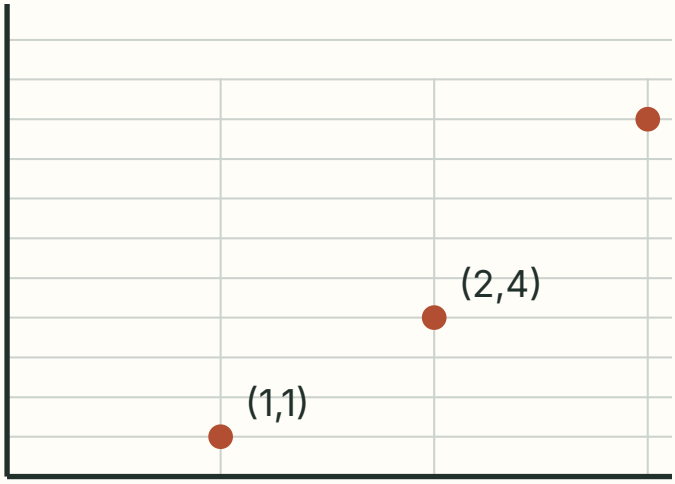
CHAPTER 6 · 6.1 · WORKED EXAMPLE 2

Describing sequences

WORKED EXAMPLE 6.1.2

Problem. The sequence 1, 4, 9, 16, … has second differences of 2. Find the nth term.

Squares on a grid



Mathematical reasoning. Constant second differences indicate a quadratic rule, and the terms are perfect squares.

1. Differences: 3, 5, 7. Second differences: 2, 2.
2. A quadratic rule has the form $an^2 + bn + c$ with $2a = 2$, so $a = 1$.
3. The terms are 1^2, 2^2, 3^2, 4^2, so the rule is n^2.

Final answer. nth term = n^2

Interpretation. The graph is curved, which signals that the rule is not linear.

CHAPTER 6 · INVESTIGATION 6.2

From pattern to formula

INQUIRY QUESTION

How can a pattern be turned into a formula that works for every position?

YOUR TASK

Take the rule 3n + 2. Predict the 10th term, then check it by extending a table.

n	1	2	3	4	...	10
3n + 2	5	8	11	14	...	32

CHAPTER 6 · INVESTIGATION 6.2

From pattern to formula

What do you notice?

- The rule gives 32 immediately.
- Extending the table one row at a time also reaches 32.
- The rule is faster and less error-prone for large n.

Observation. A formula replaces repeated listing.

MATHEMATICAL CONCLUSION

A formula replaces listing: it is a generalisation that holds for every position and can be tested on a term that has not yet been counted.

Your own words

CHAPTER 6 · 6.2

From pattern to formula

A formula is a rule that links variables. Building one means describing the structure in words first, then translating: 'twice the position plus one' becomes $2n + 1$.

Substituting into a formula gives a value; rearranging a formula gives a new relationship. Both skills are used throughout the chapter.

A formula can be checked by substitution, by comparing with a table, and by testing an extreme case such as $n = 0$ or $n = 1$.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 6 · 6.2 · WORKED EXAMPLE 1

From pattern to formula

WORKED EXAMPLE 6.2.1

Problem. A pattern of matchsticks uses 4 for the first shape and 7 for the second. Each extra shape adds 3 matchsticks. Find the number in shape n.

shape n	1	2	3	4
matchsticks	4	7	10	13

Matchstick patterns

Mathematical reasoning. The pattern is linear with difference 3, and the first term is 4.

1. Difference = 3, so a = 3.
2. At n = 1: 3(1) + b = 4, so b = 1.
3. Rule: 3n + 1. Check at n = 4: 3(4) + 1 = 13.

Final answer. 3n + 1 matchsticks

Interpretation. The formula replaces counting for large shapes such as n = 50.

CHAPTER 6 · 6.2 · WORKED EXAMPLE 2

From pattern to formula

WORKED EXAMPLE 6.2.2

Problem. A sequence has nth term $2n^2 - 1$. Find the first four terms.

n	1	2	3	4
$2n^2 - 1$	1	7	17	31

Substituting into a rule

Mathematical reasoning. Substitute each position into the rule.

1. $n = 1$: $2(1) - 1 = 1$.
2. $n = 2$: $2(4) - 1 = 7$.
3. $n = 3$: $2(9) - 1 = 17$. $n = 4$: $2(16) - 1 = 31$.

Final answer. 1, 7, 17, 31

Interpretation. Substitution is the direct test of whether a rule produces the intended sequence.

CHAPTER 6 · INVESTIGATION 6.3

Graphs of sequences

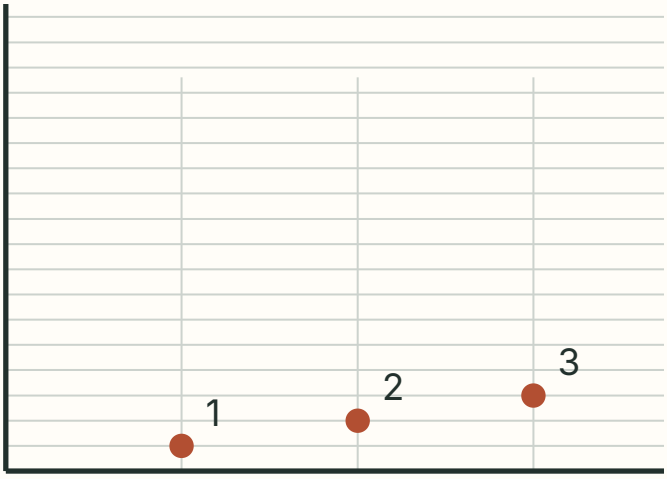
INQUIRY QUESTION

What can a graph of a sequence show that the list of terms cannot show?

YOUR TASK

Plot n, 2n and n^2 for n = 1 to 5 on the same axes and find where the order changes.

Three sequences compared



n	1	2	3	4	5
n	1	2	3	4	5
2n	2	4	6	8	10
n^2	1	4	9	16	25

CHAPTER 6 · INVESTIGATION 6.3

Graphs of sequences

What do you notice?

- 2n is always above n.
- n^2 is equal to n at n = 1 and above it from n = 2.
- The gap between n^2 and 2n grows quickly.

Observation. Different rules grow at different rates, visible as different steepness on the graph.

MATHEMATICAL CONCLUSION

Steepness and curvature compare growth rates that a list of numbers hides, so a graph shows which rule grows faster and where the order changes.

Your own words

CHAPTER 6 · 6.3

Graphs of sequences

Plotting term against position turns a sequence into a set of points. A linear rule gives points on a straight line; a quadratic rule gives points on a curve.

Reading a graph in the n direction is interpolation or extrapolation, and the same caution applies as in chapter 2: the rule must genuinely continue.

Comparing two sequences on the same axes shows which grows faster, and the crossing point shows where the order changes.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

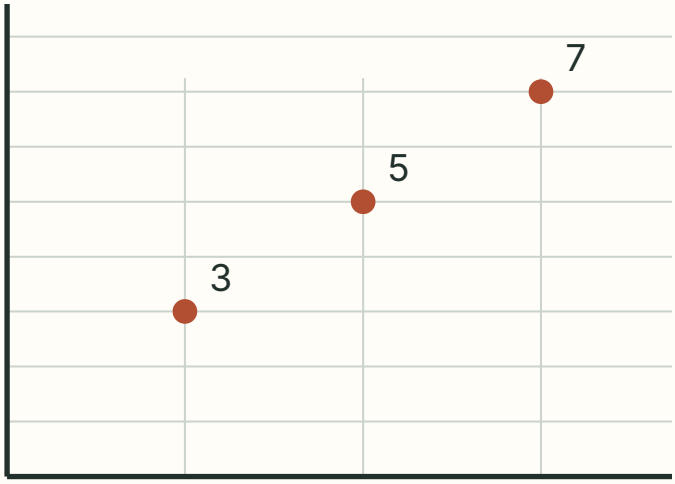
CHAPTER 6 · 6.3 · WORKED EXAMPLE 1

Graphs of sequences

WORKED EXAMPLE 6.3.1

Problem. Plot the points for the rule 2n + 1 for n = 1 to 5 and describe the graph.

A linear sequence



Mathematical reasoning. Each point has coordinates (n, term), and a linear rule gives a straight line.

1. Points: (1, 3), (2, 5), (3, 7), (4, 9), (5, 11).
2. They lie on a straight line with slope 2.
3. The line crosses the y-axis at 1.

Final answer. Five points on a straight line of slope 2

Interpretation. The slope is the difference between consecutive terms.

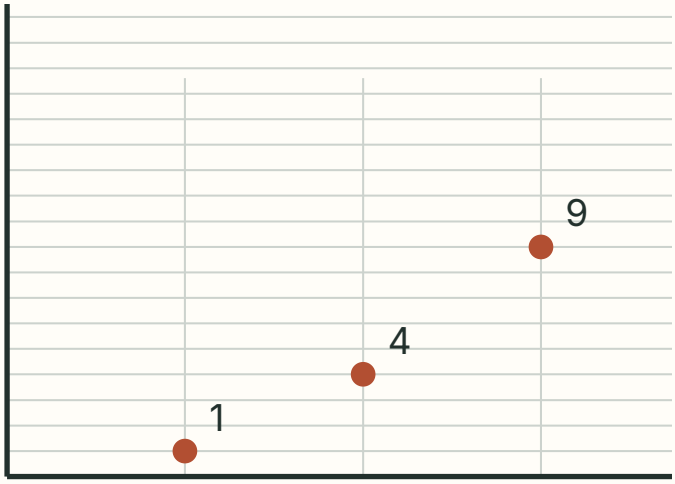
CHAPTER 6 · 6.3 · WORKED EXAMPLE 2

Graphs of sequences

WORKED EXAMPLE 6.3.2

Problem. Compare the sequences n and n^2 on the same axes and say which is larger at n = 10.

Linear against quadratic



Mathematical reasoning. n^2 grows faster than n once n is greater than 1.

1. At n = 1 both give 1.
2. At n = 2: $n^2 = 4$ is greater than n = 2.
3. At n = 10: $n^2 = 100$ is much greater than 10.

Final answer. n^2 is larger from n = 2 onwards

Interpretation. The curve separates from the line immediately, which is the visual sign of faster growth.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

6.A1 FLUENCY
Find the next term: 2, 5, 8, 11, …

6.A2 REASONING
Describe the rule for 7, 10, 13, 16, … in words.

6.A3 FLUENCY
Find the 5th term of 1, 4, 7, 10, …

6.A4 REASONING
A mosaic border uses 3, 6, 12, 24 tiles in successive frames. Is the pattern linear? Justify using the differences.

6.A5 FLUENCY
Find the first term and the difference of 20, 17, 14, 11, …

6.A6 REASONING
Explain what 'the nth term' means.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

6.A7 REASONING
Find the next two terms: 1, 2, 4, 7, 11, …

6.A8 FLUENCY
A sequence has first term 5 and difference 4. Find the 6th term.

6.A9 REASONING
Explain the difference between a term and a position.

6.A10 FLUENCY
Find the 10th term of 100, 95, 90, …

6.A11 REASONING
Is the sequence 2, 4, 8, 16, 32 growing faster than 2, 5, 8, 11, 14? Justify.

6.A12 FLUENCY
Find the next term: 81, 27, 9, 3, …

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

6.B1 FLUENCY

Write a rule for the sequence 6, 11, 16, 21, …

6.B2 FLUENCY

Find the nth term of 3, 7, 11, 15, …

6.B3 FLUENCY

Use the rule 2n + 3 to find the 20th term.

6.B4 REASONING

Investigate: does the rule 4n − 2 give the sequence 2, 6, 10, 14?

6.B5 FLUENCY

A pattern starts at 8 and adds 6 each time. Write the rule.

6.B6 REASONING

Investigate the pattern 1, 8, 27, 64. Write a rule.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

6.B7 REASONING

Explain how to test whether a rule is correct for every term.

6.B8 FLUENCY

Find the 15th term of $3n - 5$.

6.B9 FLUENCY

Write a rule for the sequence 50, 47, 44, 41, ...

6.B10 REASONING

Explain why a rule is better than a list for large n.

6.B11 FLUENCY

Find the nth term of 0, 5, 10, 15, ...

CRITERION C Communicating

Criterion C practice · Communicating

6.C1 FLUENCY
Sketch the graph of the rule n + 2 for n = 1 to 5 and describe it.

6.C2 REASONING
Explain why the graph of a quadratic rule is curved.

6.C3 FLUENCY
Two rules are n and 2n. Sketch both and say which is steeper.

6.C4 FLUENCY
Plot (n, term) for 4, 8, 12, 16 and state the slope.

6.C5 REASONING
Explain how a graph shows which sequence grows faster.

6.C6 APPLICATION
A graph of a sequence passes through (3, 10) and (4, 13). Find the rule.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

6.C7 REASONING
Explain what the crossing point of two sequence graphs means.

6.C8 FLUENCY
Plot the number of tiles in a square design, n^2, for n = 1 to 4, and state the second difference.

6.C9 APPLICATION
A sequence graph is a curve through (1, 1), (2, 4), (3, 9). Predict the 5th point.

6.C10 REASONING
Explain why a straight-line graph means a linear rule.

6.C11 APPLICATION
Sketch the rules 2n and n^2 for a pattern of strips and squares, and find where n^2 becomes larger.

6.C12 REASONING
Explain the connection between a sequence table and its graph.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

6.D1 APPLICATION

A staircase has 1, 3, 6, 10 blocks. How many blocks in the 12th staircase?

6.D2 APPLICATION

A pattern of tables seats 4, 6, 8 people. How many people at 15 tables?

6.D3 APPLICATION

A ball is dropped from 100 cm in a craft demonstration and rebounds to half the height each time. Find the height after the 4th bounce.

6.D4 APPLICATION

A savings account starts at 200 dollars and gains 15 dollars a week. Find the amount after 30 weeks.

6.D5 APPLICATION

A pattern of hexagons uses 6, 11, 16 sticks. Find the number for 20 hexagons.

6.D6 APPLICATION

Compare the rules $3n + 1$ and n^2 for a repeating border design, and find the smallest n where n^2 is larger.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

6.D7 APPLICATION

A sequence has nth term $n^2 - n$. Find the first five terms and the 10th term.

6.D8 APPLICATION

A phone battery falls 8% per hour from 100%. Write the rule and find the charge after 5 hours.

6.D9 REASONING

A graph shows two sequences crossing at n = 7. Explain what this means.

6.D10 APPLICATION

A pattern of dots has 5, 9, 13, 17. Find the 25th term.

6.D11 APPLICATION

A population grows from 500 by 10% a year. Find the population after 3 years.

6.D12 REASONING

Explain why 2^n eventually grows faster than 100n, using a graph.

CHAPTER 6 · APPLICATION AND ASSESSMENT

Applying chapter 6

Find the pattern, express it as a rule, then test the rule on a term you have not counted.

F1 · The tile pattern

A pattern of tiles has 5, 9, 13, 17 tiles in the first four figures. Find the number in figure 20, explain your rule, and show how you know the rule is correct.

F2 · The savings plan

A student saves 25 dollars a week and already has 180 dollars. Write a rule for the total after n weeks, draw the graph, and find when the total reaches 1 000 dollars.

Criteria C and D - use a graph or a table to justify your answer.

CHAPTER 6 · REVIEW

What you should now be able to do

Key ideas

- A sequence is an ordered list; a rule gives any term.
- Constant differences indicate a linear rule.
- Constant second differences indicate a quadratic rule.
- A formula is a generalisation that must hold for every position.
- Graphs of sequences show growth rates that lists hide.

Vocabulary

sequence · term · position · nth term · difference · second difference · rule · generalisation · linear · quadratic

CHAPTER 6 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

6.R1 Find the next term: 5, 11, 17, 23, …

6.R2 Write a rule for 8, 13, 18, 23, …

6.R3 Find the 12th term of 4n + 1.

6.R4 How many blocks in the 10th staircase?

6.R5 Explain why 2, 6, 18, 54 is not linear.

6.R6 A rule is 3n − 2. Find n when the term is 25.

CHAPTER 6 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

A numerical pattern describes relationships between position and value, and the logic of generalisation allows one rule to describe and predict every term, which is how patterns in design and nature are created, extended and reproduced.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of patterns and the related concepts of relationships and logic appear in your work?
- How is the global context of personal and cultural expression, and its exploration of artistry, craft, creation and beauty, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a rule found from a few terms, and what would you still like to ask?

CHAPTER 6 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Patterns appear in design, nature and sport. Generalise one you can measure.

Service as Action

COMMUNITY NEED	A community centre needs seating that can be rearranged for different events.
MATHEMATICAL CONNECTION	Patterns and generalisation to predict seating for any number of tables.
PROPOSED ACTION	Build a seating rule and a chart for the centre's tables.
EXPECTED IMPACT	Events can be planned quickly with the right number of seats.
STUDENT REFLECTION	Explain how the rule replaced counting for each event.

Community Project

PROJECT IDEA	A school garden planting plan.
NEED	The garden's planting rows need a predictable number of seedlings.
RESEARCH QUESTION	How many seedlings are needed for any number of rows, and what is the rule?
MATHEMATICS	Sequences, nth-term rules and graphs.
ACTION OR OUTCOME	A planting chart and a rule for the garden team.
BENEFICIARIES	The garden team and the school community.
EVIDENCE	The rule, the chart and a photograph of the planted rows.

Personal Project connection · MYP 5

PERSONAL INTEREST	Music, gaming or collecting.
LEARNING GOAL	To generalise a pattern from a personal interest.
PRODUCT OR OUTCOME	A rule and a prediction for a collection or score.
MATHEMATICS	Sequences, nth-term rules and graphs.
PROCESS	Collect the data, find the differences, write the rule and test it.
EVIDENCE	The data table, the rule and the prediction.

CHAPTER 7 · PART III · GEOMETRY AND SPACE

07

2D geometry: angles, perimeter and area

Angles describe turns; perimeter measures the boundary and area measures the surface. Formulas come from the properties of each shape.

ENGAGE

A rectangular room measures 6 m by 4 m. How much skirting board runs around the walls, and how much carpet covers the floor?

Framing the inquiry

Key concept: Form
Related concepts: Logic · Representation
Global context: Globalisation and sustainability
Global context exploration: Urban and rural planning

Statement of inquiry: The form of a two-dimensional shape, and the logic of how its parts relate, are represented by formulas for perimeter and area, and that representation allows spaces to be measured, planned and costed when land, flooring and materials are limited.

Questions to guide us

- **Factual:** What is the difference between perimeter and area, and which formula gives each for a rectangle, a triangle and a circle?
- **Conceptual:** Why does the angle sum of a triangle always equal 180°, and how do the properties of a shape produce its perimeter and area formulas?
- **Debatable:** When a community space is planned with a limited budget, is the cheapest shape per square metre always the best choice?

ATL · INFORMATION LITERACY. In order for a student to access information to be informed, students must collect, record and verify data, and make connections between sources of information.

CHAPTER 7 · INVESTIGATION 7.1

Angles and their relationships

INQUIRY QUESTION

What relationships exist between angles formed by lines, and why do they always hold?

YOUR TASK

Draw three different triangles, measure their angles, and add them up. Then draw a quadrilateral and do the same.

shape	angle 1	angle 2	angle 3	angle 4	total
triangle 1					
triangle 2					
quadrilateral					

CHAPTER 7 · INVESTIGATION 7.1

Angles and their relationships

What do you notice?

- Each triangle totals close to 180°, allowing for measuring error.
- The quadrilateral totals close to 360°.
- Splitting a quadrilateral along a diagonal gives two triangles.

Observation. The angle sum depends on the number of sides.

MATHEMATICAL CONCLUSION

Angle relationships come from straight lines, full turns and parallel lines, and they combine to give the angle sum of a polygon with n sides as $(n - 2) \times 180°$.

Your own words

CHAPTER 7 · 7.1

Angles and their relationships

An angle measures a turn. A full turn is 360°, a half turn 180°, and a right angle 90°.

Angles on a straight line add to 180°, and angles around a point add to 360°. Vertically opposite angles are equal because they share the same two lines.

When a transversal crosses parallel lines, corresponding angles are equal, alternate angles are equal, and co-interior angles add to 180°.

The angle sum of a triangle is 180°, which can be shown by tearing off the corners and placing them along a straight line.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

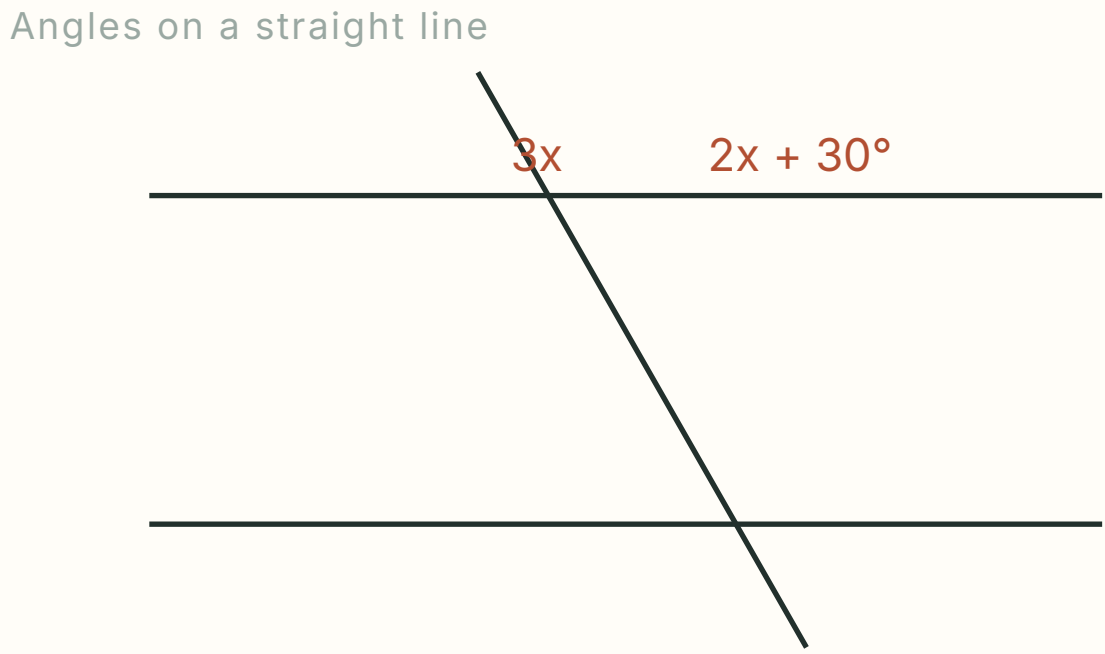
In your own words

CHAPTER 7 · 7.1 · WORKED EXAMPLE 1

Angles and their relationships

WORKED EXAMPLE 7.1.1

Problem. Two angles on a straight line are 3x and 2x + 30. Find x.



Mathematical reasoning. Angles on a straight line add to 180°, so I write an equation.

1. 3x + (2x + 30) = 180.
2. 5x = 150, so x = 30.
3. The angles are 90° and 90°.

Final answer. x = 30

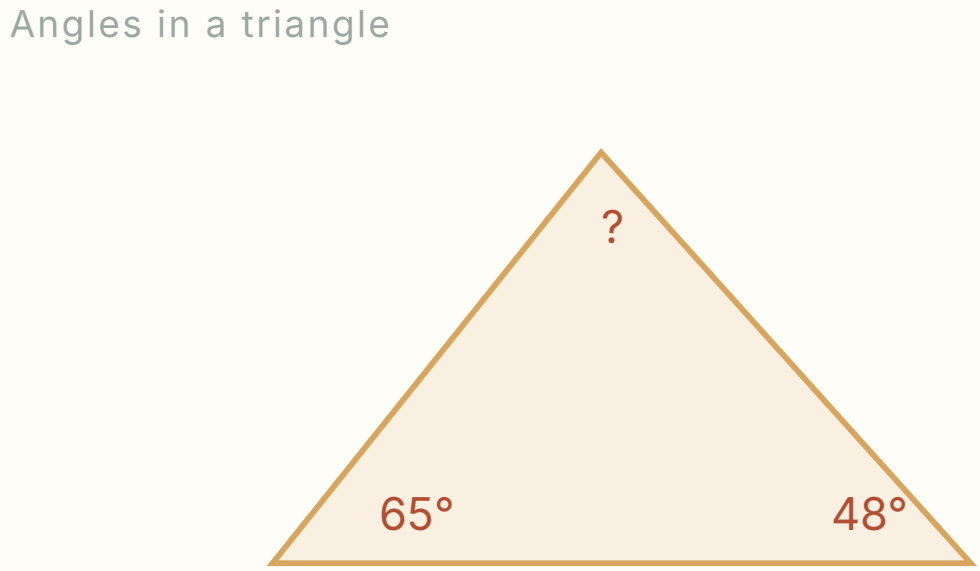
Interpretation. The relationship gives an equation, and the equation gives the value.

CHAPTER 7 · 7.1 · WORKED EXAMPLE 2

Angles and their relationships

WORKED EXAMPLE 7.1.2

Problem. Find the third angle of a triangle whose other two angles are 65° and 48°.



Mathematical reasoning. The three angles of a triangle add to 180°.

1. 65 + 48 = 113.
2. 180 − 113 = 67.

Final answer. 67°

Interpretation. The angle sum is a property of all triangles, whatever their shape.

CHAPTER 7 · INVESTIGATION 7.2

Perimeter of 2D shapes

INQUIRY QUESTION

How is the boundary of a shape measured, and how does the perimeter formula follow from the shape's properties?

YOUR TASK

Measure the perimeter of three objects by wrapping a string around them and measuring the string. Compare with calculations for the shapes you can identify.

object	shape	measured perimeter	calculated perimeter
object 1			
object 2			
object 3			

CHAPTER 7 · INVESTIGATION 7.2

Perimeter of 2D shapes

What do you notice?

- A string measures the boundary directly, whatever the shape.
- Calculated perimeters depend on identifying the shape correctly.
- Small differences come from measuring error and from the string's thickness.

Observation. Perimeter is a length, so it is measured in length units.

MATHEMATICAL CONCLUSION

Perimeter is the total boundary length, and each formula is a shortcut that follows from the properties of that shape.

Your own words

CHAPTER 7 · 7.2

Perimeter of 2D shapes

Perimeter is the total distance around a shape. For a rectangle it is $2(l + w)$, and for any polygon it is the sum of the side lengths.

The perimeter of a circle is called the circumference: $C = \pi d = 2\pi r$. The diameter is twice the radius.

When a shape is described by a formula, the perimeter formula follows from the same structure: a square of side s has perimeter $4s$.

Perimeter problems often give the perimeter and ask for a side. Write the formula, then solve the resulting equation.

> FORMAL DEVELOPMENT
>
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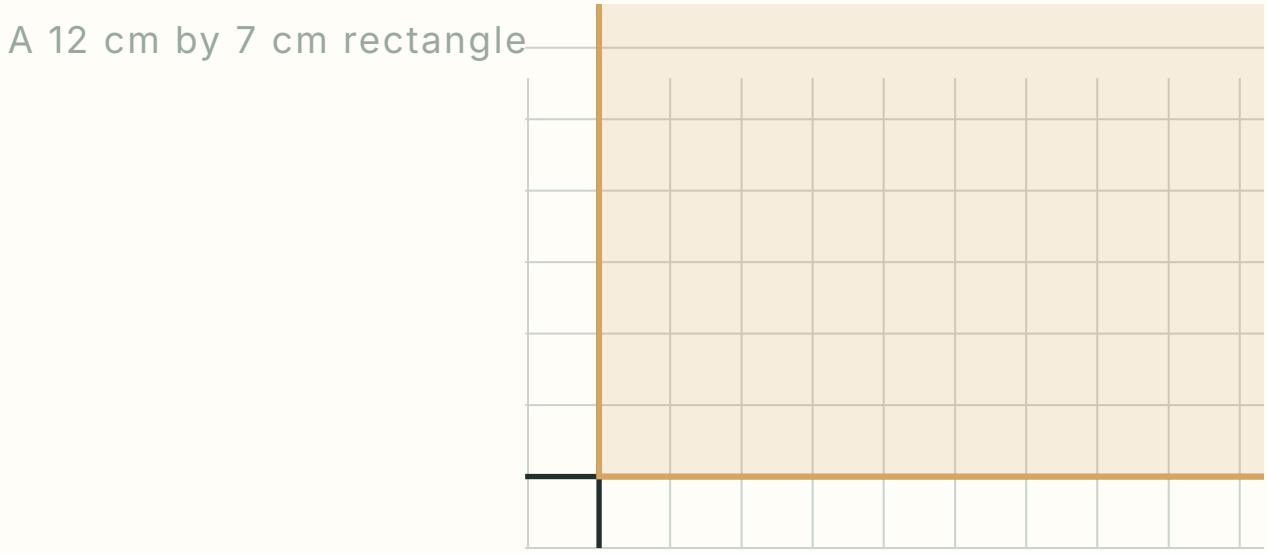
In your own words

CHAPTER 7 · 7.2 · WORKED EXAMPLE 1

Perimeter of 2D shapes

WORKED EXAMPLE 7.2.1

Problem. A rectangle is 12 cm long and 7 cm wide. Find its perimeter.



Mathematical reasoning. Perimeter is twice the sum of length and width.

1. $2(12 + 7) = 2 \times 19 = 38$.

Final answer. 38 cm

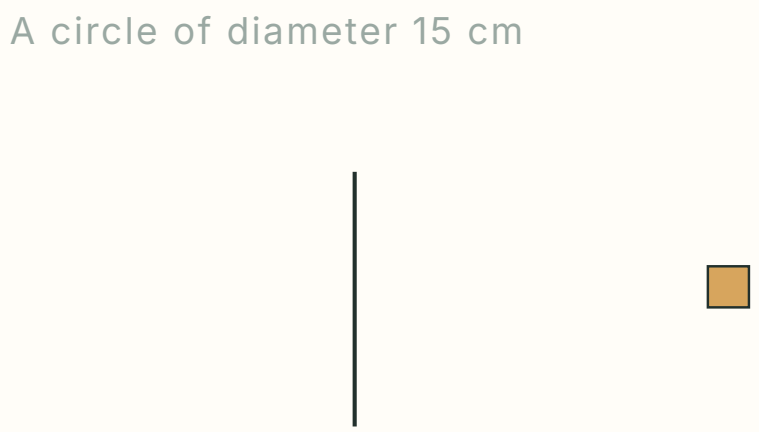
Interpretation. Adding length and width counts two sides, so doubling counts all four.

CHAPTER 7 · 7.2 · WORKED EXAMPLE 2

Perimeter of 2D shapes

WORKED EXAMPLE 7.2.2

Problem. A circle has diameter 15 cm. Find its circumference to one decimal place.



Mathematical reasoning. Circumference is π times the diameter.

1. $C = \pi d = \pi \times 15$.
2. $C \approx 47.1$ cm.

Final answer. 47.1 cm

Interpretation. Using $\pi \approx 3.14$ gives 47.1 cm; a calculator gives 47.12 cm.

CHAPTER 7 · INVESTIGATION 7.3

Area of 2D shapes

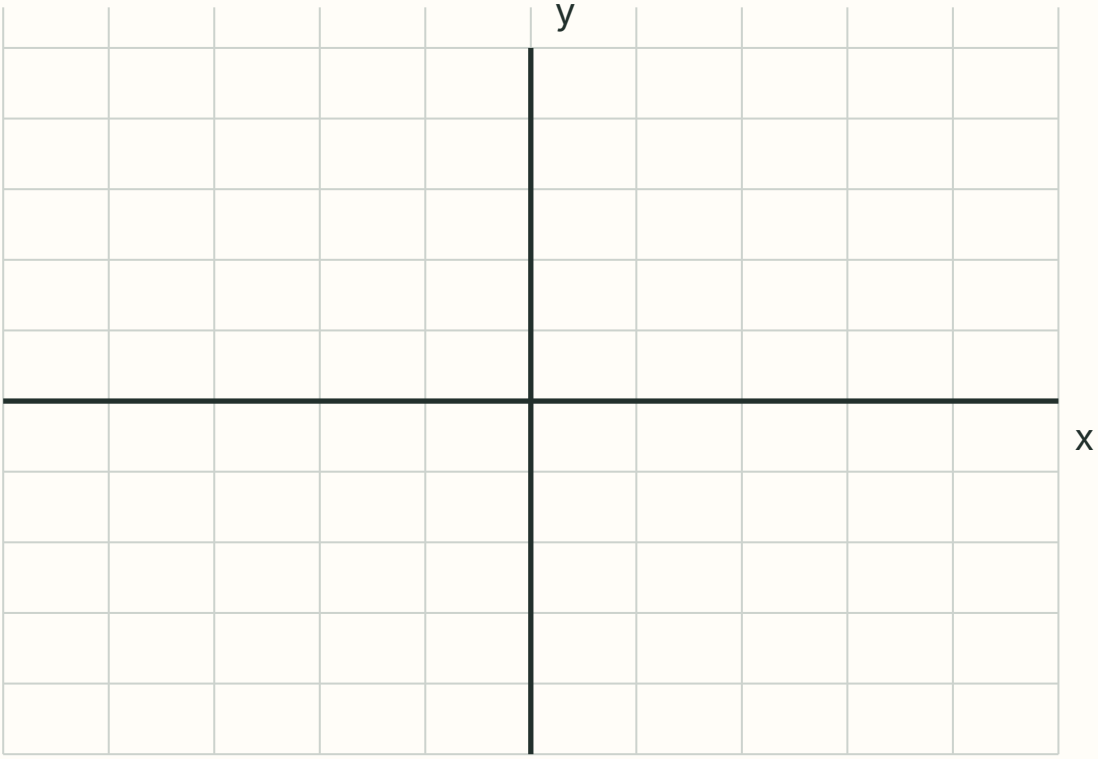
INQUIRY QUESTION

How is the surface covered by a shape measured, and where does each area formula come from?

YOUR TASK

Cut a parallelogram along its height and rearrange the pieces into a rectangle. Record the dimensions of each and compare the areas.

Parallelogram rearranged into a rectangle



shape	base	height	area
parallelogram			
rectangle			

CHAPTER 7 · INVESTIGATION 7.3

Area of 2D shapes

What do you notice?

- The base of the rectangle equals the base of the parallelogram.
- The height of the rectangle equals the height of the parallelogram.
- The areas are equal because no paper was added or removed.

Observation. A parallelogram has the same area formula as a rectangle.

MATHEMATICAL CONCLUSION

Every area formula can be derived by cutting and rearranging into a rectangle, so area of a parallelogram = base × height.

Your own words

CHAPTER 7 · 7.3

Area of 2D shapes

Area measures the surface covered. A rectangle of length l and width w has area lw; a triangle has area ½ × base × height; a parallelogram has base × height.

The triangle formula comes from the rectangle: two identical triangles make a rectangle of the same base and height.

A circle has area πr^2, and a trapezium has area $\frac{1}{2}(a + b)h$. Each formula can be derived from a simpler one by cutting and rearranging.

Compound shapes are split into parts whose areas can be calculated, then added or subtracted.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

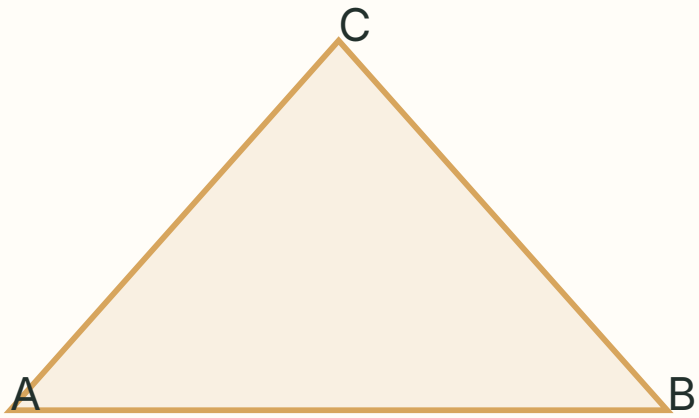
CHAPTER 7 · 7.3 · WORKED EXAMPLE 1

Area of 2D shapes

WORKED EXAMPLE 7.3.1

Problem. Find the area of a triangle with base 10 cm and height 6 cm.

Triangle with base 10 cm and height 6 cm



Mathematical reasoning. Two identical triangles form a rectangle of base 10 and height 6.

1. Rectangle area = $10 \times 6 = 60$ cm^2.
2. Triangle area = $\tfrac{1}{2} \times 60 = 30$ cm^2.

Final answer. 30 cm²

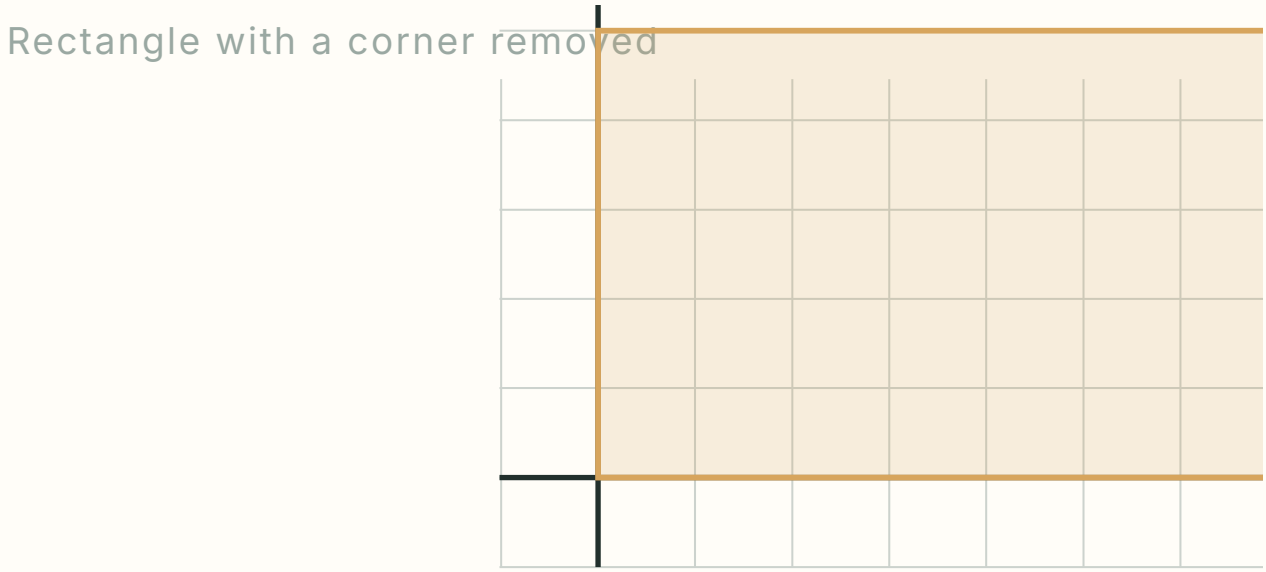
Interpretation. The half comes from the two copies needed to make the rectangle.

CHAPTER 7 · 7.3 · WORKED EXAMPLE 2

Area of 2D shapes

WORKED EXAMPLE 7.3.2

Problem. A compound shape is a rectangle 8 cm by 5 cm with a 3 cm by 2 cm rectangle removed from one corner. Find the remaining area.



Mathematical reasoning. Subtract the removed part from the whole rectangle.

1. Rectangle area = $8 \times 5 = 40$ cm^2.
2. Removed area = $3 \times 2 = 6$ cm^2.
3. Remaining area = $40 - 6 = 34$ cm^2.

Final answer. 34 cm²

Interpretation. Compound areas are handled by adding or subtracting simple areas.

CHAPTER 7 · INVESTIGATION 7.4

Using properties to solve problems

INQUIRY QUESTION

How can the properties of a shape be used to find a missing measurement?

YOUR TASK

List the properties of a square, a rectangle, a rhombus and a parallelogram in a table, and mark which properties are shared.

property	square	rectangle	rhombus	parallelogram
our right angles	yes	yes	no	no
posite sides equal	yes	yes	yes	yes
all sides equal	yes	no	yes	no

CHAPTER 7 · INVESTIGATION 7.4

Using properties to solve problems

What do you notice?

- The shared properties explain why the area formulas are related.
- The square is the most specialised of the four.
- The parallelogram is the most general.

Observation. Sharing properties lets one formula cover several shapes.

MATHEMATICAL CONCLUSION

Quadrilaterals form a hierarchy in which the more specialised shape has more properties, and those properties supply the equations that find a missing measurement.

Your own words

CHAPTER 7 · 7.4

Using properties to solve problems

Properties are the reason formulas work: opposite sides of a parallelogram are equal, diagonals of a rectangle are equal, and the diagonals of a rhombus bisect at right angles.

Naming a shape precisely is part of the mathematics. A square is a rectangle, a rhombus and a parallelogram, but a rectangle is not necessarily a square.

When a problem gives a perimeter and a relationship between sides, write both as equations and solve them together.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

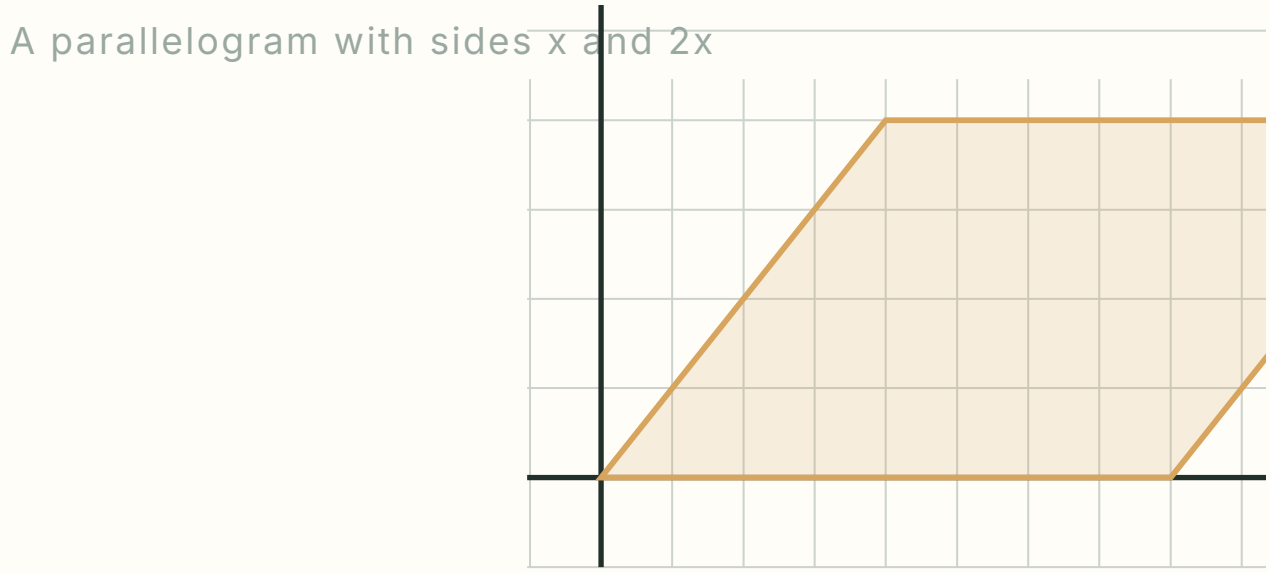
In your own words

CHAPTER 7 · 7.4 · WORKED EXAMPLE 1

Using properties to solve problems

WORKED EXAMPLE 7.4.1

Problem. A parallelogram has perimeter 40 cm. One side is twice the other. Find the side lengths.



Mathematical reasoning. Opposite sides of a parallelogram are equal, so the perimeter is 2(x + 2x).

1. Let the shorter side be x, so the longer is 2x.
2. 2(x + 2x) = 6x = 40, so x = 6.67.
3. The sides are 6.67 cm and 13.33 cm.
4. Check: 2(6.67 + 13.33) = 40.

Final answer. 6.67 cm and 13.33 cm

Interpretation. The property of equal opposite sides is what makes the equation possible.

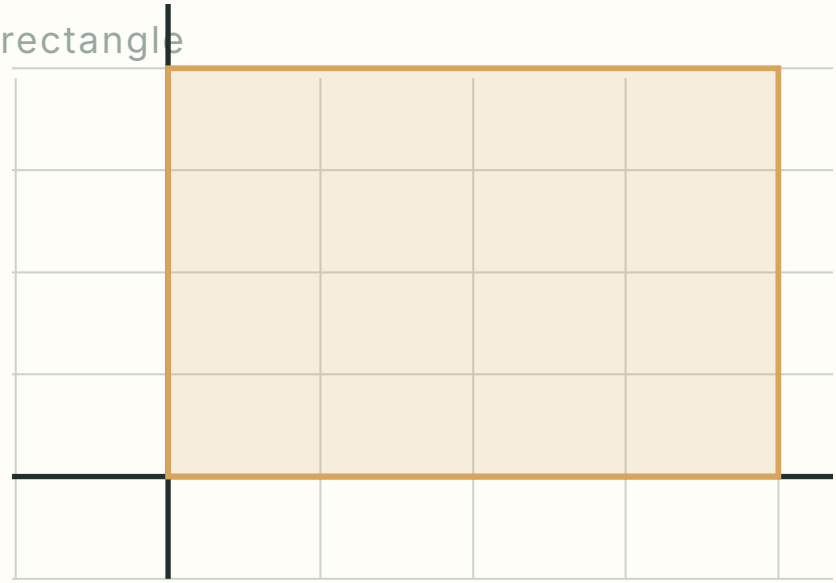
CHAPTER 7 · 7.4 · WORKED EXAMPLE 2

Using properties to solve problems

WORKED EXAMPLE 7.4.2

Problem. Explain why every square is a rectangle but not every rectangle is a square.

A square is a special rectangle



Mathematical reasoning. A rectangle needs four right angles; a square also needs four equal sides.

1. A square has four right angles, so it meets the rectangle condition.
2. A rectangle may have unequal adjacent sides.
3. So the square is a subset of the rectangles.

Final answer. A square satisfies every rectangle condition and adds one more

Interpretation. Precise naming prevents false claims in later proofs.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

7.A1 FLUENCY

Angles on a straight line add to how many degrees?

7.A2 FLUENCY

Find the angle that is complementary to 35°.

7.A3 FLUENCY

Two angles are vertically opposite and one is 72°. Find the other.

7.A4 FLUENCY

Find the third angle of a triangle with angles 90° and 40°.

7.A5 FLUENCY

Angles around a point add to how many degrees?

7.A6 FLUENCY

A quadrilateral has angles 90°, 85° and 100°. Find the fourth.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

7.A7 REASONING
Name the angle relationship between two equal angles on opposite sides of a transversal.

7.A8 FLUENCY
Find the angle supplementary to 110°.

7.A9 REASONING
Explain why the angles of a triangle add to 180°.

7.A10 FLUENCY
Co-interior angles on parallel lines add to how many degrees?

7.A11 FLUENCY
Find the angle sum of a pentagon.

7.A12 FLUENCY
An isosceles triangle has a vertex angle of 40°. Find each base angle.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

7.B1 FLUENCY

Find the perimeter of a rectangle 9 cm by 4 cm.

7.B2 FLUENCY

A square has perimeter 32 cm. Find its side.

7.B3 FLUENCY

A circular flower bed of radius 7 m is edged with fencing. Find the length of edging needed.

7.B4 APPLICATION

Investigate: a rectangle has perimeter 24 cm and length twice its width. Find the dimensions.

7.B5 REASONING

Explain the difference between perimeter and circumference.

7.B6 FLUENCY

Find the perimeter of an equilateral triangle of side 5.5 cm.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

7.B7 REASONING

Investigate the pattern: perimeters of squares of side 1, 2, 3, 4.

7.B8 FLUENCY

A circular pond of diameter 20 m is surrounded by a path. Find the circumference of the pond.

7.B9 REASONING

Explain why the perimeter of a rectangle is not simply length × width.

7.B10 FLUENCY

A shape has sides 3, 4, 5, 6 and 7 cm. Find its perimeter.

7.B11 FLUENCY

A square field has perimeter 400 m. Find its side.

CRITERION C Communicating

Criterion C practice · Communicating

7.C1 FLUENCY

Find the area of a rectangle 11 cm by 6 cm.

7.C2 FLUENCY

Find the area of a triangle with base 8 cm and height 5 cm.

7.C3 FLUENCY

Find the area of a circle of radius 4 cm.

7.C4 REASONING

Explain why the triangle area formula has a factor of one half.

7.C5 FLUENCY

Find the area of a parallelogram with base 9 cm and height 4 cm.

7.C6 REASONING

Investigate: cut a parallelogram from a site plan and rearrange it. What shape do you get, and what does this show about its area formula?

CRITERION C Communicating

Criterion C practice · Communicating (continued)

7.C7 FLUENCY

Find the area of a trapezium with parallel sides 6 and 10 cm and height 4 cm.

7.C8 REASONING

Explain why area is measured in square units.

7.C9 FLUENCY

A compound shape is two rectangles of area 12 and 18 cm^2. Find the total area.

7.C10 FLUENCY

Find the area of a square of side 7.5 cm.

7.C11 REASONING

Explain the connection between the area of a circle and its radius.

7.C12 FLUENCY

A rectangle's area is 48 cm^2 and its width is 6 cm. Find its length.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

7.D1 APPLICATION

A rectangle has length 3 cm more than its width and perimeter 34 cm. Find the area.

7.D2 APPLICATION

A square and a rectangle have the same perimeter of 36 cm. The rectangle is 10 cm by 8 cm. Which has the larger area?

7.D3 APPLICATION

A circle has circumference 31.4 cm. Find its area.

7.D4 APPLICATION

A triangular garden has base 12 m and height 7 m. Find its area.

7.D5 APPLICATION

A room is 5 m by 4 m. Find the cost of flooring at 12 dollars per m^2.

7.D6 REASONING

Explain, with a calculation, why doubling the radius of a circle more than doubles its area.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

7.D7 APPLICATION

A parallelogram has sides 8 and 5 cm and height 4 cm to the longer side. Find its area.

7.D8 APPLICATION

A picture frame is a rectangle 30 cm by 20 cm with a border 2 cm wide. Find the area of the border.

7.D9 APPLICATION

A rhombus has diagonals 10 cm and 8 cm. Find its area.

7.D10 REASONING

Explain how to find the area of an L-shaped compound figure.

7.D11 APPLICATION

A circular pond has diameter 6 m. Find the area of a 1 m wide path around it.

7.D12 REASONING

Explain why a shape with a fixed perimeter can have different areas.

CHAPTER 7 · APPLICATION AND ASSESSMENT

Applying chapter 7

Identify the shape, choose the formula, and state the units of the answer.

G1 · The classroom carpet

A classroom is 8 m by 6 m. Carpet costs 18 dollars per m^2 and skirting costs 4 dollars per m. Find the total cost and explain the two different formulas you used.

G2 · The garden design

A triangular garden has two sides of 10 m and an included angle of 60°. A rectangular garden has the same perimeter. Find the area of each and say which is larger.

Criteria B and D · state the units for area and for length separately.

CHAPTER 7 · REVIEW

What you should now be able to do

Key ideas

- Angles on a straight line add to 180°; angles around a point add to 360°.
- The angle sum of an n-sided polygon is $(n - 2) \times 180°$.
- Perimeter is boundary length; area is surface covered.
- $C = \pi d$ and $A = \pi r^2$ for a circle.
- Compound areas are found by adding or subtracting simple areas.
- Properties of shapes justify the formulas.

Vocabulary

angle · right angle · parallel · transversal · corresponding · alternate · co-interior · perimeter · circumference · area · compound shape

CHAPTER 7 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

7.R1 Find the fourth angle of a quadrilateral with 95°, 80° and 105°.

7.R2 Find the perimeter of a rectangle 13 cm by 8 cm.

7.R3 Find the area of a triangle with base 9 cm and height 6 cm.

7.R4 Find the circumference of a circle of radius 5 cm.

7.R5 Find the area of a trapezium with sides 7 and 11 cm and height 5 cm.

7.R6 Explain the difference between perimeter and area.

CHAPTER 7 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

The form of a two-dimensional shape, and the logic of how its parts relate, are represented by formulas for perimeter and area, and that representation allows spaces to be measured, planned and costed when land, flooring and materials are limited.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of logic and representation appear in your work?
- How is the global context of globalisation and sustainability, and its exploration of urban and rural planning, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a formula when a real space is irregular, and what would you still like to ask?

CHAPTER 7 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Measurement and design are everywhere in a school building. Measure a real space.

Service as Action

COMMUNITY NEED	A community hall needs its floor refinished and its walls repainted.
MATHEMATICAL CONNECTION	Perimeter and area to calculate the materials needed.
PROPOSED ACTION	Measure the hall and produce a materials list and cost estimate.
EXPECTED IMPACT	The renovation is completed within budget with no wasted materials.
STUDENT REFLECTION	Explain how the measurements changed the plan.

Community Project

PROJECT IDEA	A school playground redesign.
NEED	The playground surface is uneven and unsafe in places.
RESEARCH QUESTION	How much surface material and edging are needed for a safe redesign?
MATHEMATICS	Perimeter, area and compound shapes.
ACTION OR OUTCOME	A measured plan and a materials list for the redesign.
BENEFICIARIES	Students and staff using the playground.
EVIDENCE	The plan, the materials list and the measurements.

Personal Project connection · MYP 5

PERSONAL INTEREST	Interior design, art or sport court marking.
LEARNING GOAL	To design and cost a small space.
PRODUCT OR OUTCOME	A scale plan of a room or court with area and perimeter calculations.
MATHEMATICS	Perimeter, area and scale drawing.
PROCESS	Measure, draw to scale, calculate, then cost the materials.
EVIDENCE	The plan, the calculations and the cost estimate.

CHAPTER 8 · PART III · GEOMETRY AND SPACE

08

3D geometry: volume and surface area

Volume measures the space inside a solid; surface area measures the material needed to cover it. Both follow from the properties of the solid.

ENGAGE

Two boxes hold the same volume. One is long and thin, the other nearly cubic. Which needs less cardboard to make?

Framing the inquiry

Key concept: Form
Related concepts: Logic · Representation
Global context: Globalisation and sustainability
Global context exploration: Human impact on the environment

Statement of inquiry: The form of a solid, and the logic of how its faces and cross-sections relate, are represented by formulas for volume and surface area, and that representation allows the material used for containers to be measured and reduced.

Questions to guide us

- **Factual:** What is the difference between volume and surface area, and which formula gives each for a cuboid, a cylinder and a prism?
- **Conceptual:** Why do two solids with the same volume have different surface areas, and how does a net reveal the surface area of a solid?
- **Debatable:** Is the most efficient shape for packaging always the best choice for the environment, and how would you use mathematics to argue your case?

ATL · TRANSFER. In order for a student to apply skills and knowledge in unfamiliar situations, students must select a structure that fits the new situation, apply it independently, and test the result for sense.

CHAPTER 8 · INVESTIGATION 8.1

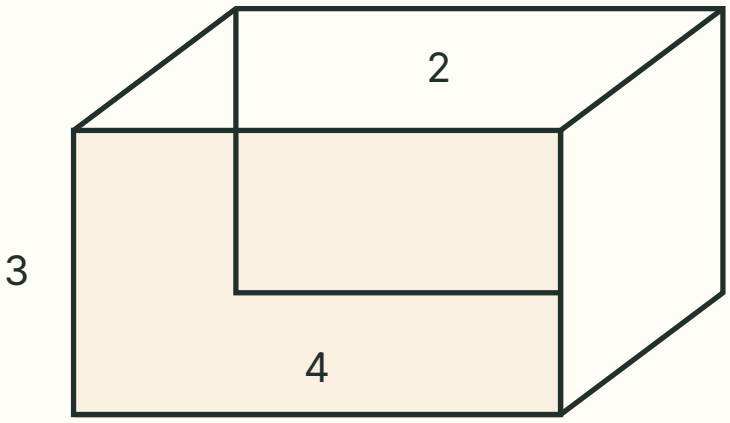
Naming and drawing solids

INQUIRY QUESTION

How can a solid be described so that its faces, edges and vertices can be counted without seeing all of them?

YOUR TASK

Build the net of a cuboid, fold it into the solid, then unfold it and count faces, edges and vertices.



Net folded into a cuboid

solid	faces	edges	vertices
cuboid	6	12	8
cube	6	12	8
your solid			

CHAPTER 8 · INVESTIGATION 8.1

Naming and drawing solids

What do you notice?

- Every cuboid has the same counts, whatever its dimensions.
- A cube has square faces instead of rectangular ones.
- The net folds without gaps, which is why it works.

Observation. Faces, edges and vertices are fixed by the type of solid.

MATHEMATICAL CONCLUSION

A solid is described by its faces, edges and vertices, and a net lays every face out flat, so a cuboid always has 6 faces, 12 edges and 8 vertices.

Your own words

CHAPTER 8 · 8.1

Naming and drawing solids

A solid has faces, edges and vertices. A cuboid has 6 faces, 12 edges and 8 vertices; a cube is a cuboid with all edges equal.

A prism has two identical parallel faces joined by rectangles. A pyramid has one base and triangular faces meeting at an apex. A cylinder has two circular faces and a curved surface.

A net is a flat pattern that folds into a solid. Drawing the net is the most reliable way to count faces and to find the surface area.

Plans and elevations are two-dimensional views: the plan looks down, and the elevation looks from the side.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

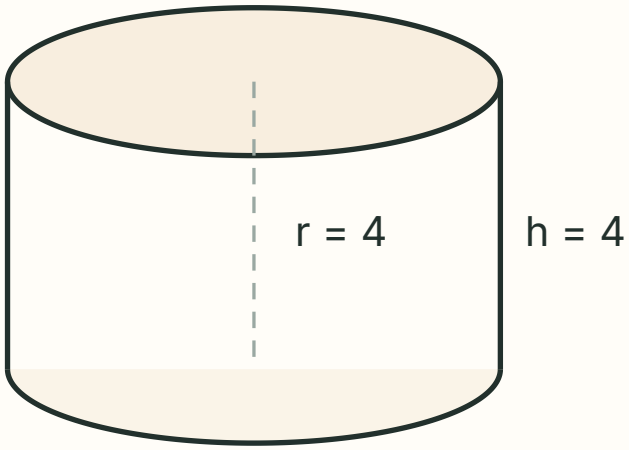
In your own words

CHAPTER 8 · 8.1 · WORKED EXAMPLE 1

Naming and drawing solids

WORKED EXAMPLE 8.1.1

Problem. A solid has 6 rectangular faces, 12 edges and 8 vertices, and all edges are equal. Name it.



A cube

Mathematical reasoning. Six square faces with equal edges and 8 vertices define a cube.

1. Six faces, all square.
2. 12 edges, all the same length.
3. 8 vertices where three edges meet.

Final answer. a cube

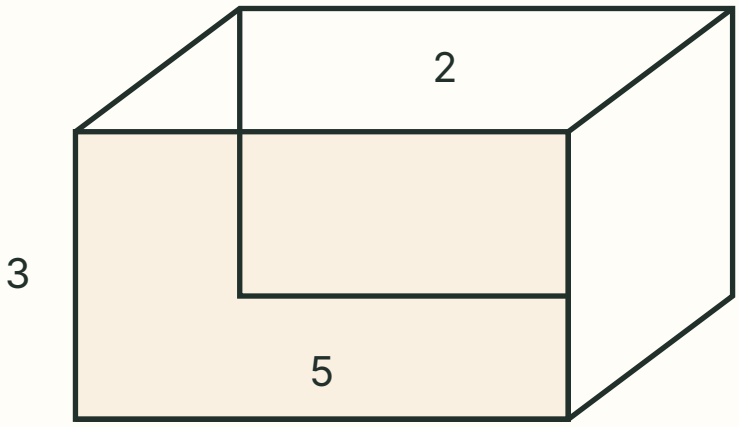
Interpretation. A cube is the most specialised cuboid, so it inherits every cuboid property.

CHAPTER 8 · 8.1 · WORKED EXAMPLE 2

Naming and drawing solids

WORKED EXAMPLE 8.1.2

Problem. Draw a description of the net of a cuboid 5 cm by 3 cm by 2 cm.



A 5 × 3 × 2 cm cuboid

Mathematical reasoning. The net has three pairs of identical rectangles.

1. Two faces of 5 × 3.
2. Two faces of 5 × 2.
3. Two faces of 3 × 2.
4. Arrange them so that each pair folds to meet.

Final answer. three pairs of rectangles: two 5 × 3, two 5 × 2 and two 3 × 2

Interpretation. The net makes the surface area a sum of six rectangle areas.

CHAPTER 8 · INVESTIGATION 8.2

Volume of prisms and cylinders

INQUIRY QUESTION

How can the space inside very different solids be measured with one rule?

YOUR TASK

Fill a cuboid box with 1 cm cubes, then with 2 cm cubes. Record how many of each fit.

cube size	along length	along width	along height	total
1 cm				
2 cm				

CHAPTER 8 · INVESTIGATION 8.2

Volume of prisms and cylinders

What do you notice?

- A 2 cm cube occupies the space of eight 1 cm cubes.
- The number of cubes scales with the volume ratio 8:1.
- Partial cubes at the edges show why units must match.

Observation. Volume counts unit cubes, so it scales with the cube of the linear size.

MATHEMATICAL CONCLUSION

One rule covers every prism — volume = area of cross-section × length — of which volume of a cuboid = length × width × height is the special case.

Your own words

CHAPTER 8 · 8.2

Volume of prisms and cylinders

Volume measures the space inside a solid and is measured in cubic units. A cuboid of length l, width w and height h has volume lwh.

Any prism has volume equal to the area of its cross-section times its length. This single rule covers cuboids, triangular prisms and cylinders.

A cylinder has volume $\pi r^2 h$, because its cross-section is a circle. The radius and the height must use the same unit.

One cubic centimetre is one millilitre, and 1 000 cm^3 is one litre. Volume conversions use powers of ten, exactly like length.

> FORMAL DEVELOPMENT
>
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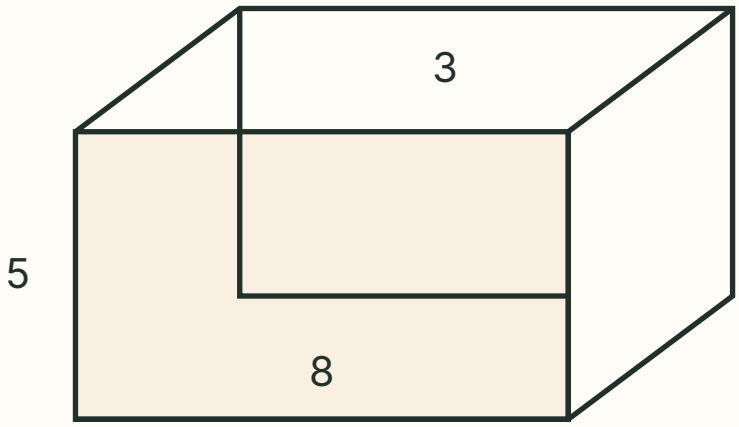
In your own words

CHAPTER 8 · 8.2 · WORKED EXAMPLE 1

Volume of prisms and cylinders

WORKED EXAMPLE 8.2.1

Problem. Find the volume of a cuboid 8 cm by 5 cm by 3 cm.



An 8 × 5 × 3 cm cuboid

Mathematical reasoning. Volume is length × width × height.

1. $8 \times 5 = 40$ cm^2 for the base.
2. $40 \times 3 = 120$ cm^3.

Final answer. 120 cm³

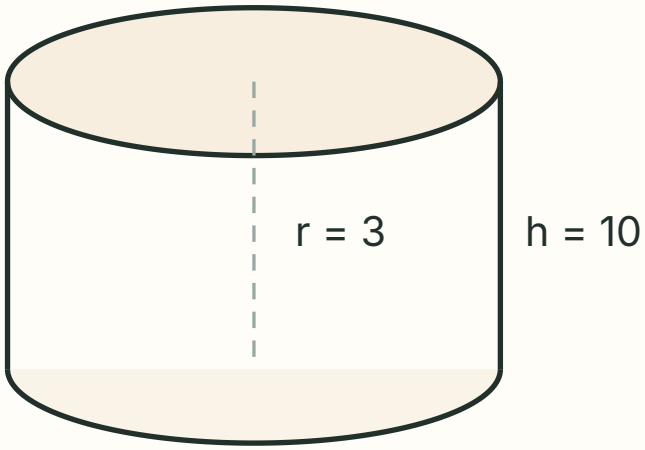
Interpretation. The base area times the height counts how many layers fit inside.

CHAPTER 8 · 8.2 · WORKED EXAMPLE 2

Volume of prisms and cylinders

WORKED EXAMPLE 8.2.2

Problem. A cylinder has radius 3 cm and height 10 cm. Find its volume.



A cylinder of radius 3 cm

Mathematical reasoning. The cross-section is a circle, so volume = $\pi r^2 h$.

1. Base area = $\pi \times 3^2 = 28.27$ cm^2.
2. Volume = $28.27 \times 10 = 282.7$ cm^3.

Final answer. 282.7 cm³

Interpretation. The same cross-section-times-length rule works for every prism.

CHAPTER 8 · INVESTIGATION 8.3

Surface area of solids

INQUIRY QUESTION

How can the material needed to cover a solid be measured, and why does the same volume need different amounts of material?

YOUR TASK

Compare the surface areas of two cuboids with volume 24 cm^3: one $2 \times 3 \times 4$ and one $1 \times 4 \times 6$. Which uses less material?

dimensions	volume	surface area
$2 \times 3 \times 4$	24 cm^3	52 cm^2
$1 \times 4 \times 6$	24 cm^3	68 cm^2

CHAPTER 8 · INVESTIGATION 8.3

Surface area of solids

What do you notice?

- Both boxes hold the same volume.
- The more cubic box has the smaller surface area.
- The flatter box needs more material for the same capacity.

Observation. For a fixed volume, the shape closest to a cube uses the least surface area.

MATHEMATICAL CONCLUSION

Surface area is the sum of the face areas, so it depends on shape and not only on volume: for a fixed volume the shape closest to a cube uses the least material.

Your own words

CHAPTER 8 · 8.3

Surface area of solids

Surface area is the total area of every face. For a cuboid it is $2(lw + lh + wh)$, and for a cube of side s it is $6s^2$.

A cylinder has surface area $2\pi r^2 + 2\pi rh$: two circles plus a rectangle that wraps around. The rectangle has height h and width equal to the circumference $2\pi r$.

Surface area problems in packaging ask for the least material, which means finding the shape with the smallest surface area for a fixed volume.

Nets are the tool for irregular solids: draw the net, find each face's area, then add.

> FORMAL DEVELOPMENT
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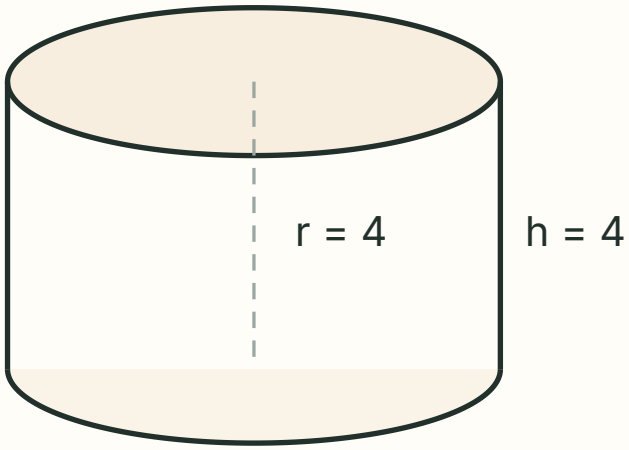
In your own words

CHAPTER 8 · 8.3 · WORKED EXAMPLE 1

Surface area of solids

WORKED EXAMPLE 8.3.1

Problem. Find the surface area of a cube of side 4 cm.



A cube of side 4 cm

Mathematical reasoning. A cube has six identical square faces.

1. One face = $4 \times 4 = 16$ cm^2.
2. Six faces = $6 \times 16 = 96$ cm^2.

Final answer. 96 cm²

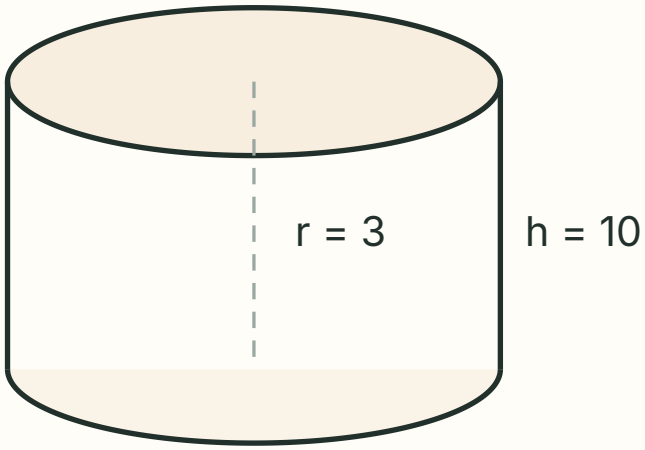
Interpretation. The formula $6s^2$ is just six copies of s^2.

CHAPTER 8 · 8.3 · WORKED EXAMPLE 2

Surface area of solids

WORKED EXAMPLE 8.3.2

Problem. Find the surface area of a cylinder of radius 3 cm and height 10 cm.



A cylinder of radius 3 cm

Mathematical reasoning. The curved surface unrolls into a rectangle of height 10 and width $2\pi r$.

1. Two circles: $2 \times \pi \times 9 = 56.55$ cm^2.
2. Curved surface: $2\pi \times 3 \times 10 = 188.50$ cm^2.
3. Total: $56.55 + 188.50 = 245.05$ cm^2.

Final answer. 245.05 cm²

Interpretation. Unrolling the curved surface turns it into a rectangle, which is why the formula works.

CHAPTER 8 · INVESTIGATION 8.4

Solving 3D problems

INQUIRY QUESTION

How can volume, surface area and edge length be used together to solve a real problem about a container?

YOUR TASK

Find a real box, measure it, and compute its volume, surface area and edge total. Then compare with a second box of similar volume.

box	volume	surface area	edge total
box 1			
box 2			

CHAPTER 8 · INVESTIGATION 8.4

Solving 3D problems

What do you notice?

- Similar volumes can come with very different surface areas.
- The flatter box has the larger surface area.
- The edge total is easy to check by running a tape around the box.

Observation. Measuring a real solid links the formulas to physical material.

MATHEMATICAL CONCLUSION

Volume, surface area and edge length are three distinct measures of one solid, and a real problem usually needs more than one of them.

Your own words

CHAPTER 8 · 8.4

Solving 3D problems

A three-step method handles most problems: find the missing dimension from a formula, compute the required measure, then check the units.

Converting between volume and capacity uses 1 cm^3 = 1 ml and 1 litre = 1 000 cm^3. Density problems link mass, volume and density from chapter 3.

Real problems combine measures: a water tank needs its volume for capacity and its surface area for the material.

> FORMAL DEVELOPMENT
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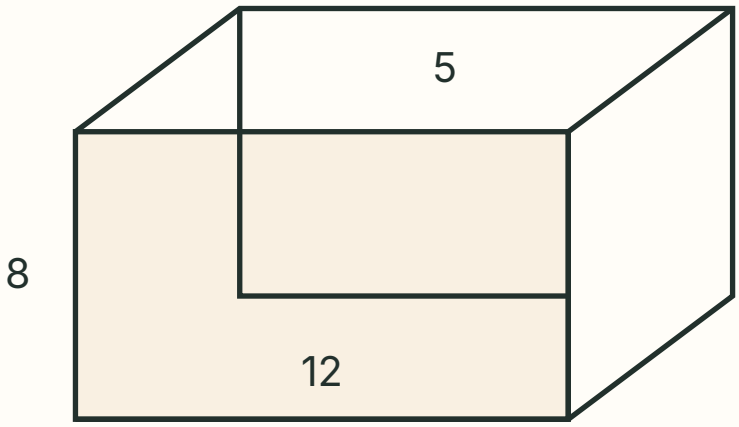
In your own words

CHAPTER 8 · 8.4 · WORKED EXAMPLE 1

Solving 3D problems

WORKED EXAMPLE 8.4.1

Problem. A tank is a cuboid 1.2 m by 0.8 m by 0.5 m. Find its capacity in litres.



A tank in decimetres

Mathematical reasoning. Find the volume in cubic metres, then convert.

1. Volume = $1.2 \times 0.8 \times 0.5 = 0.48$ m^3.
2. 1 m^3 = 1 000 litres, so 0.48 m^3 = 480 litres.

Final answer. 480 litres

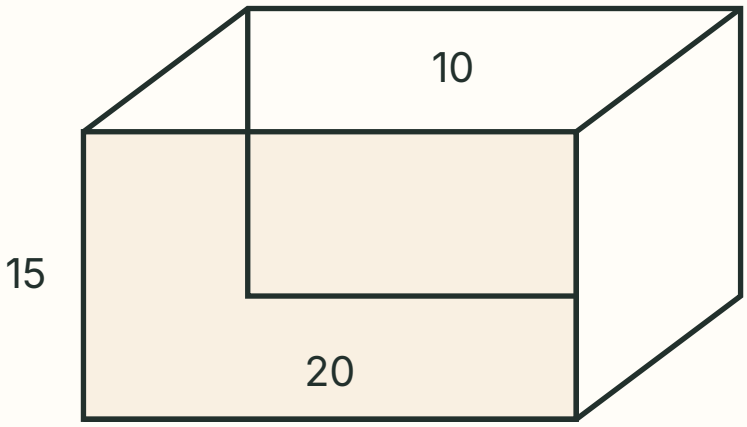
Interpretation. Converting after calculating avoids mixing units inside the formula.

CHAPTER 8 · 8.4 · WORKED EXAMPLE 2

Solving 3D problems

WORKED EXAMPLE 8.4.2

Problem. A closed cuboid box is 20 cm by 15 cm by 10 cm. Find the surface area and the length of tape needed to go round all the edges once.



A box with edges to tape

Mathematical reasoning. Surface area uses the three face areas; the edge total is 4 times the sum of the three dimensions.

1. Surface area = $2(20 \times 15 + 20 \times 10 + 15 \times 10) = 2(300 + 200 + 150) = 1\,300$ cm^2.
2. Edge total = $4(20 + 15 + 10) = 4 \times 45 = 180$ cm.

Final answer. 1 300 cm² and 180 cm of tape

Interpretation. A single solid can require several different measures at once.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

8.A1 FLUENCY

How many faces, edges and vertices does a cuboid have?

8.A2 FLUENCY

Name a solid with two identical circular faces and a curved surface.

8.A3 REASONING

What is a net, and why does a carton designer need one before the cardboard is cut?

8.A4 FLUENCY

A carton is in the shape of a triangular prism. How many faces must be printed and folded?

8.A5 REASONING

Explain the difference between a prism and a pyramid, using two containers a factory might make.

8.A6 REASONING

What does a plan view show, and how does it help when a container is packed into a box?

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

8.A7 FLUENCY

How many edges meet at each vertex of a cube?

8.A8 FLUENCY

Name the solid with 6 square faces.

8.A9 REASONING

Explain why a net is useful for finding surface area.

8.A10 FLUENCY

A roof light is in the shape of a square-based pyramid. How many faces does it have?

8.A11 REASONING

Explain what an elevation shows, and why a designer draws one before making a container.

8.A12 FLUENCY

A carton is a triangular prism. How many vertices does it have, and how many edges meet at each one?

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

8.B1 FLUENCY

Find the volume of a cuboid 7 cm by 4 cm by 3 cm.

8.B2 FLUENCY

Find the volume of a cube of side 5 cm.

8.B3 FLUENCY

Find the volume of a cylinder of radius 2 cm and height 9 cm.

8.B4 FLUENCY

Investigate: how many 1 cm cubes fit in a $6 \times 5 \times 4$ box?

8.B5 REASONING

Explain why volume is measured in cubic units.

8.B6 APPLICATION

A cuboid has volume 96 cm^3 and base area 24 cm^2. Find its height.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

8.B7 REASONING

Investigate the pattern: volumes of cubes of side 1, 2, 3, 4.

8.B8 FLUENCY

A tank holds 2 500 cm^3 of liquid. Convert this into litres and say why the smaller unit is useful for a small container.

8.B9 REASONING

Explain the cross-section rule for the volume of a prism.

8.B10 APPLICATION

Find the volume of a triangular prism with base 6 cm, height 4 cm and length 10 cm.

8.B11 FLUENCY

A tank holds 0.6 m^3. Find its capacity in litres.

CRITERION C Communicating

Criterion C practice · Communicating

8.C1 FLUENCY

Find the surface area of a cuboid 5 cm by 4 cm by 3 cm.

8.C2 FLUENCY

Find the surface area of a cube of side 6 cm.

8.C3 FLUENCY

Find the surface area of a cylinder of radius 2 cm and height 5 cm.

8.C4 REASONING

Explain why a cylinder's surface area has two parts.

8.C5 APPLICATION

A cube has surface area 150 cm^2. Find its side.

8.C6 REASONING

Investigate: which cuboid with volume 36 cm^3 has the least surface area?

CRITERION C Communicating

Criterion C practice · Communicating (continued)

8.C7 REASONING
Explain how a net helps find the surface area of a pyramid.

8.C8 FLUENCY
Find the surface area of a cuboid 10 cm by 10 cm by 2 cm.

8.C9 REASONING
Explain the difference between volume and surface area.

8.C10 CHALLENGE
A cylinder has surface area 100 cm^2 and radius 2 cm. Find its height.

8.C11 REASONING
Explain why a cube has the least surface area for its volume among cuboids.

8.C12 FLUENCY
Find the total edge length of a cuboid 8 cm by 6 cm by 4 cm.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

8.D1 APPLICATION

A box is 30 cm by 20 cm by 15 cm. Find its volume and surface area.

8.D2 APPLICATION

A cylindrical tank has radius 0.5 m and height 1.2 m. Find its capacity in litres.

8.D3 APPLICATION

A steel block of mass 7 800 g has volume 1 000 cm^3. Find its density and name the material if steel is 7.8 g/cm^3.

8.D4 CHALLENGE

A closed box has surface area 376 cm^2 and is a cube. Find its volume.

8.D5 APPLICATION

A rectangular pool is 25 m by 10 m by 2 m. Find the volume of water and the area of the four walls.

8.D6 REASONING

Explain why two boxes with the same volume can need different amounts of cardboard.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

8.D7 APPLICATION

A cylindrical can has diameter 7 cm and height 11 cm. Find the metal needed for a closed can.

8.D8 APPLICATION

A cube of side 10 cm is cut into 1 cm cubes. How many small cubes, and what is their total surface area?

8.D9 APPLICATION

A room is 6 m by 5 m by 3 m. Find the volume of air and the area to paint (walls and ceiling).

8.D10 REASONING

Explain how to find the capacity of an irregular container.

8.D11 APPLICATION

A prism has cross-section area 15 cm^2 and length 8 cm. Find its volume.

8.D12 REASONING

Explain why a thin box of volume 1 litre needs more cardboard than a cubic one.

CHAPTER 8 · APPLICATION AND ASSESSMENT

Applying chapter 8

Name the solid, choose the formula, and keep the units consistent throughout.

H1 · The water tank

A rectangular tank measures 1.5 m by 0.6 m by 0.4 m. Find its capacity in litres, the area of metal needed to build it, and the cost at 25 dollars per m^2.

H2 · The packaging choice

A company packs 1 litre of juice. Compare a cuboid 10 cm by 5 cm by 20 cm with a cylinder of radius 5 cm and height 12.7 cm. Which needs less material?

Criteria B and D · compute both surface areas before concluding.

CHAPTER 8 · REVIEW

What you should now be able to do

Key ideas

- A solid has faces, edges and vertices fixed by its type.
- A net folds into a solid and reveals all its faces.
- Volume of a prism = area of cross-section × length.
- Surface area of a cuboid = $2(lw + lh + wh)$.
- Surface area of a cylinder = $2\pi r^2 + 2\pi rh$.
- For a fixed volume, the shape closest to a cube uses the least material.

Vocabulary

face · edge · vertex · net · prism · pyramid · cylinder · cuboid · cube · volume · surface area · capacity

CHAPTER 8 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

8.R1 Find the volume of a cuboid 9 cm by 5 cm by 4 cm.

8.R2 Find the surface area of a cube of side 3 cm.

8.R3 Find the volume of a cylinder of radius 3 cm and height 7 cm.

8.R4 How many faces does a square-based pyramid have?

8.R5 Convert 3 200 cm^3 into litres.

8.R6 Explain the difference between volume and capacity.

CHAPTER 8 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

The form of a solid, and the logic of how its faces and cross-sections relate, are represented by formulas for volume and surface area, and that representation allows the material used for containers to be measured and reduced.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of logic and representation appear in your work?
- How is the global context of globalisation and sustainability, and its exploration of human impact on the environment, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of choosing a container by surface area alone, and what would you still like to ask?

CHAPTER 8 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Packaging and construction are volume problems. Optimise a real container.

Service as Action

COMMUNITY NEED	A community kitchen needs containers that hold set volumes with little waste.
MATHEMATICAL CONNECTION	Volume and surface area to design efficient containers.
PROPOSED ACTION	Design and compare three containers for 2 litres and recommend the most efficient.
EXPECTED IMPACT	Less packaging material and lower cost for the kitchen.
STUDENT REFLECTION	Explain how surface area changed the design choice.

Community Project

PROJECT IDEA	A school recycling and packaging audit.
NEED	Packaging waste from the school canteen is not measured.
RESEARCH QUESTION	Which packaging shape gives the least surface area for the volume sold?
MATHEMATICS	Volume, surface area and density.
ACTION OR OUTCOME	An audit report with a recommendation to the canteen.
BENEFICIARIES	The school and the local environment.
EVIDENCE	The measurements, the report and the recommendation.

Personal Project connection · MYP 5

PERSONAL INTEREST	Model making, gaming or storage design.
LEARNING GOAL	To design a container for a specific volume.
PRODUCT OR OUTCOME	A net, a folded model and a materials calculation.
MATHEMATICS	Volume, surface area and nets.
PROCESS	Choose a volume, design the net, build it, then check the capacity.
EVIDENCE	The net, the model and the calculations.

CHAPTER 9 · PART III · GEOMETRY AND SPACE

09

Transformations and symmetry

A transformation moves a shape without changing its size or its angles. Describing the movement precisely is the whole task.

ENGAGE

A tile pattern repeats across a wall. Each tile is the same shape in a new position. How do you describe the movement from one tile to the next?

Framing the inquiry

Key concept: Form
Related concepts: Logic · Relationships
Global context: Personal and cultural expression
Global context exploration: Artistry, craft, creation and beauty

Statement of inquiry: The form of a shape is preserved by transformations, and the logic of describing each movement precisely — and the relationships between them — allows patterns to be created, extended and recognised in art, craft and design.

Questions to guide us

- **Factual:** What are the four basic transformations, and what information is needed to describe each one completely?
- **Conceptual:** Which properties of a shape are preserved under each transformation, and how are translations, reflections and rotations related to one another?
- **Debatable:** Is a reflected or enlarged shape still the same shape, and how would you defend your answer to a designer?

ATL · COMMUNICATION. In order for a student to communicate information and ideas effectively, students must organise a structure for a purpose, use appropriate notation, and check that a reader can follow the argument.

CHAPTER 9 · INVESTIGATION 9.1

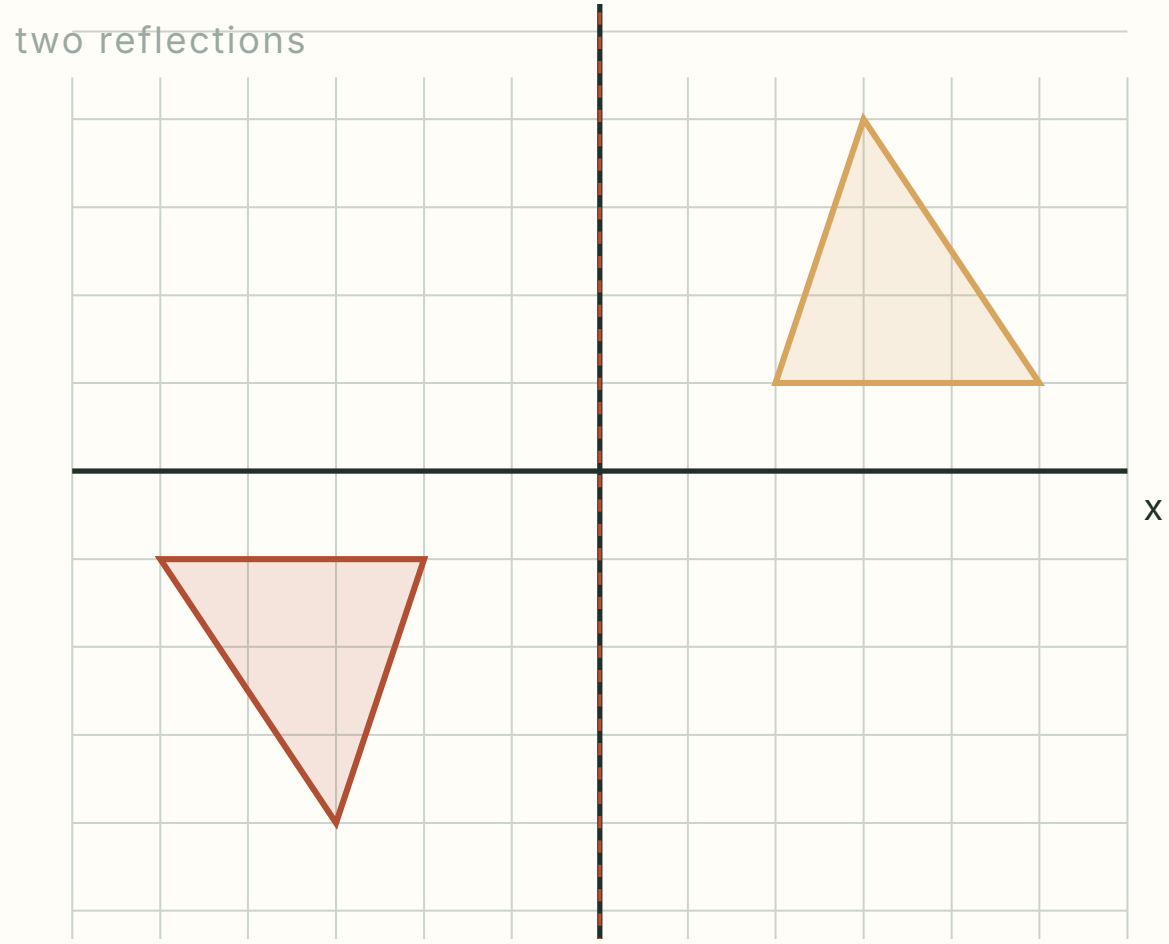
Translations and reflections

INQUIRY QUESTION

How can a movement of a shape be described so that someone else can reproduce it exactly?

YOUR TASK

Draw a shape on grid paper, reflect it in the y-axis, then reflect the image in the x-axis. Compare the final image with the original.



step	vertices
original	(2,1), (5,1), (3,4)
reflected in y	(−2,1), (−5,1), (−3,4)
reflected in x	(−2,−1), (−5,−1), (−3,−4)

CHAPTER 9 · INVESTIGATION 9.1

Translations and reflections

What do you notice?

- Both coordinates change sign after the two reflections.
- The final image is the original rotated by 180° about the origin.
- Lengths and angles are unchanged throughout.

Observation. Two reflections in perpendicular lines have the same effect as a half turn.

MATHEMATICAL CONCLUSION

A movement is fully described by a vector or a mirror line, and combining them shows that reflections in perpendicular mirror lines are equivalent to a rotation of 180°.

Your own words

CHAPTER 9 · 9.1

Translations and reflections

A translation slides every point of a shape the same distance in the same direction. It is described by a column vector: 3 right and 2 up is written (3, 2).

A reflection flips a shape in a mirror line. Every point moves the same perpendicular distance to the other side of the line.

Both transformations preserve length and angle, so the image is congruent to the original. A translation preserves orientation; a reflection reverses it.

Describing a transformation needs the mirror line or the vector. 'It moved' is not a description; 'translated by (3, 2)' is.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

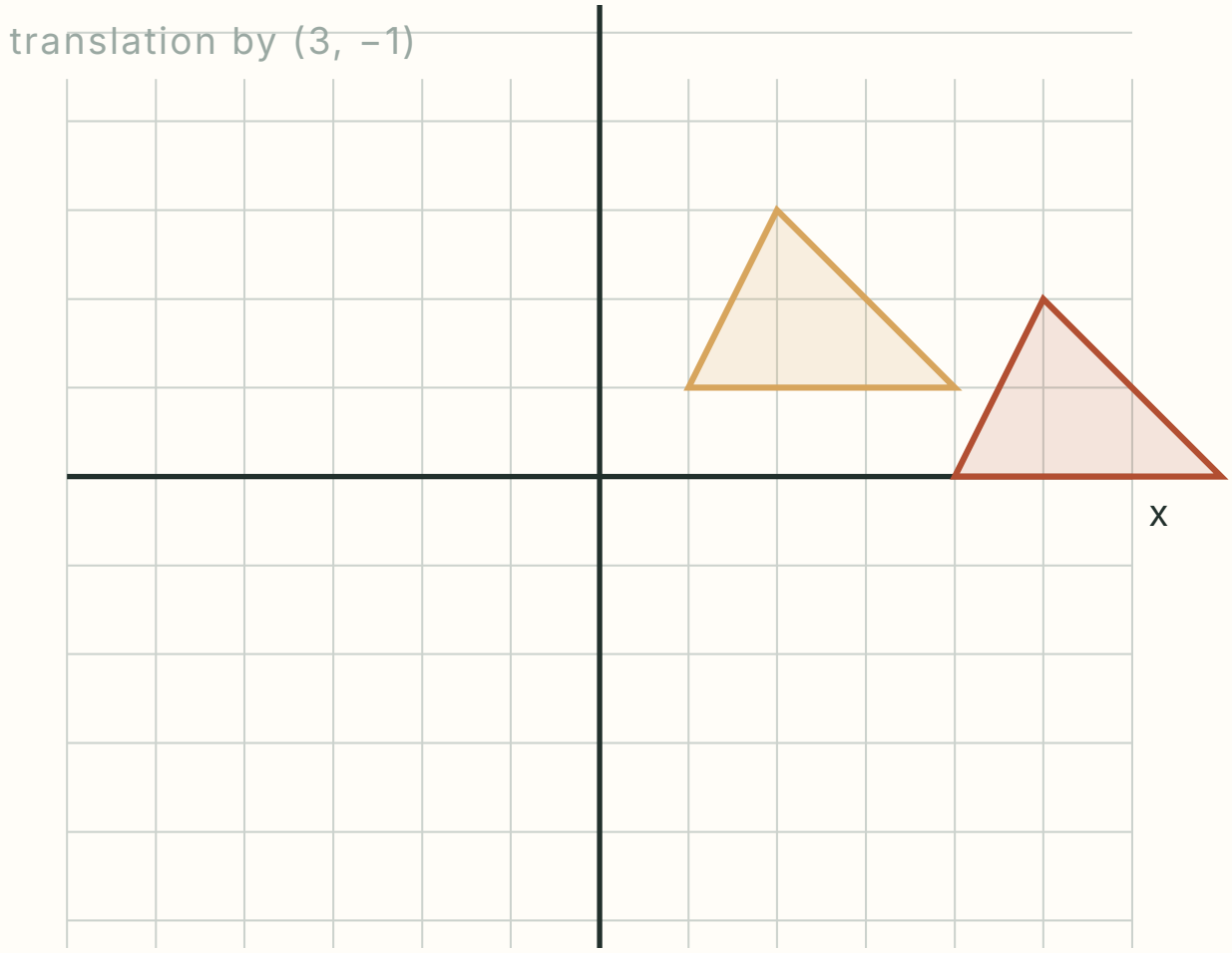
In your own words

CHAPTER 9 · 9.1 · WORKED EXAMPLE 1

Translations and reflections

WORKED EXAMPLE 9.1.1

Problem. Translate the triangle with vertices (1, 1), (4, 1), (2, 3) by the vector (3, −1). Give the image.



Mathematical reasoning. Add the vector to every vertex.

1. (1, 1) + (3, −1) = (4, 0).
2. (4, 1) + (3, −1) = (7, 0).
3. (2, 3) + (3, −1) = (5, 2).

Final answer. (4, 0), (7, 0), (5, 2)

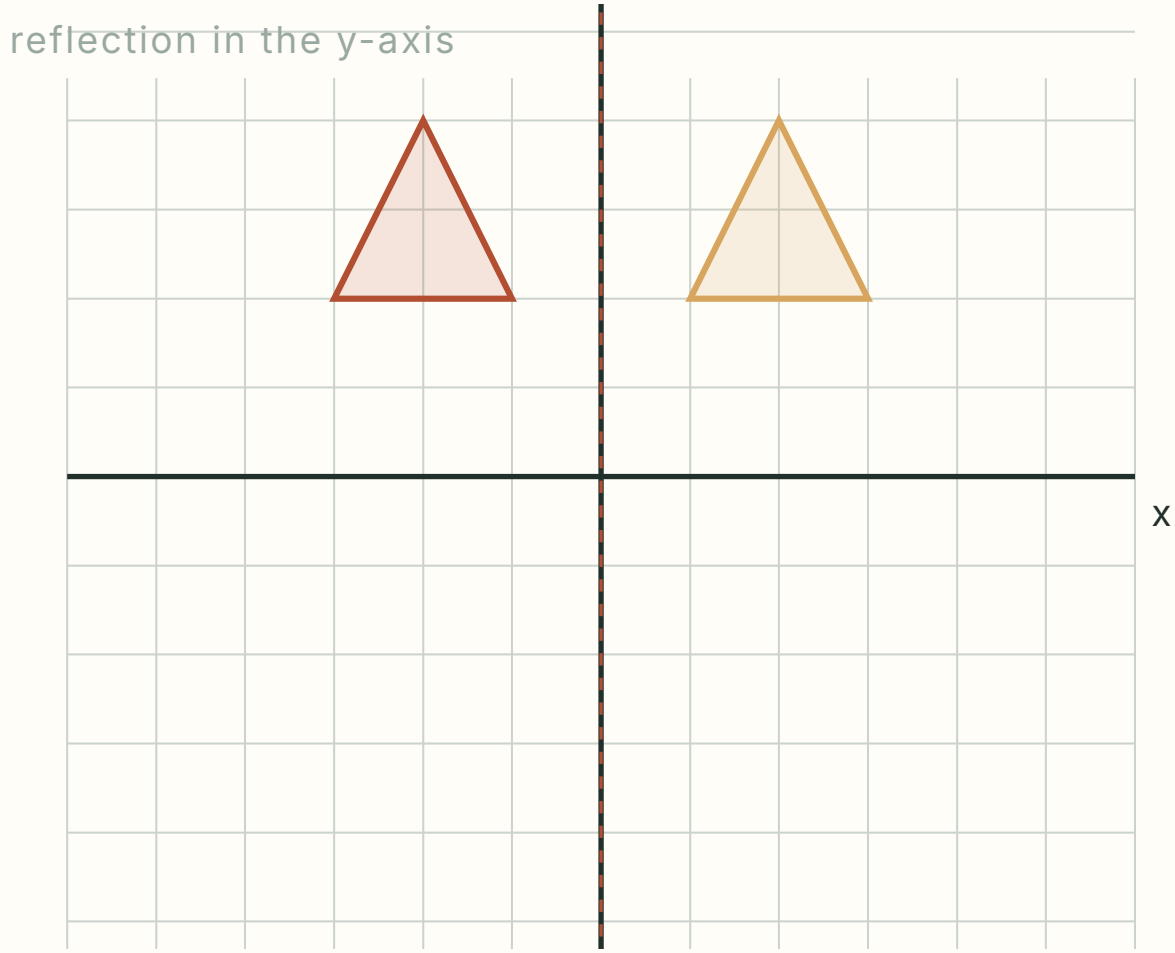
Interpretation. Every point moves identically, which is why the shape is unchanged.

CHAPTER 9 · 9.1 · WORKED EXAMPLE 2

Translations and reflections

WORKED EXAMPLE 9.1.2

Problem. Reflect the point (3, 2) in the y-axis and in the line y = x.



Mathematical reasoning. Reflection in the y-axis changes the sign of x; reflection in y = x swaps the coordinates.

1. In the y-axis: (3, 2) → (−3, 2).
2. In y = x: (3, 2) → (2, 3).

Final answer. (−3, 2) and (2, 3)

Interpretation. Each mirror line has its own rule, which is why the line must be stated.

CHAPTER 9 · INVESTIGATION 9.2

Rotations and enlargements

INQUIRY QUESTION

What changes and what stays the same when a shape is rotated or enlarged?

YOUR TASK

Enlarge the same triangle by factors 1, 2 and 3 about the origin. Record the base, height and area at each step.

factor	base	height	area
1	2	3	3
2	4	6	12
3	6	9	27

CHAPTER 9 · INVESTIGATION 9.2

Rotations and enlargements

What do you notice?

- Base and height multiply by k.
- Area multiplies by k^2.
- The shape of the triangle never changes.

Observation. Enlargement preserves angles and scales lengths and areas by k and k^2.

MATHEMATICAL CONCLUSION

Rotation preserves length and angle, while an enlargement preserves angle only: under an enlargement of scale factor k, lengths scale by k and areas by k^2.

Your own words

CHAPTER 9 · 9.2

Rotations and enlargements

A rotation turns a shape about a fixed centre through a given angle. It is described by the centre, the angle and the direction.

An enlargement changes the size. It is described by a centre and a scale factor: a factor greater than 1 enlarges, between 0 and 1 reduces, and negative reverses the direction.

Enlargement preserves angles but multiplies every length by the scale factor, so area is multiplied by its square. This is the same scaling rule as in chapter 2.

Congruent shapes have the same size and shape; similar shapes have the same shape but different sizes. Congruence comes from rotations, reflections and translations; similarity comes from enlargement.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

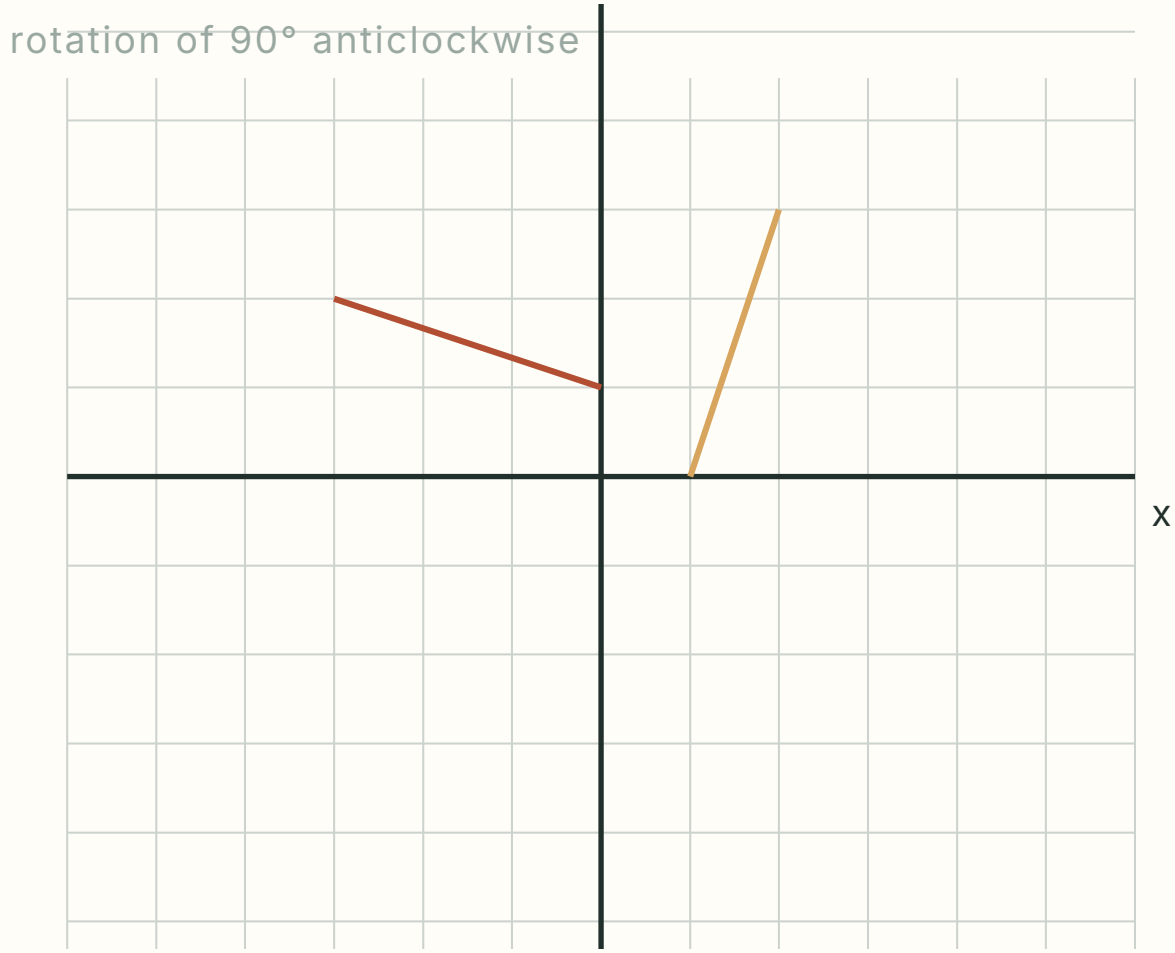
In your own words

CHAPTER 9 · 9.2 · WORKED EXAMPLE 1

Rotations and enlargements

WORKED EXAMPLE 9.2.1

Problem. Rotate the point (2, 3) by 90° anticlockwise about the origin.



Mathematical reasoning. Under a 90° anticlockwise rotation about the origin, (x, y) becomes (−y, x).

1. Apply the rule: (2, 3) → (−3, 2).

Final answer. (−3, 2)

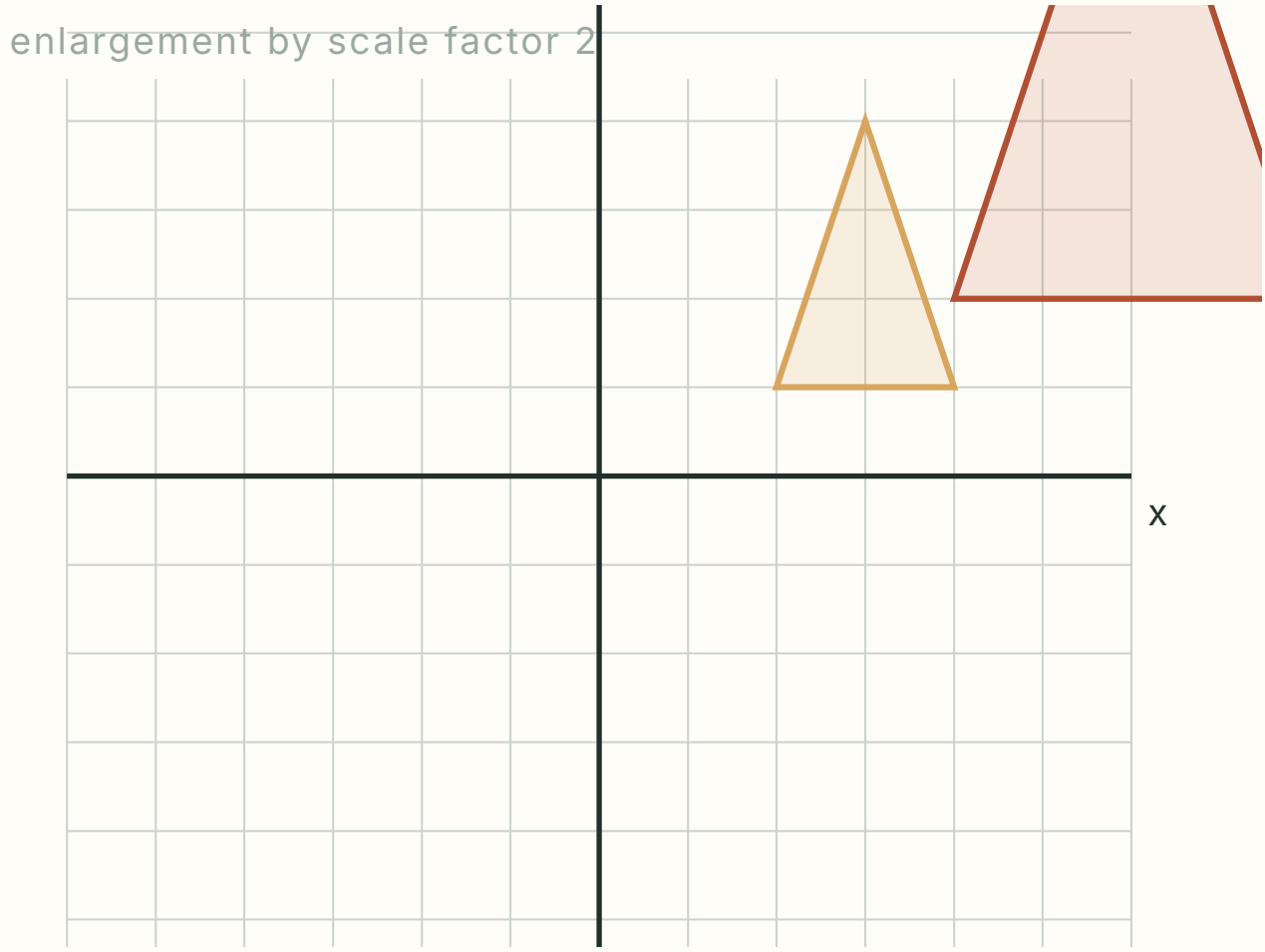
Interpretation. The distance from the origin is unchanged, which is why the rotation is an isometry.

CHAPTER 9 · 9.2 · WORKED EXAMPLE 2

Rotations and enlargements

WORKED EXAMPLE 9.2.2

Problem. Enlarge a triangle with vertices (2, 1), (4, 1), (3, 4) by scale factor 2 about the origin. Find the image and the ratio of the areas.



Mathematical reasoning. Multiply every coordinate by the scale factor, and square it for area.

1. Image vertices: (4, 2), (8, 2), (6, 8).
2. Original base 2, image base 4, so lengths double.
3. Area ratio = $2^2 = 4$, so the image has four times the area.

Final answer. (4, 2), (8, 2), (6, 8) and an area ratio of 4:1

Interpretation. The shape is similar: angles are identical but the area is four times greater.

CHAPTER 9 · INVESTIGATION 9.3

Symmetry

INQUIRY QUESTION

How can symmetry be described and counted, and what does it tell you about a shape?

YOUR TASK

Collect five capital letters. For each, record the number of lines of symmetry and the order of rotational symmetry.

letter	lines of symmetry	rotational order
A	1	1
H	2	2
N	0	2
your letter		

CHAPTER 9 · INVESTIGATION 9.3

Symmetry

What do you notice?

- Some letters have both kinds, some only one, some neither.
- Rotational order 1 means no rotational symmetry.
- The counts depend only on the shape of the letter.

Observation. Line and rotational symmetry are independent properties.

MATHEMATICAL CONCLUSION

A shape's symmetry is fully described by its lines of symmetry and its rotational order, and counting them also checks a transformation.

Your own words

CHAPTER 9 · 9.3

Symmetry

A shape has line symmetry if it can be folded along a line so that the two halves match exactly. That line is a line of symmetry.

Rotational symmetry means the shape fits onto itself after a rotation of less than 360°. The order of rotational symmetry is how many times it fits in a full turn.

A regular polygon with n sides has n lines of symmetry and rotational symmetry of order n.

Symmetry is used in design, in classification of shapes, and in checking a transformation: a result that breaks the symmetry is usually an error.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

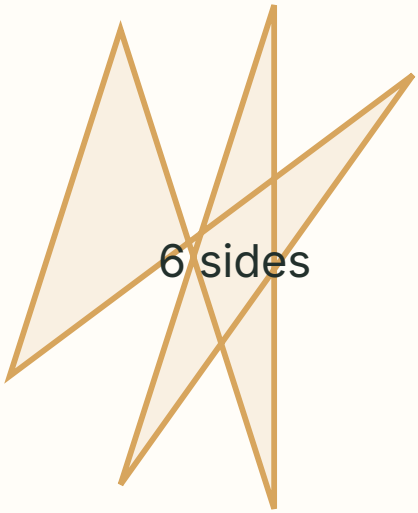
CHAPTER 9 · 9.3 · WORKED EXAMPLE 1

Symmetry

WORKED EXAMPLE 9.3.1

Problem. State the number of lines of symmetry and the order of rotational symmetry of a regular hexagon.

A regular hexagon



Mathematical reasoning. A regular n-gon has n lines of symmetry and rotational order n.

1. A regular hexagon has 6 sides.
2. Lines of symmetry: 6 (3 through opposite vertices, 3 through midpoints).
3. Rotational order: 6, with rotation angle 60°.

Final answer. 6 lines of symmetry and rotational symmetry of order 6

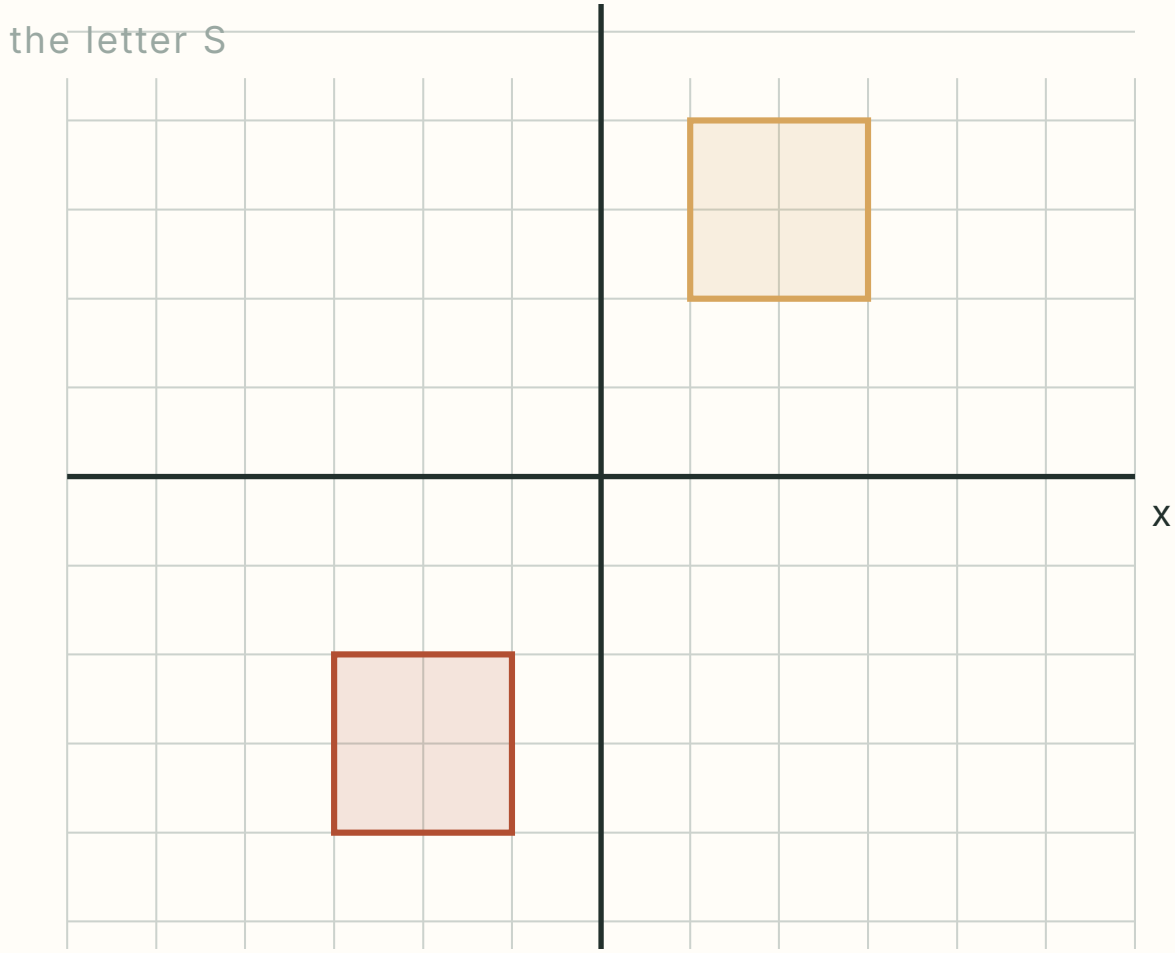
Interpretation. Regularity is what guarantees the full symmetry.

CHAPTER 9 · 9.3 · WORKED EXAMPLE 2

Symmetry

WORKED EXAMPLE 9.3.2

Problem. Does the letter 'S' have line symmetry or rotational symmetry?



Mathematical reasoning. Test folding and turning against the actual shape.

1. No line folds the S onto itself, so no line symmetry.
2. A 180° turn maps the S onto itself.
3. So the S has rotational symmetry of order 2 only.

Final answer. rotational symmetry of order 2, no line symmetry

Interpretation. A shape can have one kind of symmetry without the other.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

9.A1 FLUENCY
Describe the transformation that slides a shape 4 right and 3 down.

9.A2 FLUENCY
Translate the point (2, 5) by the vector (−3, 1).

9.A3 FLUENCY
Reflect the point (4, 3) in the y-axis.

9.A4 REASONING
What is preserved under a translation?

9.A5 FLUENCY
Reflect the point (5, 2) in the x-axis.

9.A6 REASONING
Explain the difference between a translation and a reflection.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

9.A7 FLUENCY

Reflect the point (3, 4) in the line y = x.

9.A8 FLUENCY

A point moves from (1, 1) to (4, 5). Write the translation vector.

9.A9 REASONING

Explain why a reflected triangle is congruent to the original.

9.A10 FLUENCY

Translate (0, 0) by (5, −2).

9.A11 REASONING

Explain what a mirror line does to a point, using the fold of a piece of fabric as your example.

9.A12 FLUENCY

Reflect (−2, 7) in the y-axis.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

9.B1 FLUENCY

Rotate the point (3, 0) by 90° anticlockwise about the origin.

9.B2 FLUENCY

Rotate the point (2, 4) by 180° about the origin.

9.B3 FLUENCY

Enlarge the point (2, 3) by scale factor 3 about the origin.

9.B4 REASONING

Investigate: what happens to a shape under scale factor 1?

9.B5 REASONING

Explain what a negative scale factor does to a printed motif, and how you can tell from the image.

9.B6 FLUENCY

Rotate (1, 2) by 90° clockwise about the origin.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

9.B7 REASONING

Investigate the pattern: areas of a triangle enlarged by factors 1, 2, 3.

9.B8 REASONING

State the three things needed to describe a rotation.

9.B9 FLUENCY

Enlarge the triangle (2, 2), (4, 2), (3, 5) by scale factor 0.5.

9.B10 REASONING

Explain the difference between congruent and similar shapes.

9.B11 FLUENCY

Rotate (0, 5) by 180° about the origin.

CRITERION C Communicating

Criterion C practice · Communicating

9.C1 FLUENCY

How many lines of symmetry does a square have?

9.C2 FLUENCY

State the order of rotational symmetry of an equilateral triangle.

9.C3 REASONING

Explain the difference between line and rotational symmetry.

9.C4 FLUENCY

How many lines of symmetry does a regular pentagon have?

9.C5 REASONING

Investigate: does a parallelogram have line symmetry?

9.C6 FLUENCY

State the order of rotational symmetry of a rectangle.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

9.C7 REASONING
Explain why a circle has infinitely many lines of symmetry.

9.C8 FLUENCY
A shape has rotational symmetry of order 4. What rotation angle maps it onto itself?

9.C9 REASONING
Investigate the pattern: lines of symmetry of regular polygons with 3, 4, 5, 6 sides.

9.C10 REASONING
Explain how symmetry can be used to check a transformation.

9.C11 FLUENCY
How many lines of symmetry does an isosceles triangle have?

9.C12 REASONING
Explain what rotational order 1 means.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

9.D1 APPLICATION

A triangle has vertices (1, 2), (5, 2), (3, 6). Reflect it in the y-axis, then translate by (1, −2). Give the final vertices.

9.D2 APPLICATION

A shape is enlarged by scale factor 3. Its area becomes 54 cm^2. Find the original area.

9.D3 APPLICATION

A regular octagon is rotated. What is the smallest rotation that maps it onto itself?

9.D4 REASONING

Describe a single transformation mapping (2, 3) to (5, 7).

9.D5 APPLICATION

A square of side 4 cm is enlarged by scale factor 2.5. Find the new area.

9.D6 REASONING

Explain, with a calculation, why an enlargement of factor 2 multiplies the area by 4.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

9.D7 REASONING
Two reflections in the x-axis then the y-axis are equivalent to what single transformation?

9.D8 APPLICATION
A triangle is enlarged by factor −2 about the origin. Where does (1, 1) go?

9.D9 REASONING
A shape has 5 lines of symmetry and rotational order 5. Name a possible shape.

9.D10 APPLICATION
Translate the shape (0,0), (2,0), (0,3) by (4,1) and state the area ratio.

9.D11 APPLICATION
An enlargement maps (3, 1) to (9, 3). Find the scale factor.

9.D12 REASONING
Explain why a scalene triangle has no line symmetry.

CHAPTER 9 · APPLICATION AND ASSESSMENT

Applying chapter 9

State the transformation completely: type, then all the values needed to fix it.

I1 · The tile pattern

A tile has vertices (1, 1), (3, 1), (2, 3). Describe the transformations that produce a row of four tiles, and state the vertices of the last tile.

I2 · The logo design

A logo is made by enlarging a shape by scale factor 2 and then reflecting it. State the area ratio of the final logo to the original and justify your answer.

Criteria B and D · explain vector operations and check the result.

CHAPTER 9 · REVIEW

What you should now be able to do

Key ideas

- A translation is described by a column vector.
- A reflection needs a mirror line; it reverses orientation.
- A rotation needs a centre, an angle and a direction.
- An enlargement needs a centre and a scale factor; lengths scale by k and areas by k^2.
- Congruence comes from isometries; similarity comes from enlargement.
- A regular n-gon has n lines of symmetry and rotational order n.

Vocabulary

transformation · translation · vector · reflection · mirror line · rotation · centre of rotation · enlargement · scale factor · congruent · similar · line of symmetry · rotational symmetry

CHAPTER 9 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

9.R1 Translate (3, 4) by (−2, 5).

9.R2 Reflect (6, 1) in the x-axis.

9.R3 Rotate (2, 5) by 180° about the origin.

9.R4 How many lines of symmetry does a regular hexagon have?

9.R5 An enlargement of factor 4 changes an area of 5 cm^2 to what?

9.R6 Explain the difference between congruent and similar.

CHAPTER 9 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

The form of a shape is preserved by transformations, and the logic of describing each movement precisely — and the relationships between them — allows patterns to be created, extended and recognised in art, craft and design.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of logic and relationships appear in your work?
- How is the global context of personal and cultural expression, and its exploration of artistry, craft, creation and beauty, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of describing a design by transformations alone, and what would you still like to ask?

CHAPTER 9 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Symmetry and transformation govern pattern, design and structure. Find an example in your community.

Service as Action

COMMUNITY NEED	A community centre wall needs a mural that can be painted by volunteers.
MATHEMATICAL CONNECTION	Transformations to repeat a motif across the wall.
PROPOSED ACTION	Design a motif and a transformation rule so volunteers can paint the full mural.
EXPECTED IMPACT	A completed mural produced consistently by a team.
STUDENT REFLECTION	Explain how the transformation rule kept the pattern accurate.

Community Project

PROJECT IDEA	A school symmetry and pattern audit.
NEED	The school's decorative patterns are inconsistent and hard to maintain.
RESEARCH QUESTION	Which transformations generate the patterns already used in the school?
MATHEMATICS	Translations, reflections, rotations and symmetry.
ACTION OR OUTCOME	A pattern catalogue with the transformation that generates each one.
BENEFICIARIES	The school and future design projects.
EVIDENCE	The catalogue, the diagrams and the symmetry counts.

Personal Project connection · MYP 5

PERSONAL INTEREST	Art, graphic design or game level design.
LEARNING GOAL	To create a design using at least two transformations.
PRODUCT OR OUTCOME	A design or level plan built from a transformed motif.
MATHEMATICS	Transformations, congruence and similarity.
PROCESS	Draw the motif, apply transformations, then describe each one precisely.
EVIDENCE	The design and the written description of each transformation.

CHAPTER 10 · PART IV · PROBABILITY AND DATA

10

Probability and simulation

Probability measures how likely an outcome is. Experiments, sample spaces and simulations turn chance into numbers that can be reasoned about.

ENGAGE

A game gives a point when two dice show the same number. Is the game fair? How could you find out without playing it a thousand times?

Framing the inquiry

Key concept: Relationships
Related concepts: Logic · Representation
Global context: Scientific and technical innovation
Global context exploration: Randomness, chance, risk and uncertainty

Statement of inquiry: The relationship between an experiment and the outcomes it can produce is described by probability, and the logic of counting those outcomes — with representation as a sample space or a simulation — allows randomness to be quantified and risk to be assessed rather than guessed.

Questions to guide us

- **Factual:** What does a probability of 0.5 mean, and how are experimental and theoretical probabilities calculated?
- **Conceptual:** Why can experimental results differ from theoretical probabilities, and how does the sample space connect the outcomes to the probabilities?
- **Debatable:** Is a simulation ever better than a real experiment, and what would you need to know about the risk before relying on it?

ATL · CRITICAL THINKING. In order for a student to draw reasonable conclusions and generalisations, students must test conclusions against evidence, revise them where necessary, and develop theories.

CHAPTER 10 · INVESTIGATION 10.1

Describing probability

INQUIRY QUESTION

How can chance be described as a number, and what does that number mean?

YOUR TASK

Toss a coin 20 times and record the number of heads. Repeat three times and compare with the theoretical probability of 1/2.

trial	heads	tails	experimental P(head
1			
2			
3			

CHAPTER 10 · INVESTIGATION 10.1

Describing probability

What do you notice?

- Each trial gives a different experimental probability.
- The theoretical probability stays at 1/2.
- More tosses bring the experimental value closer to 1/2.

Observation. Experimental probability varies; theoretical probability is fixed by the outcomes.

MATHEMATICAL CONCLUSION

Probability is a number from 0 to 1 on a scale from impossible to certain, and experimental probability approaches theoretical probability as the number of trials grows.

Your own words

CHAPTER 10 · 10.1

Describing probability

The probability of an event is a number from 0 to 1. Zero means impossible, 1 means certain, and 0.5 means equally likely to happen or not.

Theoretical probability comes from counting equally likely outcomes: P(event) = number of favourable outcomes ÷ total number of outcomes. Experimental probability comes from repeating a trial.

Probabilities of all outcomes of one experiment add to 1, so the probability of 'not A' is 1 − P(A).

Probability can be written as a fraction, a decimal or a percentage, but the scale from 0 to 1 is the reference for all of them.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

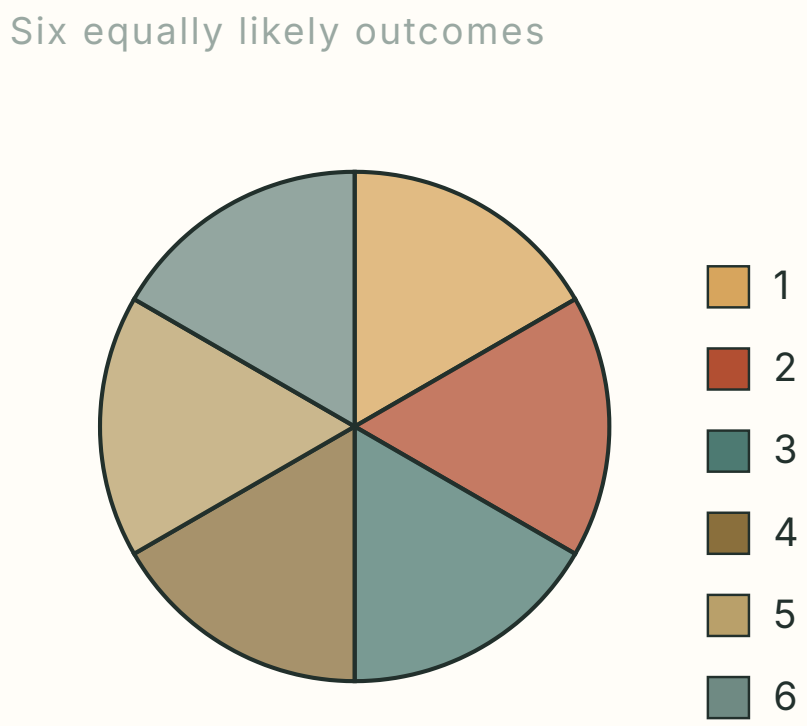
In your own words

CHAPTER 10 · 10.1 · WORKED EXAMPLE 1

Describing probability

WORKED EXAMPLE 10.1.1

Problem. A fair six-sided die is rolled. Find the probability of rolling a number greater than 4.



Mathematical reasoning. There are six equally likely outcomes, and two of them are favourable.

1. Outcomes: 1, 2, 3, 4, 5, 6.
2. Favourable: 5 and 6, so 2 outcomes.
3. $P = 2/6 = 1/3$.

Final answer. 1/3

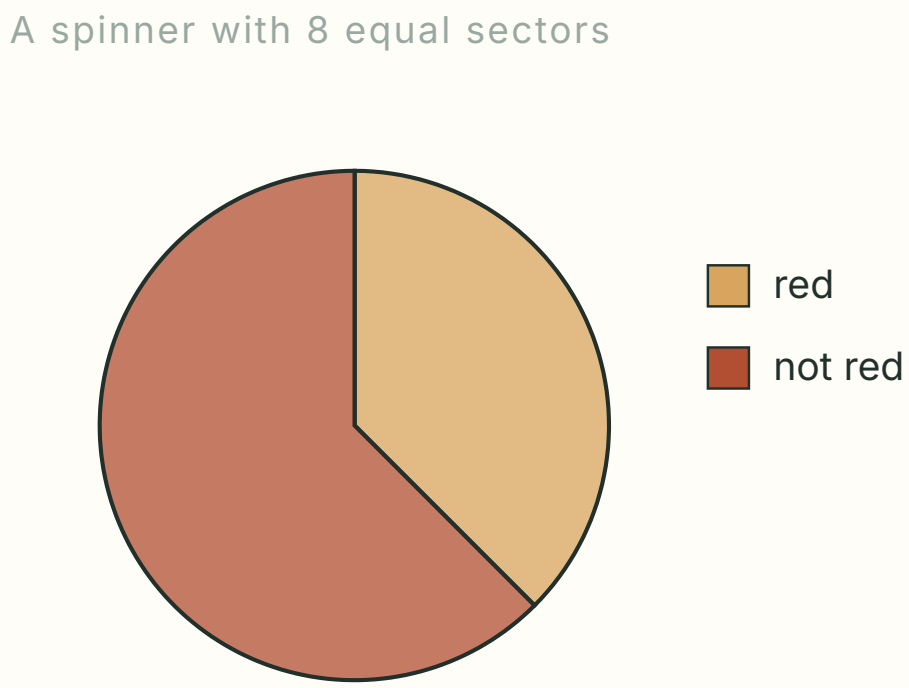
Interpretation. The fraction counts favourable outcomes against all equally likely outcomes.

CHAPTER 10 · 10.1 · WORKED EXAMPLE 2

Describing probability

WORKED EXAMPLE 10.1.2

Problem. A spinner has 8 equal sectors, 3 of them red. Find the probability of not landing on red.



Mathematical reasoning. The probability of 'not red' is 1 minus the probability of red.

1. P(red) = 3/8.
2. P(not red) = 1 − 3/8 = 5/8.
3. There are 5 non-red sectors, which confirms the result.

Final answer. 5/8

Interpretation. Complementary probabilities always add to 1.

CHAPTER 10 · INVESTIGATION 10.2

Sample spaces and listing outcomes

INQUIRY QUESTION

How can every possible outcome be counted so that none is missed?

YOUR TASK

List the sample space for two coins. Then list the sample space for a coin and a four-sided die, and compare the sizes.

experiment	outcomes	size
two coins	HH, HT, TH, TT	4
coin and 4-sided die	H1, H2, H3, H4, T1, T2, T3, T4	8

CHAPTER 10 · INVESTIGATION 10.2

Sample spaces and listing outcomes

What do you notice?

- Two independent choices multiply the number of outcomes.
- HT and TH are different outcomes when the coins are distinct.
- The sample space size is the denominator of every probability.

Observation. The size of a sample space is the product of the numbers of outcomes for each part.

MATHEMATICAL CONCLUSION

Listing the full sample space prevents missing favourable outcomes, and its size gives the denominator of every theoretical probability.

Your own words

CHAPTER 10 · 10.2

Sample spaces and listing outcomes

A sample space is the set of all possible outcomes. Listing it completely is the first step in any probability calculation.

For two events, a table or a grid lists all combinations. For two dice there are 36 outcomes, and 6 of them are doubles.

An outcome is a single result; an event is a set of outcomes. The probability of an event is the sum of the probabilities of its outcomes.

Missing an outcome is the most common error, so the count of the sample space is always worth checking.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

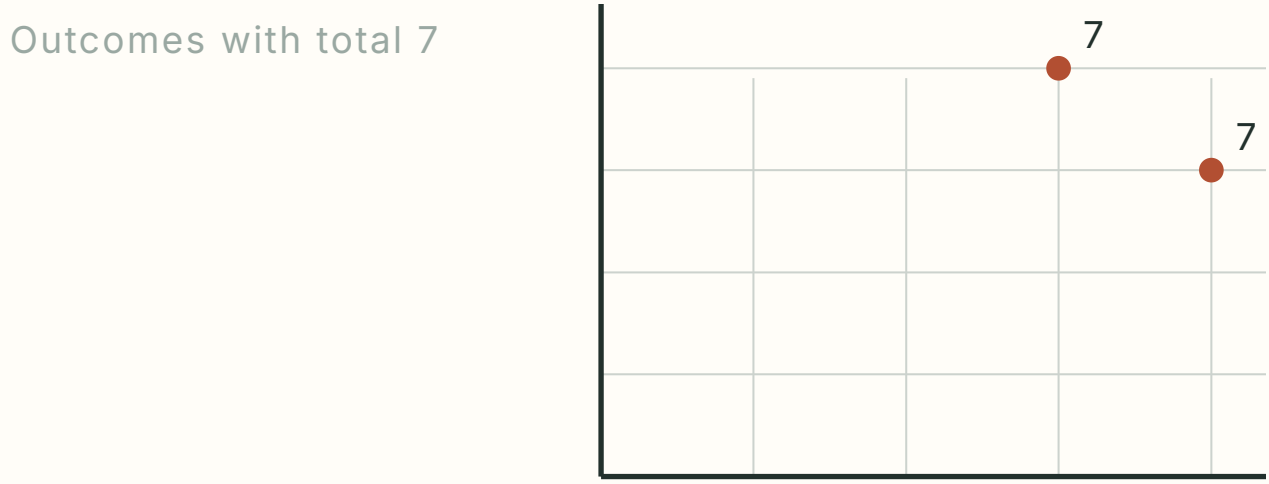
In your own words

CHAPTER 10 · 10.2 · WORKED EXAMPLE 1

Sample spaces and listing outcomes

WORKED EXAMPLE 10.2.1

Problem. Two dice are rolled. Find the probability that the total is 7.



Mathematical reasoning. There are 36 equally likely outcomes, and six of them give a total of 7.

1. Sample space: $6 \times 6 = 36$ outcomes.
2. Totals of 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) — six outcomes.
3. $P(\text{total } 7) = 6/36 = 1/6$.

Final answer. 1/6

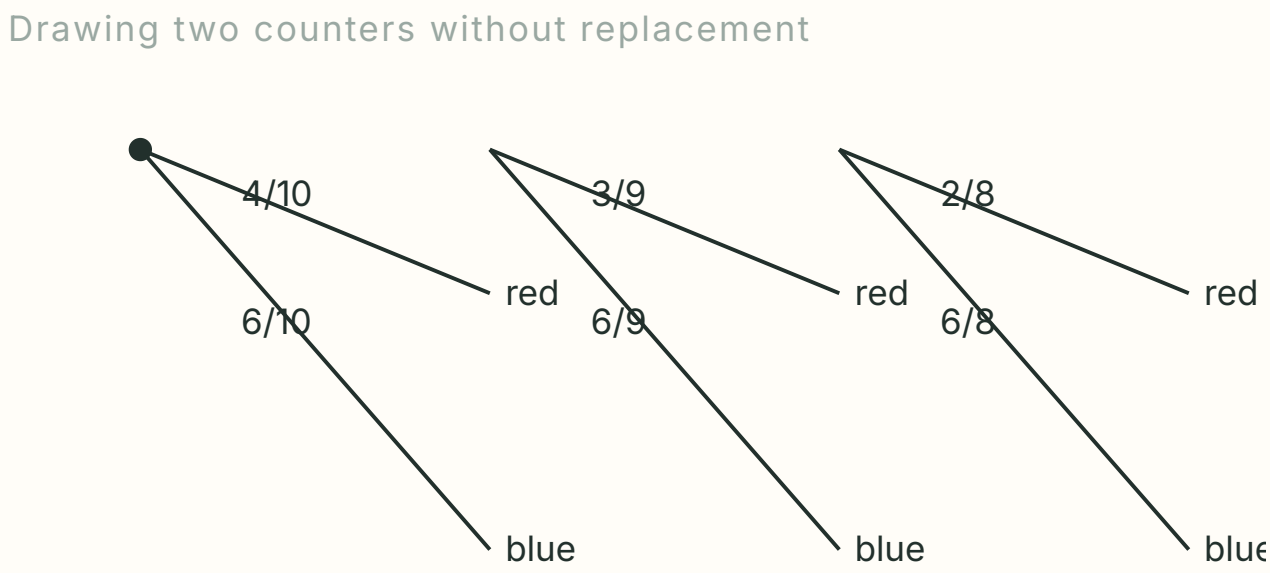
Interpretation. The grid shows the six favourable outcomes at a glance.

CHAPTER 10 · 10.2 · WORKED EXAMPLE 2

Sample spaces and listing outcomes

WORKED EXAMPLE 10.2.2

Problem. A bag holds 4 red and 6 blue counters. Two are drawn without replacement. Find the probability that both are red.



Mathematical reasoning. Without replacement, the probabilities change after the first draw.

1. P(first red) = 4/10.
2. P(second red | first red) = 3/9.
3. P(both red) = 4/10 × 3/9 = 12/90 = 2/15.

Final answer. 2/15

Interpretation. The second probability is smaller because one red counter has already been removed.

CHAPTER 10 · INVESTIGATION 10.3

Probability rules

INQUIRY QUESTION

How do the probabilities of events combine, and what rules govern them?

YOUR TASK

Roll two dice 60 times and record the totals. Compare the experimental probabilities of a total of 7 with the theoretical value of 1/6.

total	frequency	experimental P	theoretical P
7			1/6
other			5/6

CHAPTER 10 · INVESTIGATION 10.3

Probability rules

What do you notice?

- The experimental probability of 7 is close to but rarely exactly 1/6.
- Sixty trials is a small sample, so variation is expected.
- The theoretical value is exact because all 36 outcomes are equally likely.

Observation. Experimental results fluctuate around the theoretical probability.

MATHEMATICAL CONCLUSION

The addition rule covers mutually exclusive outcomes, the multiplication rule covers independent stages, and more trials reduce the gap between experimental and theoretical probability.

Your own words

CHAPTER 10 · 10.3

Probability rules

For mutually exclusive events, P(A or B) = P(A) + P(B). If they are not mutually exclusive, the overlap must be subtracted.

For independent events, P(A and B) = P(A) × P(B). Dependence means the first event changes the probability of the second.

These two rules cover most probability calculations, and choosing between them depends on whether the question says 'or' or 'and'.

A tree diagram makes the structure visible: add along the branches for 'or', multiply along a path for 'and'.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

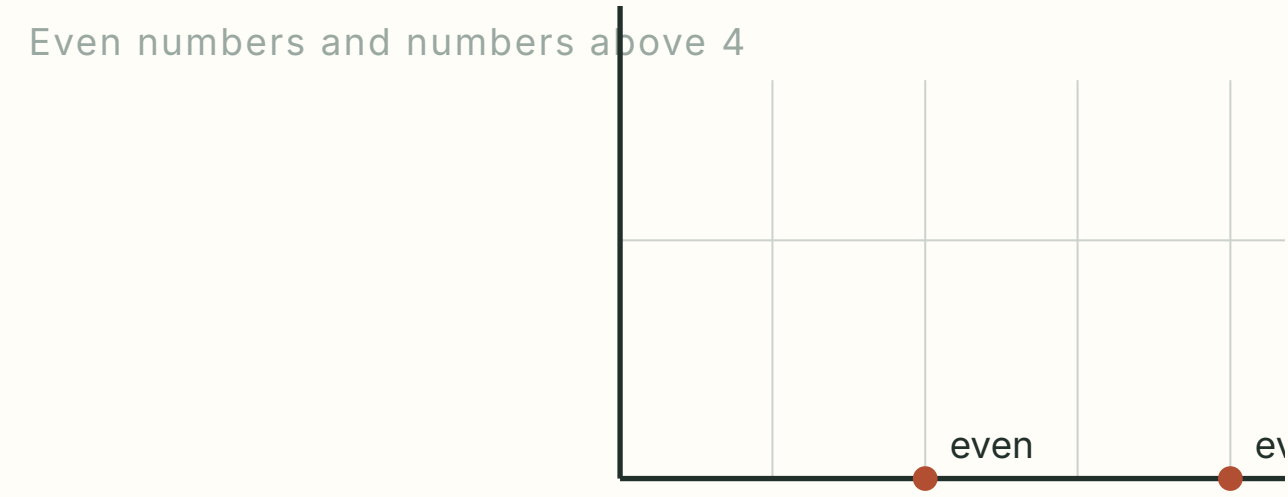
In your own words

CHAPTER 10 · 10.3 · WORKED EXAMPLE 1

Probability rules

WORKED EXAMPLE 10.3.1

Problem. A die is rolled. Find P(even or greater than 4).



Mathematical reasoning. The events overlap at 6, so the overlap must be subtracted.

1. P(even) = 3/6. P(greater than 4) = 2/6.
2. Overlap: P(6) = 1/6.
3. P(even or > 4) = 3/6 + 2/6 − 1/6 = 4/6 = 2/3.

Final answer. 2/3

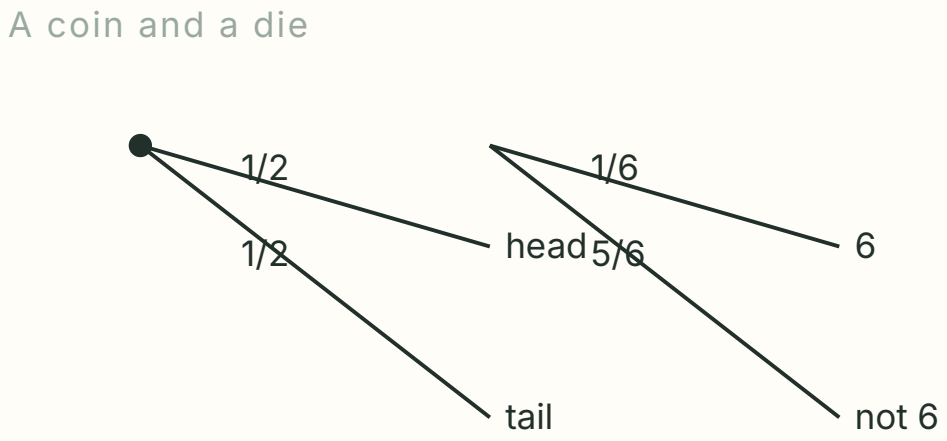
Interpretation. Ignoring the overlap would double-count the outcome 6.

CHAPTER 10 · 10.3 · WORKED EXAMPLE 2

Probability rules

WORKED EXAMPLE 10.3.2

Problem. A coin is tossed and a die is rolled. Find P(head and 6).



Mathematical reasoning. The events are independent, so the probabilities multiply.

1. P(head) = 1/2. P(6) = 1/6.
2. P(head and 6) = 1/2 × 1/6 = 1/12.

Final answer. 1/12

Interpretation. Independence is what allows the multiplication.

CHAPTER 10 · INVESTIGATION 10.4

Simulation and fair games

INQUIRY QUESTION

How can a real experiment be replaced by a model, and when is a game fair?

YOUR TASK

Simulate 50 spins of a five-sector spinner using random digits, and compare the experimental frequency of each sector with the theoretical 1/5.

sector	theoretical P	experimental frequency
1	1/5	
2	1/5	
3	1/5	
4	1/5	
5	1/5	

CHAPTER 10 · INVESTIGATION 10.4

Simulation and fair games

What do you notice?

- Digits 0 and 1 give sector 1, and so on, for five sectors.
- Fifty trials give frequencies near 10 for each sector.
- The total frequency must be 50, which checks the simulation.

Observation. A simulation reproduces theoretical probabilities when the rule matches them.

MATHEMATICAL CONCLUSION

A simulation is a model of a real experiment, and its rule must reproduce the probabilities; a game is fair when every player has the same chance of winning.

Your own words

CHAPTER 10 · 10.4

Simulation and fair games

A simulation uses random numbers to model a situation that is impractical to repeat. A random number table, dice, coins or a calculator can all generate the outcomes.

A fair game gives each player the same expected chance of winning. Expected value is the average result over many repetitions, and it decides whether a game is worth playing.

Simulations are also used to test designs: how long a queue gets, how often a machine fails, how a disease spreads.

Every simulation needs a stated rule linking random numbers to outcomes, and enough trials for the result to settle.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

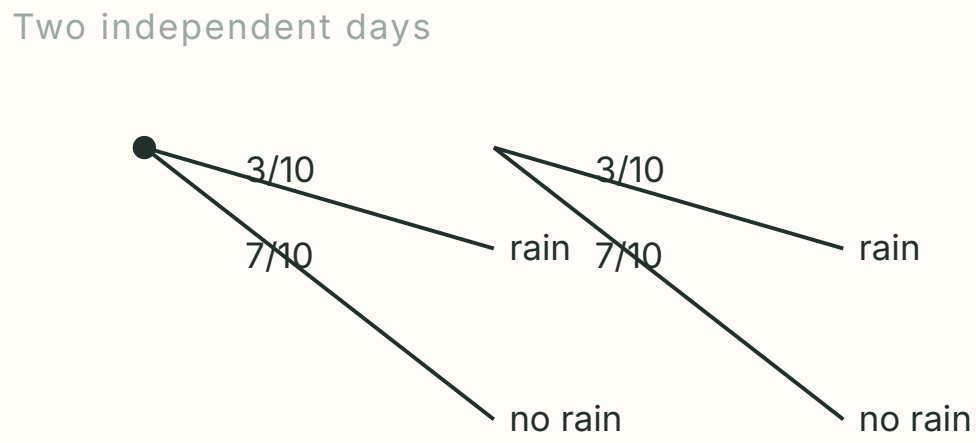
In your own words

CHAPTER 10 · 10.4 · WORKED EXAMPLE 1

Simulation and fair games

WORKED EXAMPLE 10.4.1

Problem. Simulate a 30% chance of rain using single digits. Describe the rule and the probability of rain on two consecutive days.



Mathematical reasoning. Digits 0, 1 and 2 give 3 favourable digits out of 10, so the chance is 30%.

1. Digits 0, 1, 2 mean rain; 3–9 mean no rain.
2. P(rain) = 3/10.
3. P(rain on both days) = 3/10 × 3/10 = 9/100.

Final answer. digits 0–2 for rain; 9/100 for two consecutive rainy days

Interpretation. The simulation rule must match the target probability exactly.

CHAPTER 10 · 10.4 · WORKED EXAMPLE 2

Simulation and fair games

WORKED EXAMPLE 10.4.2

Problem. A game costs 1 dollar to play and pays 5 dollars when a die shows 6. Find the expected value and say whether the game is fair.

outcome	probability	value
6	1/6	5 dollars
not 6	5/6	0 dollars

Expected value of the game

Mathematical reasoning. Expected value is the sum of each value multiplied by its probability, minus the cost.

1. Expected payout = $1/6 \times 5 = 0.83$ dollars.
2. Cost = 1 dollar.
3. Expected value = $0.83 - 1 = -0.17$ dollars.

Final answer. −0.17 dollars, so the game is not fair to the player

Interpretation. A negative expected value means a loss on average, however lucky a single game feels.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

10.A1 FLUENCY
State the range of possible probabilities.

10.A2 FLUENCY
A bag holds 3 red and 7 blue counters. Find P(red).

10.A3 FLUENCY
A fair coin is tossed. Find P(heads).

10.A4 FLUENCY
If P(rain) = 0.3, find P(no rain).

10.A5 REASONING
Explain the difference between theoretical and experimental probability.

10.A6 FLUENCY
A die is rolled. Find P(6).

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

10.A7 FLUENCY

A weather model gives a 0.25 chance of rain. Convert this into a fraction and a percentage and explain which is clearer to a reader.

10.A8 REASONING

Explain what a probability of 1 means.

10.A9 FLUENCY

A spinner has 5 equal sectors. Find P(any one sector).

10.A10 FLUENCY

A class has 12 boys and 18 girls. A student is chosen at random. Find P(girl).

10.A11 REASONING

Explain why probabilities cannot exceed 1.

10.A12 FLUENCY

A pack of cards holds 2 green, 3 yellow and 5 red cards. Find P(not red) and explain what the answer means for a player.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

10.B1 FLUENCY
List the sample space for two coins.

10.B2 FLUENCY
How many outcomes are there when two dice are rolled?

10.B3 FLUENCY
Two dice are rolled. Find P(doubles).

10.B4 REASONING
Investigate: are HT and TH the same outcome for two coins?

10.B5 FLUENCY
A lock code uses two digits from 0 to 9. How many codes are possible, and how does this affect the risk of guessing it?

10.B6 REASONING
Explain the difference between an outcome and an event.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

10.B7 FLUENCY
List the outcomes of a spinner with 4 sectors labelled A, B, C, D.

10.B8 REASONING
Investigate the pattern: sample space sizes for 1, 2, 3 coins.

10.B9 FLUENCY
A die and a coin are used. Find the sample space size.

10.B10 REASONING
Explain why listing the sample space helps avoid errors.

10.B11 FLUENCY
Two dice are rolled. Find P(total 2).

CRITERION C Communicating

Criterion C practice · Communicating

10.C1 FLUENCY
A die is rolled. Find P(even or 5).

10.C2 APPLICATION
A card is drawn from a 52-card pack. Find P(heart or king).

10.C3 FLUENCY
Two coins are tossed. Find P(two heads).

10.C4 REASONING
Explain when to add probabilities and when to multiply.

10.C5 REASONING
A bag holds 5 red and 5 blue counters. Two are drawn without replacement. Explain in words how to find P(both red) and evaluate it.

10.C6 REASONING
Investigate: why does P(even or odd) equal 1 for a fair die, and what does this tell you about certain events?

CRITERION C Communicating

Criterion C practice · Communicating (continued)

10.C7 REASONING
A fair die is rolled twice. Explain how to find P(6 then not 6) and evaluate it, saying why the two rolls are independent.

10.C8 REASONING
Explain what mutually exclusive means.

10.C9 FLUENCY
A spinner has 3 red and 3 blue sectors. Find P(red then blue in two spins).

10.C10 REASONING
Explain what independent events are, using a risk example such as two separate days of bad weather.

10.C11 REASONING
A bag holds 4 white and 6 black. One is drawn. Find P(white or black).

10.C12 APPLICATION
Two dice are rolled. Find P(total greater than 10).

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

10.D1 APPLICATION
A game costs 2 dollars and pays 10 dollars with probability 1/5. Find the expected value.

10.D2 REASONING
A coin is tossed until the first head. Describe a simulation using random digits.

10.D3 REASONING
A factory finds 3% of items faulty. Simulate this with two digits and state the rule.

10.D4 APPLICATION
A lottery ticket costs 1 dollar. The chance of winning 100 dollars is 1/100. Find the expected value.

10.D5 REASONING
Explain why a simulation needs many trials.

10.D6 APPLICATION
A game pays 4 dollars for a head and nothing for a tail, and costs 1 dollar. Find the expected value.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

10.D7 REASONING
A basketball player scores with probability 0.7. Simulate 10 shots using digits.

10.D8 REASONING
A queue simulation shows an average wait of 4 minutes over 200 trials. Explain what this means.

10.D9 REASONING
Two players each roll a die. The higher roll wins. Explain why the game is fair.

10.D10 APPLICATION
A game pays 6 dollars when two dice total 7 and costs 1 dollar. Find the expected value.

10.D11 REASONING
Explain how a random digit table can model a 25% chance.

10.D12 APPLICATION
A raffle sells 200 tickets and you buy 3. Find your probability of winning and the expected value of a 50-dollar prize.

CHAPTER 10 · APPLICATION AND ASSESSMENT

Applying chapter 10

List the outcomes first, then compute. Experimental results are compared with theory, not confused with it.

J1 · The school fair game

A game charges 2 dollars. A die is rolled: a 6 pays 10 dollars, an even number pays 1 dollar, otherwise nothing. Find the expected value and advise whether the game should be run at the fair.

J2 · The weather simulation

The chance of rain on a given day is 40%. Design a simulation using random digits, run 50 simulated days, and compare the experimental probability with the theoretical value.

Criteria B and D · state the sample space and justify your choice. compare.

CHAPTER 10 · REVIEW

What you should now be able to do

Key ideas

- Probability is a number from 0 to 1.
- Theoretical probability counts equally likely outcomes.
- Experimental probability comes from trials and varies.
- Complementary probabilities add to 1.
- Add for mutually exclusive 'or'; multiply for independent 'and'.
- A simulation models a real experiment with a stated rule.

Vocabulary

probability · outcome · event · sample space · theoretical probability · experimental probability · mutually exclusive · independent · simulation · expected value · fair game

CHAPTER 10 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

10.R1 A bag holds 4 red and 6 blue. Find P(red).

10.R2 List the sample space for a coin and a die.

10.R3 Two dice are rolled. Find P(total 7).

10.R4 A die is rolled. Find P(prime or even).

10.R5 Design a simulation for a 20% chance using digits.

10.R6 A game costs 1 dollar and pays 5 dollars with probability 1/4. Find the expected value.

CHAPTER 10 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> The relationship between an experiment and the outcomes it can produce is described by probability, and the logic of counting those outcomes — with representation as a sample space or a simulation — allows randomness to be quantified and risk to be assessed rather than guessed.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of logic and representation appear in your work?
- How is the global context of scientific and technical innovation, and its exploration of randomness, chance, risk and uncertainty, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a model or a simulation of a real situation, and what would you still like to ask?

CHAPTER 10 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Probability supports decisions under uncertainty. Use it to advise a real choice.

Service as Action

COMMUNITY NEED	A community group runs a raffle and needs to know whether it is fair to participants.
MATHEMATICAL CONNECTION	Probability and expected value to assess the raffle.
PROPOSED ACTION	Compute the expected value and advise the group on ticket pricing.
EXPECTED IMPACT	Participants understand the odds and the group raises funds fairly.
STUDENT REFLECTION	Explain how expected value informed the advice.

Community Project

PROJECT IDEA	A school canteen queue simulation.
NEED	Queues at lunchtime are long and unpredictable.
RESEARCH QUESTION	How many service points are needed to keep the average wait under three minutes?
MATHEMATICS	Probability, simulation and averages.
ACTION OR OUTCOME	A simulation report with a recommendation to the school.
BENEFICIARIES	Students and canteen staff.
EVIDENCE	The simulation, the results and the recommendation.

Personal Project connection · MYP 5

PERSONAL INTEREST	Games, sport or gaming design.
LEARNING GOAL	To design and test a fair game.
PRODUCT OR OUTCOME	A game with a computed expected value and a fairness statement.
MATHEMATICS	Probability, expected value and simulation.
PROCESS	List outcomes, compute probabilities, then simulate to check.
EVIDENCE	The rules, the calculations and the simulation results.

CHAPTER 11 · PART IV · PROBABILITY AND DATA

11

Univariate data

A single data set can be displayed, summarised and compared. The choice of display and of average changes what the data appear to say.

ENGAGE

A class records the number of text messages sent by each student in a day. How would you describe the class's habits in one sentence?

Framing the inquiry

Key concept: Logic
Related concepts: Representation · Relationships
Global context: Globalisation and sustainability
Global context exploration: Population and demography

Statement of inquiry: The logic of organising a data set, and the relationships between its values, are represented by displays, measures of centre and measures of spread, and that representation allows a sample to be reported honestly and used to describe a larger population.

Questions to guide us

- **Factual:** What do the mean, median and mode measure, and what does each of the range and the IQR measure?
- **Conceptual:** Why can two data sets with the same mean look very different, and how does the choice of display change what the data appear to say?
- **Debatable:** Can an average ever be misleading when it is used to describe a whole population, and how would you argue for a fairer report?

ATL · INFORMATION LITERACY. In order for a student to access information to be informed, students must collect, record and verify data, and make connections between various sources of information.

CHAPTER 11 · INVESTIGATION 11.1

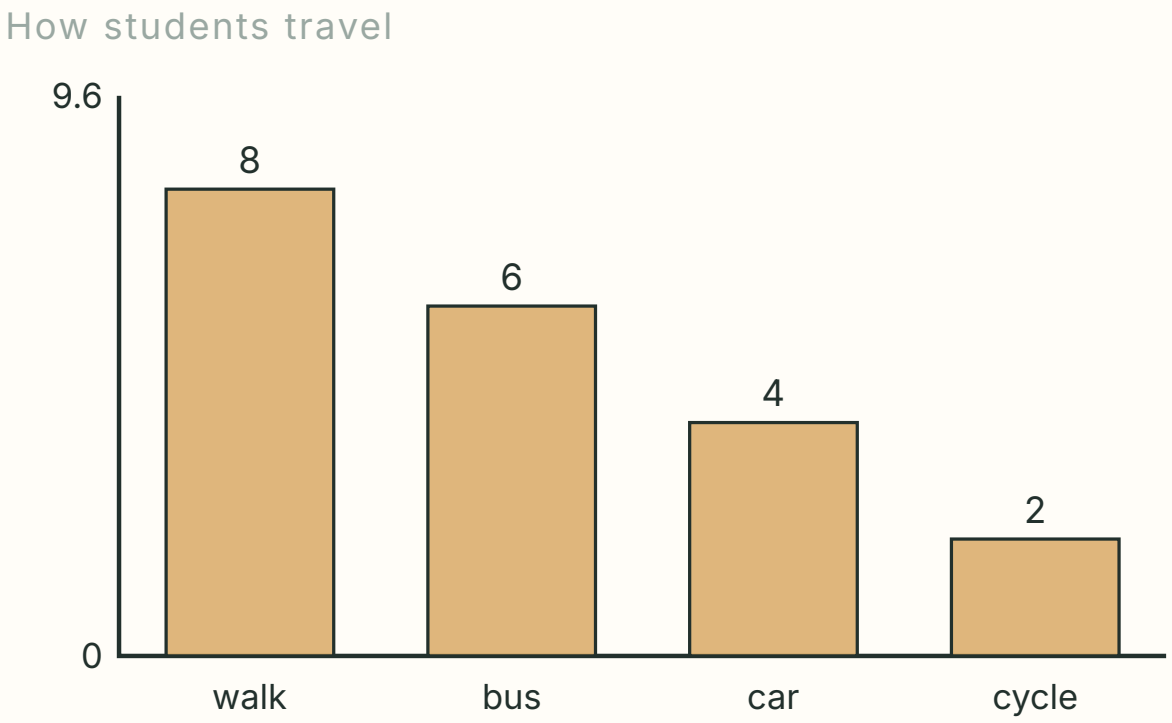
Collecting and displaying data

INQUIRY QUESTION

How can a collection of data be organised and displayed so that its story is visible?

YOUR TASK

Survey 20 students on how they travel to school. Build a frequency table and a bar chart, then comment on the most and least common answers.



travel	tally	frequency
walk		
bus		
car		
cycle		

CHAPTER 11 · INVESTIGATION 11.1

Collecting and displaying data

What do you notice?

- The frequencies must add to 20.
- The tallest bar is the mode of the data.
- The display makes the comparison immediate.

Observation. A frequency table turns raw data into a countable summary.

MATHEMATICAL CONCLUSION

Data are organised in a table and displayed in a chart, and the display must match the data type and the question being asked.

Your own words

CHAPTER 11 · 11.1

Collecting and displaying data

Data are collected for a purpose. A survey question must be clear, unbiased and answerable, and the sample must be described.

A frequency table organises raw data; a bar chart compares categories; a histogram shows the distribution of grouped numerical data; a pie chart shows parts of a whole.

The display must match the data type: categorical data suit bar and pie charts, continuous numerical data suit histograms and line graphs.

Every display needs labelled axes, a scale and units. A display without units cannot be interpreted.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

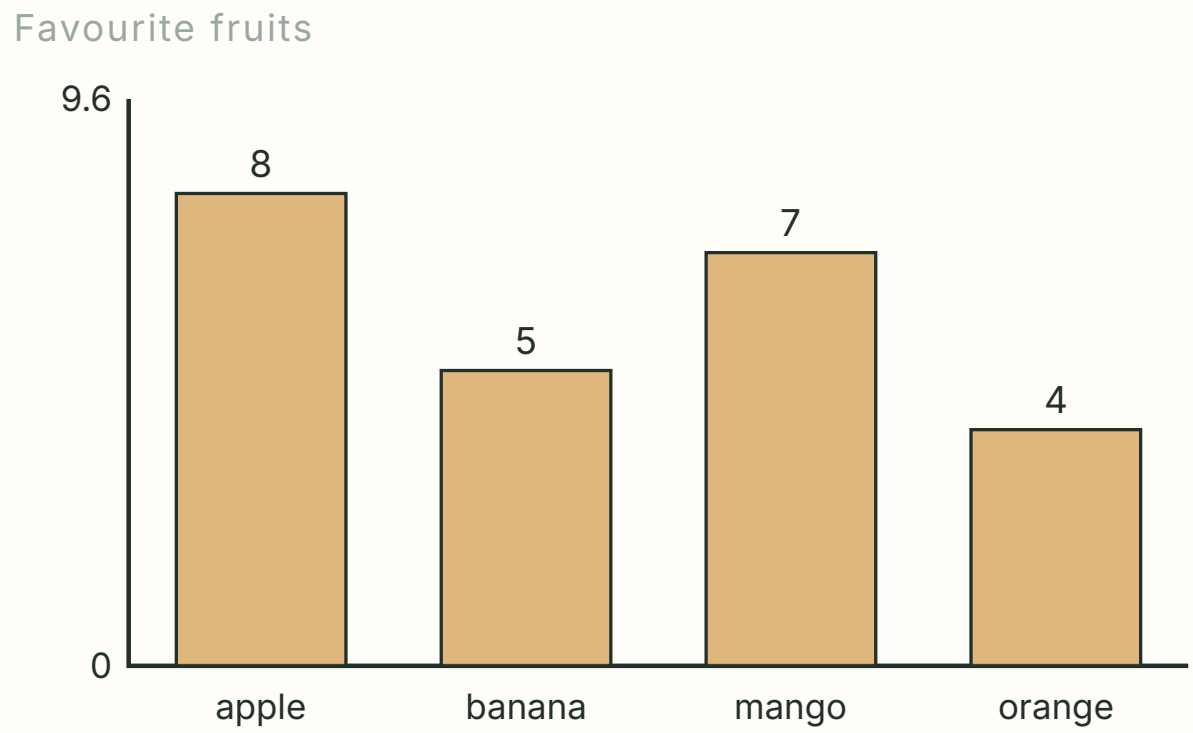
In your own words

CHAPTER 11 · 11.1 · WORKED EXAMPLE 1

Collecting and displaying data

WORKED EXAMPLE 11.1.1

Problem. A class records favourite fruits: 8 apple, 5 banana, 7 mango, 4 orange. Draw a suitable display.



Mathematical reasoning. The data are categorical, so a bar chart shows the comparison directly.

1. Label the categories on the horizontal axis.
2. Use a consistent vertical scale in number of students.
3. Draw one bar per category with height equal to the frequency.

Final answer. A bar chart with four bars of heights 8, 5, 7 and 4

Interpretation. A pie chart would also work but makes exact comparison harder.

CHAPTER 11 · 11.1 · WORKED EXAMPLE 2

Collecting and displaying data

WORKED EXAMPLE 11.1.2

Problem. Twelve students record the number of books read: 2, 3, 3, 4, 5, 5, 5, 6, 7, 8, 9, 12. Build a frequency table.

books	tally	frequency
2	|	1
3	||	2
4	|	1
5	|||	3
6	|	1
7	|	1
8	|	1
9	|	1
12	|	1

Books read by twelve students

Mathematical reasoning. Count how often each value occurs and record it.

1. List each distinct value.
2. Tally the occurrences.
3. Check that the frequencies add to 12.

Final answer. A frequency table with total frequency 12

Interpretation. The check on the total catches any missed tally.

CHAPTER 11 · INVESTIGATION 11.2

Measures of centre

INQUIRY QUESTION

How can the middle of a data set be summarised by a single number, and which single number is the fairest?

YOUR TASK

Roll a die 20 times, record the values, then compute the mean, median and mode. Compare with the theoretical mean of 3.5.

statistic	value
mean	
median	
mode	
theoretical mean	3.5

CHAPTER 11 · INVESTIGATION 11.2

Measures of centre

What do you notice?

- The experimental mean is close to 3.5 but rarely exactly 3.5.
- The mode depends on which value happened most often.
- The median is the average of the 10th and 11th ordered values.

Observation. Experimental statistics vary around the theoretical values.

MATHEMATICAL CONCLUSION

Each average summarises a different aspect of the data, so the mean uses every value, the median resists extremes and the mode suits categories.

Your own words

CHAPTER 11 · 11.2

Measures of centre

The mean is the sum of the values divided by how many there are. It uses every value, so it is affected by extreme values.

The median is the middle value when the data are ordered. With an even number of values it is the mean of the two middle values, and it is unaffected by extremes.

The mode is the most frequent value. It is the only average that works for categorical data.

Choosing an average is a decision: the mean for a balanced summary, the median when there are outliers, the mode for categories.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 11 · 11.2 · WORKED EXAMPLE 1

Measures of centre

WORKED EXAMPLE 11.2.1

Problem. Find the mean, median and mode of 4, 7, 7, 9, 13.



Mathematical reasoning. Order the data, then apply each definition.

1. Mean: $(4 + 7 + 7 + 9 + 13) \div 5 = 40 \div 5 = 8$.
2. Median: the third of five ordered values is 7.
3. Mode: 7 occurs twice.

Final answer. mean 8, median 7, mode 7

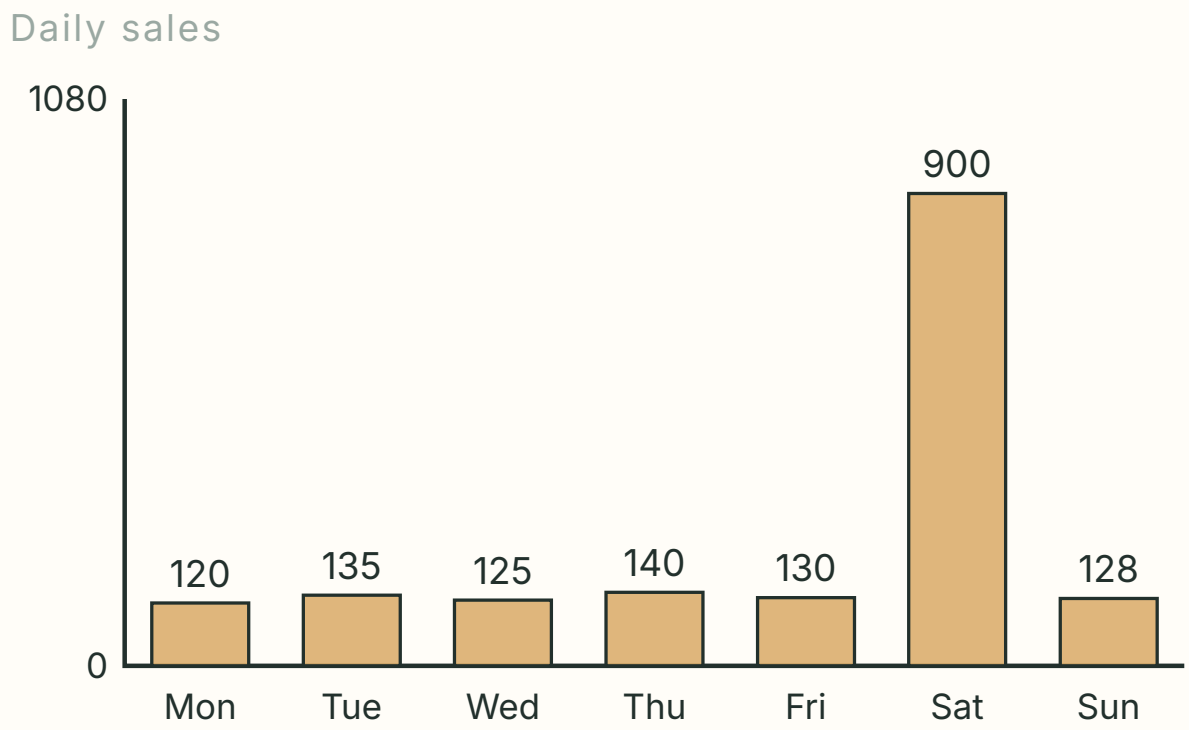
Interpretation. The mean is pulled up by the 13, which is why it exceeds the median.

CHAPTER 11 · 11.2 · WORKED EXAMPLE 2

Measures of centre

WORKED EXAMPLE 11.2.2

Problem. A shop's daily sales for a week are 120, 135, 125, 140, 130, 900 and 128 dollars. Which average best describes a typical day?



Mathematical reasoning. The value 900 is an outlier, so the median describes a typical day better.

1. Mean = 1 678 ÷ 7 ≈ 240 dollars, which is above six of the seven days.
2. Ordered data: 120, 125, 128, 130, 135, 140, 900. Median = 130 dollars.
3. The median of 130 dollars represents a typical day.

Final answer. The median, 130 dollars

Interpretation. One unusual day distorts the mean but not the median.

CHAPTER 11 · INVESTIGATION 11.3

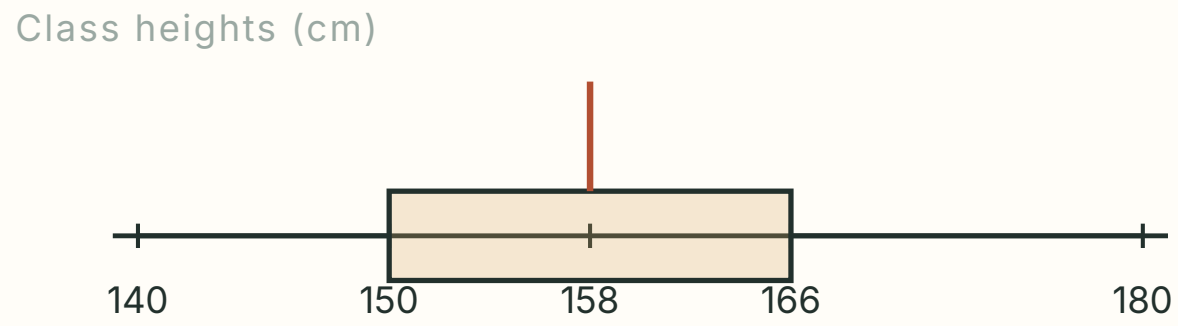
Measures of spread

INQUIRY QUESTION

How can the spread of a data set be summarised, and why does spread matter as much as centre?

YOUR TASK

Measure the height of every student in the class to the nearest centimetre. Compute the range, the quartiles and the IQR, and draw a box plot.



statistic	value
minimum	
Q1	
median	
Q3	
maximum	
range	
IQR	

CHAPTER 11 · INVESTIGATION 11.3

Measures of spread

What do you notice?

- The quartiles divide the ordered data into four equal groups.
- The box covers the middle half of the data.
- The whiskers reach the extremes.

Observation. The box plot displays five summary values in a single picture.

MATHEMATICAL CONCLUSION

Spread is measured by the range and the IQR, and a box plot shows centre, spread and extremes together.

Your own words

CHAPTER 11 · 11.3

Measures of spread

The range is the largest value minus the smallest. It is simple but uses only two values.

The interquartile range is the range of the middle half: Q3 − Q1. Quartiles split the ordered data into four equal parts.

A box plot shows the minimum, Q1, median, Q3 and maximum in one display, which makes comparing two data sets straightforward.

Comparing data sets needs both centre and spread. Two classes can have the same mean but very different spreads.

> FORMAL DEVELOPMENT
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

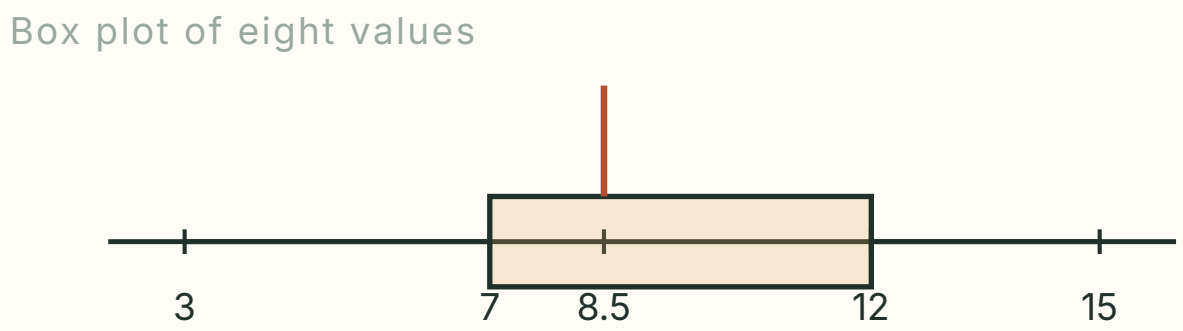
In your own words

CHAPTER 11 · 11.3 · WORKED EXAMPLE 1

Measures of spread

WORKED EXAMPLE 11.3.1

Problem. Find the range and interquartile range of 3, 5, 7, 8, 9, 11, 13, 15.



Mathematical reasoning. Split the ordered data into quarters and take the middle half.

1. Range = 15 − 3 = 12.
2. Lower half: 3, 5, 7, 8, so Q1 = 6.
3. Upper half: 9, 11, 13, 15, so Q3 = 12.
4. IQR = 12 − 6 = 6.

Final answer. range 12 and interquartile range 6

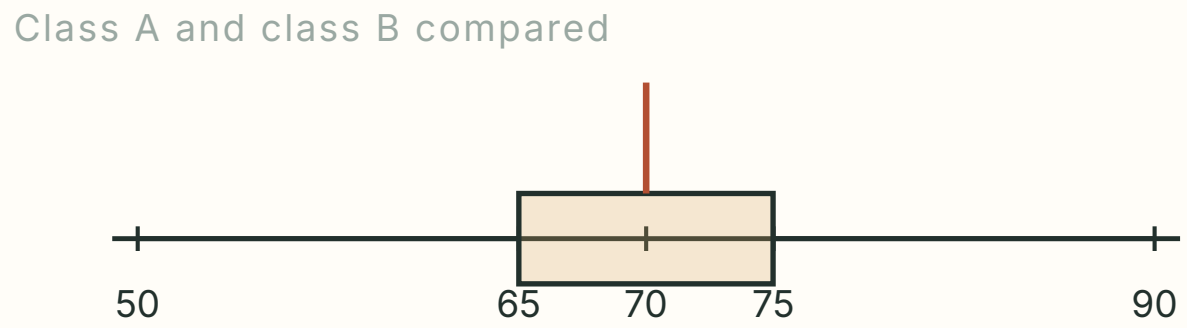
Interpretation. The IQR ignores the extremes, so it describes the bulk of the data.

CHAPTER 11 · 11.3 · WORKED EXAMPLE 2

Measures of spread

WORKED EXAMPLE 11.3.2

Problem. Two classes have mean test score 70. Class A has range 40 and class B has range 10. Compare the classes.



Mathematical reasoning. Same centre but different spread means the classes differ in consistency.

1. Both means are 70, so the centres match.
2. Class A's range of 40 shows a wide spread of scores.
3. Class B's range of 10 shows the scores cluster tightly.

Final answer. Both classes average 70, but class B is far more consistent

Interpretation. A mean alone hides the spread, so both measures are reported together.

CHAPTER 11 · INVESTIGATION 11.4

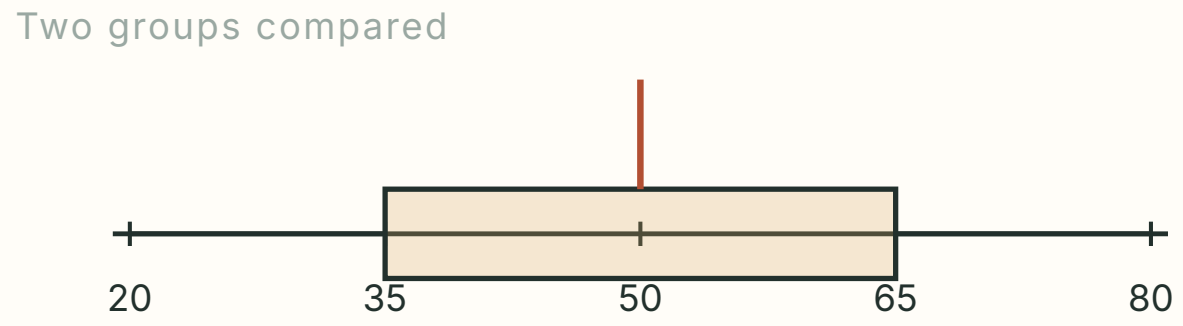
Interpreting and reporting data

INQUIRY QUESTION

How can a report about data be made honest, and what must it state before it draws a conclusion?

YOUR TASK

Collect two data sets on the same question from different groups, display them on box plots and write a comparison.



comparison	group 1	group 2
median		
IQR		
range		

CHAPTER 11 · INVESTIGATION 11.4

Interpreting and reporting data

What do you notice?

- The medians compare the centres.
- The IQRs compare the spread of the middle half.
- A conclusion needs both pieces of information.

Observation. A comparison is complete only when centre and spread are both reported.

MATHEMATICAL CONCLUSION

Report the centre, the spread and the sample before drawing a conclusion, because a conclusion about a population rests on the logic of how the sample was chosen.

Your own words

CHAPTER 11 · 11.4

Interpreting and reporting data

A report on data states the question, the sample, the display, the statistics and the conclusion, in that order.

Averages can mislead when the sample is biased, when outliers are hidden, or when the wrong average is chosen. A critical reader asks which average and which sample.

Predictions from data are statements about a group, not about an individual. A 70% chance applies to the group, not to one person.

Comparing two data sets uses a common scale, and states both the difference in centre and the difference in spread.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

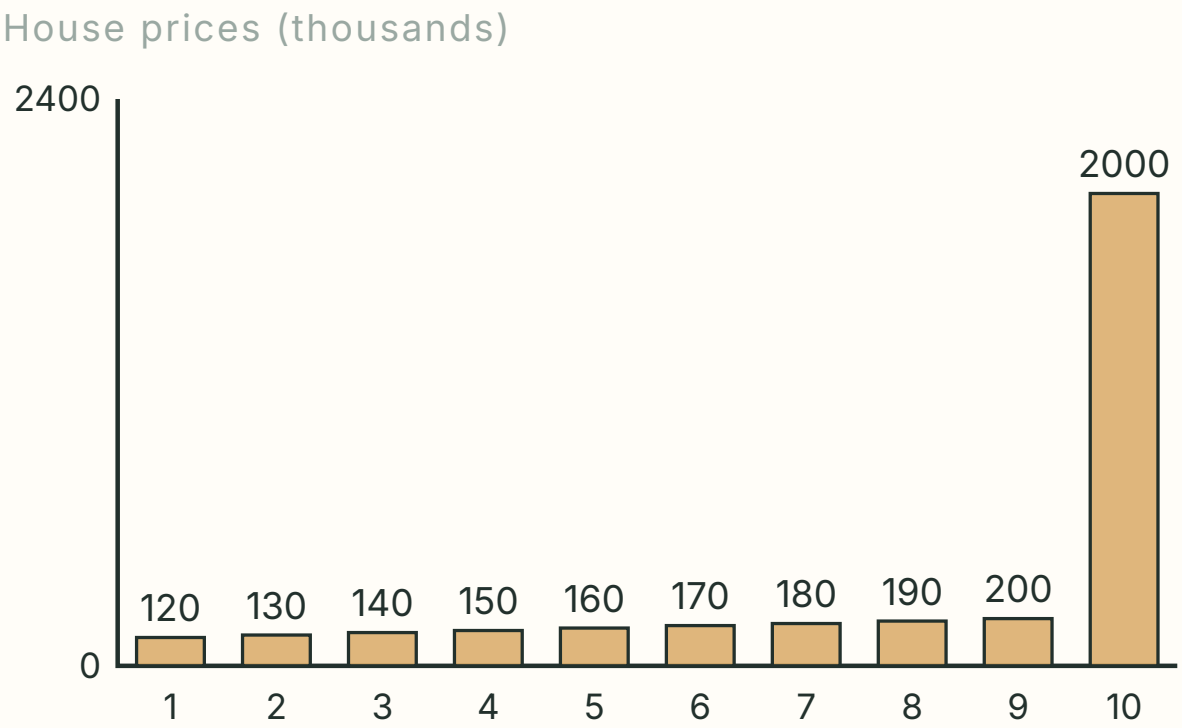
In your own words

CHAPTER 11 · 11.4 · WORKED EXAMPLE 1

Interpreting and reporting data

WORKED EXAMPLE 11.4.1

Problem. A newspaper reports 'average house price 250 000 dollars' from a sample of 10 houses, one of which sold for 2 000 000 dollars. Explain the problem.



Mathematical reasoning. One extreme value in a small sample distorts the mean.

1. The mean is pulled far above the other nine values.
2. The median would be near 165 000 dollars.
3. The sample of 10 is also too small to represent the market.

Final answer. The mean is distorted by one extreme value, and the sample is too small

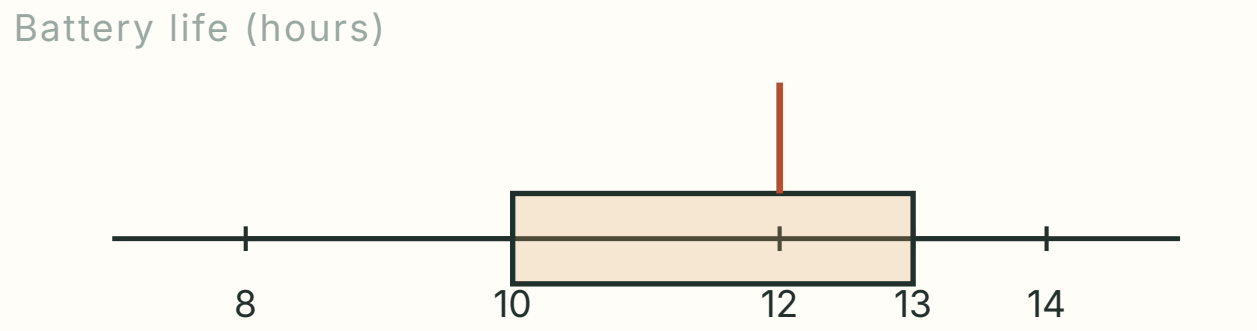
Interpretation. A critical reader asks which average was used and how large the sample was.

CHAPTER 11 · 11.4 · WORKED EXAMPLE 2

Interpreting and reporting data

WORKED EXAMPLE 11.4.2

Problem. Two phone batteries last 8–14 hours and 10–12 hours. Compare them using centre and spread.



Mathematical reasoning. Report both the centre and the spread for each battery.

1. Battery A: median about 12 hours, range 6 hours.
2. Battery B: median about 11 hours, range 2 hours.
3. A lasts longer on average but is less consistent.

Final answer. Battery A lasts longer; battery B is more consistent

Interpretation. The better choice depends on which quality the user values more.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

11.A1 FLUENCY
Name a display suitable for categorical data.

11.A2 FLUENCY
A frequency table has frequencies 3, 5, 7 and 5. How many values in total?

11.A3 REASONING
What must every display include?

11.A4 FLUENCY
Name a display for continuous numerical data.

11.A5 REASONING
Explain the difference between a bar chart and a histogram.

11.A6 REASONING
A survey asks 'Do you agree that homework is useless?' Explain the problem.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

11.A7 REASONING
What does the mode of a categorical data set mean?

11.A8 REASONING
A pie chart shows 4 sectors. What do the sectors represent?

11.A9 REASONING
A frequency table's total is 25 but the survey had 30 responses. What went wrong?

11.A10 REASONING
Explain why units must be stated on a data display.

11.A11 FLUENCY
Name the display that shows parts of a whole.

11.A12 REASONING
Explain what a sample is.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

11.B1 FLUENCY

Find the mean of 6, 8, 10, 12.

11.B2 FLUENCY

Find the median of 3, 7, 9, 12, 15.

11.B3 FLUENCY

Find the mode of 4, 4, 5, 6, 6, 6, 7.

11.B4 REASONING

Investigate: find the mean and median of 2, 4, 6, 8, 100.

11.B5 FLUENCY

Find the median of 5, 8, 11, 14.

11.B6 REASONING

Explain when the median is a better average than the mean.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

11.B7 FLUENCY

Find the mean of 10, 12, 14, 16, 18.

11.B8 REASONING

Investigate the pattern: means of 1–5, 2–6, 3–7.

11.B9 REASONING

Which average can be used for categorical data?

11.B10 FLUENCY

Find the mean of 2.5, 3.5 and 4.

11.B11 REASONING

Explain why the mean uses every value but the median does not.

CRITERION C Communicating

Criterion C practice · Communicating

11.C1 FLUENCY
Find the range of 12, 15, 18, 22, 30.

11.C2 REASONING
Explain what the interquartile range measures.

11.C3 FLUENCY
Find the quartiles of 4, 6, 8, 10, 12, 14.

11.C4 REASONING
Explain the advantage of a box plot over a list of values.

11.C5 REASONING
Two data sets have the same mean but different ranges. What does this mean?

11.C6 REASONING
Investigate: a survey of 7 households records 1, 3, 5, 7, 9, 11, 13 people. Find the IQR and explain what it says about the middle of the sample.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

11.C7 REASONING
Explain why the range can be misleading.

11.C8 REASONING
What does a box plot's box represent?

11.C9 APPLICATION
Find the range and IQR of 5, 5, 6, 7, 8, 9, 20.

11.C10 REASONING
Explain why comparing data sets needs both centre and spread.

11.C11 FLUENCY
A box plot has whiskers from 10 to 50 and a box from 20 to 40. Find the IQR.

11.C12 REASONING
Explain what a long whisker on a box plot suggests.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

11.D1 REASONING

A shop's sales are 40, 45, 50, 55, 500 dollars. Which average best describes a typical day?

11.D2 REASONING

Two classes have means 65 and 72 with IQRs 4 and 20. Compare them.

11.D3 REASONING

A survey of 10 students is used to describe a school of 1 000. Explain the risk.

11.D4 REASONING

A report says 'average wage 30 000 dollars' but most workers earn 22 000. Explain how this is possible.

11.D5 REASONING

A health survey finds that a treatment works for 70% of patients. Explain what this says about one patient chosen at random, and what it does not say.

11.D6 REASONING

Compare two box plots: A has median 60 and IQR 10; B has median 60 and IQR 25.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

11.D7 REASONING

A school report gives the mean only. What extra information would you ask for?

11.D8 REASONING

A class mean rises from 60 to 68. The IQR rises from 5 to 18. Interpret both changes.

11.D9 APPLICATION

A pie chart shows 40% walking, 35% bus, 25% car for 200 students. Find each number.

11.D10 REASONING

Explain how a biased sample can make a conclusion false.

11.D11 REASONING

A data set has mean 50 and median 50, but range 90. Describe the shape.

11.D12 REASONING

A teacher says 'the average improved by 5 marks'. Give two questions a critical reader should ask.

CHAPTER 11 · APPLICATION AND ASSESSMENT

Applying chapter 11

Choose a display that answers the question, then choose the average that fits the purpose. Justify both choices.

K1 · The school survey

A class surveys 30 students on hours of sleep. The results are summarised by a box plot. Write a report stating the centre, the spread and one conclusion, and justify your choice of display.

K2 · The two classes

Two classes have mean scores of 68 and 71 with IQRs of 6 and 15. Advise the school on which class needs more support and justify your answer.

Criteria B and D · report the mean and spread before advising on your sample.

CHAPTER 11 · REVIEW

What you should now be able to do

Key ideas

- A display must match the data type and the question.
- The mean uses every value; the median is unaffected by extremes; the mode suits categories.
- The range and the IQR measure spread.
- A box plot shows minimum, Q1, median, Q3 and maximum together.
- Comparing data sets needs both centre and spread.
- A critical reader asks which average, which sample and what spread.

Vocabulary

data · frequency · categorical · continuous · bar chart · histogram · pie chart · box plot · mean · median · mode · range · quartile · interquartile range · outlier · sample

CHAPTER 11 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

11.R1 Find the mean of 12, 15, 18, 21.

11.R2 Find the median and mode of 3, 5, 5, 7, 9.

11.R3 Find the range and IQR of 4, 6, 8, 10, 12, 14, 40.

11.R4 Name a display for continuous data.

11.R5 Explain why the mean can be misleading.

11.R6 Two data sets share a mean but differ in IQR. What does this mean?

CHAPTER 11 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> The logic of organising a data set, and the relationships between its values, are represented by displays, measures of centre and measures of spread, and that representation allows a sample to be reported honestly and used to describe a larger population.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of logic and the related concepts of representation and relationships appear in your work?
- How is the global context of globalisation and sustainability, and its exploration of population and demography, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a sample when it is used to describe a population, and what would you still like to ask?

CHAPTER 11 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Data collection serves decisions. Collect data on a question that matters in your community.

Service as Action

COMMUNITY NEED	A community health post needs to know how far residents travel for care.
MATHEMATICAL CONNECTION	Data collection, averages and spread to summarise travel distances.
PROPOSED ACTION	Survey residents and report the median distance and spread to the health post.
EXPECTED IMPACT	The health post can plan outreach based on real travel data.
STUDENT REFLECTION	Explain how the choice of average affected the recommendation.

Community Project

PROJECT IDEA	A school travel survey.
NEED	The school wants to reduce traffic congestion at the gate.
RESEARCH QUESTION	How do students travel to school, and what is the distribution of journey times?
MATHEMATICS	Data collection, displays, averages and spread.
ACTION OR OUTCOME	A survey report with a recommendation to the school.
BENEFICIARIES	Students, families and school staff.
EVIDENCE	The questionnaire, the data table and the report.

Personal Project connection · MYP 5

PERSONAL INTEREST	Sport, gaming or personal habits.
LEARNING GOAL	To collect and summarise data on a personal interest.
PRODUCT OR OUTCOME	A data report with a display and at least two averages.
MATHEMATICS	Data collection, averages, spread and displays.
PROCESS	Define the question, collect the data, display it, then summarise.
EVIDENCE	The data, the display and the written summary.

CHAPTER 1 · CRITERION A · KNOWING AND UNDERSTANDING

1.A1 **2:3**
Divide both parts by the HCF, 4.

1.A2 **6:9:15 = 2:3:5**
Add the parts for the total, then simplify all three so the sharing rule is clear.

1.A3 **3:4**
HCF(45, 60) = 15, so the rule is 3 parts books to 4 parts pencils.

1.A4 **Yes, both are 2:3**
6:9 = 2:3 and 10:15 = 2:3, so each class receives the same share of each fruit.

1.A5 **3:4**
Total = 50; 30:50 = 3:4.

1.A6 **Part-part compares 5 with 7; part-whole compares 5 with 12, which is the council example.**
The whole is 5 + 7 = 12.

1.A7 **2:3**
Multiply both parts by 10, then divide by 2.

1.A8 **12:30 = 2:5**
Everyone = 30.

1.A9 **27**
The multiplier is 3, so 9 × 3 = 27.

1.A10 **40**
32 ÷ 4 = 8, so 5 × 8 = 40.

1.A11 **1000 g**
400 ÷ 2 = 200, so 5 × 200 = 1000 g.

1.A12 **No — the group with 4 parts receives more.**
Swapping the parts swaps which group receives the larger share.

CHAPTER 1 · CRITERION B · INVESTIGATING PATTERNS

1.B1	**10:16 (k=2), 15:24 (k=3), 20:32 (k=4)** Multiply both parts by the same non-zero k, so both groups gain equally.
1.B2	**Yes, 3:5, and it is still fair** k may be any non-zero number, including a fraction, and the comparison is unchanged.
1.B3	**14:18, because it is 7:9** The other three all simplify to 3:4.
1.B4	**10:15 and 12:18; both parts increase by 2 and 3** The difference pattern keeps the ratio fixed.
1.B5	**5:15 (5 shaded and 15 unshaded)** Each figure adds the same unit to both parts.
1.B6	**The comparison is the same but the amounts are 100 times larger.** Equivalent ratios differ by a multiplier, so the rule is the same and the size is not.
1.B7	**False — 2:3 and 4:6 are equivalent but 5 is not 10.** Equivalence is about the comparison, not about the totals.
1.B8	**No — each situation has exactly one simplest form.** Simplifying by the HCF is unique.
1.B9	**k = 5** The second part is multiplied by 5, so the first part must be too.
1.B10	**Every value in one row is the matching value in the other multiplied by a fixed number.** The fixed multiplier is the unit rate.
1.B11	**12:20** Both parts grow by the same multiplier, here 3 each row.

CHAPTER 1 · CRITERION C · COMMUNICATING

1.C1 **25**
40 ÷ 8 = 5, so the group with 5 parts receives 25 of the 40 items.

1.C2 **Find how many times 3 fits into 18 (six), then multiply 7 by six.**
The multiplier is found in one row and applied to the other, so the comparison is preserved.

1.C3 **15 pages**
20 ÷ 4 = 5 steps; 5 × 3 = 15.

1.C4 **7:9, 14:18, 21:27, 28:36**
Each row multiplies both parts by the same number, so the sharing rule never changes.

1.C5 **Any justified choice; a table keeps all pairs visible at once.**
Justification must name the reader of the representation.

1.C6 **3 cm : 15 km**
15 ÷ 5 = 3, so every 1 cm on the map stands for 5 km on the ground.

1.C7 **0.6, or 3/5 of a part per part**
Dividing the second part by the first gives the amount of the second quantity per one of the first.

1.C8 **6 and 4**
The multiplier is 2 throughout, so one share costs 2 dollars.

1.C9 **Each table row is a point (x, y) on the graph.**
Tables and graphs are two notations for one relationship.

1.C10 **22.5 litres**
9 ÷ 2 = 4.5; 4.5 × 5 = 22.5.

1.C11 **one part = 12**
84 ÷ 7 = 12.

1.C12 **2.5**
10 ÷ 4 = 2.5.

CHAPTER 1 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

1.D1 **75 and 45**
120 ÷ 8 = 15; 5 × 15 = 75 and 3 × 15 = 45.

1.D2 **1500 dollars**
2400 ÷ 8 = 300; 5 × 300 = 1500.

1.D3 **288**
120 ÷ 5 = 24; 7 × 24 = 168; 168 + 120 = 288.

1.D4 **3:2; 54 and 36 dollars**
60:40 = 3:2; 90 ÷ 5 = 18.

1.D5 **The ratio gives parts of 51.43 minutes, so it cannot be applied to whole minutes without changing the rule.**
360 ÷ 7 is not a whole number, so the stated ratio needs adjusting.

1.D6 **Neither — both are 3/5, so the representation is equal.**
15/25 = 0.6 and 18/30 = 0.6, so the two classes are represented in the same proportion.

1.D7 **150 ml**
500 ÷ 10 = 50; 3 × 50 = 150.

1.D8 **40, 60 and 100 dollars**
200 ÷ 20 = 10; fair only if the agreed rule is hours contributed — equal shares would give 66.67 each.

1.D9 **400 ml**
2 000 ÷ 5 = 400 ml.

1.D10 **The 3-litre offer (1.60 per litre against 1.70)**
3.40 ÷ 2 = 1.70 and 4.80 ÷ 3 = 1.60.

1.D11 **12 voted originally (with 5 yes and 3 no scaling to 15 and 9 before the change)**
Let the original counts be 5x and 3x; then (5x + 12) = 3(3x) gives x = 3.

1.D12 **They agree only when every part of the ratio is 1.**
5:3 gives shares of 5/8 and 3/8, not 1/2 and 1/2.

CHAPTER 2 · CRITERION A · KNOWING AND UNDERSTANDING

2.A1 **Yes, k = 3.5**
Every quotient is 3.5.

2.A2 **c = 4x, so c = 36 dollars**
The cost is the mass multiplied by the unit price.

2.A3 **No, because $y \neq 0$ when x = 0**
A proportional relationship passes through the origin.

2.A4 **c = 6l**
Cost is 6 times the length.

2.A5 **96 litres**
8 × 12 = 96.

2.A6 **It is the price of one kilogram, so every price is that number times the mass.**
The constant is the unit rate, and it fixes every other pair of values.

2.A7 **24.5 litres**
0.07 × 350 = 24.5.

2.A8 **Not proportional — 25 is not 28, so the price per kg is not constant.**
A proportional price list has the same cost per kilogram in every row.

2.A9 **67.5, so 67 whole flyers**
15 × 4.5 = 67.5, and a whole flyer cannot be half printed.

2.A10 **No**
It does not pass through the origin.

2.A11 **h = 10 + 2w; 26 cm**
The relationship is additive, not proportional.

2.A12 **k = 2.5 dollars per kg**
The constant of proportionality is the unit price.

CHAPTER 2 · CRITERION B · INVESTIGATING PATTERNS

2.B1 **60 km/h**
240 ÷ 4 = 60, so the cost per kilometre can be found from the speed and the hourly cost.

2.B2 **5 kg for 12 dollars (2.40 per kg against 2.50)**
Convert both to price per kg.

2.B3 **1.5 km per minute**
90 ÷ 60 = 1.5; a short trip is better measured in minutes.

2.B4 **15 dollars per hour; 225 dollars**
15 × 15 = 225.

2.B5 **It puts both offers on the same basis, one unit of the product.**
Comparison requires a common unit.

2.B6 **0.75 dollars per apple**
6 ÷ 8 = 0.75.

2.B7 **12.5 km per litre**
150 ÷ 12 = 12.5.

2.B8 **The distance, 120 km**
Speed × time is constant, so the relationship is not proportional.

2.B9 **875 g; 125 g per loaf**
500 ÷ 4 = 125 g per loaf, then 125 × 7 = 875 g.

2.B10 **It is the value of y when x = 1, which is exactly k in y = kx.**
The unit rate fixes the whole relationship.

2.B11 **5 dollars per litre**
1 litre = 1 000 ml, so multiply by 10.

CHAPTER 2 · CRITERION C · COMMUNICATING

2.C1 **A line through the origin with slope 3**
The slope is the constant of proportionality, here the unit price.

2.C2 **Zero quantity costs zero, so (0, 0) must lie on the line.**
A proportional relationship has no fixed charge, so the line starts at the origin.

2.C3 **The line of slope 5**
The steeper line rises further for the same mass, so its unit price is higher.

2.C4 **24**
$4 \times 6 = 24$.

2.C5 **Interpolation reads between known points; extrapolation reads beyond them.**
Both assume the relationship continues unchanged.

2.C6 **45 dollars per km**
$90 \div 2 = 45$.

2.C7 **y = 0, 2.5, 5, 7.5, 10**
Multiply each x by 2.5.

2.C8 **It would charge something for nothing, so there is a fixed fee as well as a unit price.**
A non-zero intercept means a cost at zero quantity, which is not proportional.

2.C9 **4.8 litres**
$0.4 \times 12 = 4.8$.

2.C10 **The slope is the unit rate that every row of the table shares.**
Each table row divides to the same value, which is the slope of the line.

2.C11 **The 2 dollars per item line**
A larger unit price gives a steeper line through the origin.

2.C12 **y = 0.6x, for example a price of 0.60 dollars per item**
$3 \div 5 = 0.6$, and the constant is the unit price.

CHAPTER 2 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

2.D1 **9 m**
$4.5 \times 200 = 900$ cm $= 9$ m.

2.D2 **1:40**
$480 \div 12 = 40$.

2.D3 **3456 cm^3; 64:1**
$6 \times 4 = 24$ cm, so $24^3 = 13\,824$ cm^3 and $13\,824 \div 216 = 64$.

2.D4 **2 592 dollars**
$2\,400 \times 1.08 = 2\,592$.

2.D5 **60 dollars; multiplier 0.80**
$75 \times 0.80 = 60$.

2.D6 **Perimeter ×2, area ×4**
Perimeter is proportional to the side, area to the square of the side.

2.D7 **15 cm**
The scale factor is $20 \div 8 = 2.5$, so $6 \times 2.5 = 15$ cm.

2.D8 **40.25 dollars**
$35 \times 1.15 = 40.25$.

2.D9 **No — it is linear with a starting value of 500**
$V = 500 - 8t$ does not pass through the origin.

2.D10 **2**
Areas scale by k^2, so $k = \sqrt{(80 \div 20)} = 2$.

2.D11 **20%**
$3 \div 15 = 0.2$, so the increase is 20%.

2.D12 **The area grows by 4, not 2, because area is proportional to the square of the side.**
A side of 2 m gives 4 m^2 and a side of 4 m gives 16 m^2, which is four times as much.

CHAPTER 3 · CRITERION A · KNOWING AND UNDERSTANDING

3.A1 **75 km/h**
300 ÷ 4 = 75.

3.A2 **A comparison of two different kinds of quantity, e.g. 60 km/h**
The two units must be different for a rate.

3.A3 **30 bottles per minute**
240 ÷ 8 = 30.

3.A4 **5 litres per minute**
45 ÷ 9 = 5.

3.A5 **Each speed applies for a different time, so the totals decide the average.**
Total distance ÷ total time is the definition.

3.A6 **18 km/h**
40 min = 2/3 h; 12 ÷ (2/3) = 18.

3.A7 **16 dollars per hour**
96 ÷ 6 = 16.

3.A8 **28 litres**
350 ÷ 100 = 3.5; 3.5 × 8 = 28.

3.A9 **270 labels**
18 × 15 = 270.

3.A10 **A ratio compares like quantities; a rate compares different kinds.**
km/h compares distance with time, which are different.

3.A11 **4 minutes**
1/6 + 1/12 = 1/4 of the tank per minute.

3.A12 **60 km/h**
150 ÷ 2.5 = 60.

CHAPTER 3 · CRITERION B · INVESTIGATING PATTERNS

3.B1 **5 400 m**
× 1 000.

3.B2 **0.025 m^2**
÷ 10 000.

3.B3 **210 minutes**
× 60.

3.B4 **25 m/s**
90 ÷ 3.6 = 25.

3.B5 **2 500 litres**
1 m^3 = 1 000 litres.

3.B6 **Lengths scale by 10, areas by 10^2 and volumes by 10^3.**
The unit is squared or cubed.

3.B7 **72 km/h**
45 × 1.6 = 72.

3.B8 **77 coins**
15.40 ÷ 0.20 = 77 exactly, so the count is exact.

3.B9 **0.75 g/cm^3**
1 kg/L = 1 g/cm^3.

3.B10 **It keeps the arithmetic exact and avoids repeating decimals.**
Converting afterwards can introduce rounding error.

3.B11 **1 200 kg**
× 1 000.

CHAPTER 3 · CRITERION C · COMMUNICATING

3.C1 **3 g/cm³**
900 ÷ 300 = 3.

3.C2 **It compares mass per unit volume, so objects of different sizes can be compared.**
Density normalises for size.

3.C3 **50 Pa**
200 ÷ 4 = 50.

3.C4 **The same force acts on a much smaller area, so the pressure is larger.**
Pressure = force ÷ area.

3.C5 **The one with the smaller volume**
Density = mass ÷ volume.

3.C6 **300 g**
1.2 × 250 = 300.

3.C7 **They are inversely proportional.**
Doubling the area halves the pressure.

3.C8 **8 g/cm³**
Volume 64 cm³; 512 ÷ 64 = 8.

3.C9 **1 g/cm³**
1 000 kg = 1 000 000 g and 1 m³ = 1 000 000 cm³.

3.C10 **Each material has its own density, so measuring it identifies the material.**
Compare the measured value with known values.

3.C11 **60 N**
300 × 0.2 = 60.

3.C12 **Wide tyres increase the area, reducing the pressure on the road.**
Pressure = force ÷ area.

CHAPTER 3 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

3.D1 **48 km/h**
Total 240 km; times 2 h and 3 h; 240 ÷ 5 = 48.

3.D2 **800 km**
60 ÷ 7.5 = 8; 8 × 100 = 800.

3.D3 **4 000 Pa**
12 000 ÷ 3 = 4 000.

3.D4 **6 km**
The gap grows at 3 km/h; 3 × 2 = 6.

3.D5 **3 125 g**
750 ÷ 6 = 125 per person; × 25.

3.D6 **4 100 000 kg**
4 000 × 1 025 = 4 100 000.

3.D7 **3.97 m/s**
10 000 m ÷ 2 520 s ≈ 3.97.

3.D8 **6.67 litres per minute**
0.4 m^3 = 400 L per hour; ÷ 60 ≈ 6.67.

3.D9 **The average hides how the speed varied during the race.**
Average speed uses only totals.

3.D10 **25 minutes**
300 ÷ 12 = 25.

3.D11 **22.5 litres**
5 × 4.5 = 22.5.

3.D12 **0.5 g/cm^3**
Volume 200 cm^3; 100 ÷ 200 = 0.5.

CHAPTER 4 · CRITERION A · KNOWING AND UNDERSTANDING

4.A1 **−7, −3, 0, 2, 5**
Read from left to right; the further left, the lower the point.

4.A2 **+12**
Same distance from the reference, on the other side of it.

4.A3 **−5 is greater; −9 is deeper**
−5 lies to the right of −9, so it is greater, but it is less far below the reference.

4.A4 **0**
A number and its opposite sum to zero.

4.A5 **The reference point between positive and negative values**
Zero is the freezing point of water on the Celsius scale.

4.A6 **−8**
Below the reference is negative.

4.A7 **4, 1, −1, −2, −6**
Read from right to left on the number line.

4.A8 **−7**
The opposite of an opposite is the original number.

4.A9 **Each point has one mirror image in zero**
The mirror image in zero is unique, so the opposite is unique.

4.A10 **Neither — it is the reference level itself**
Zero is the reference point, not a direction away from it.

4.A11 **7 floors**
5 − (−2) = 7.

4.A12 **−7 m and +5 m**
−1 − 6 = −7 or −1 + 6 = 5.

CHAPTER 4 · CRITERION B · INVESTIGATING PATTERNS

4.B1 **7 + (−10) = −3**
Move 10 units left from 7 on the number line.

4.B2 **−8 − (−3) = −5**
Subtracting a negative moves right.

4.B3 **−12 + 5 = −7 °C**
Move 5 units right from −12.

4.B4 **No, both give 5**
Addition is commutative.

4.B5 **−5 m**
Move 11 units left from 6.

4.B6 **5 and 7; add 2 each time**
The common difference is +2.

4.B7 **It is the same as adding the opposite, which moves right.**
a − (−b) = a + b, so the value grows.

4.B8 **(−5) + (−5) + (−5) = −15**
Three moves of −5.

4.B9 **−2 °C**
−6 + 4 = −2.

4.B10 **Start at 3 and move 8 units left, landing on −5.**
Left moves subtract, and crossing zero changes the sign.

4.B11 **−9**
Move 7 units left from −2, so the point is 9 m below the reference.

CHAPTER 4 · CRITERION C · COMMUNICATING

4.C1 **The products fall by 4 each time, so they continue below zero.**
Different signs give a negative product, and the pattern forces the sign.

4.C2 **(−4) × 6 = −24 m**
Each minute is a move of −4, so six moves give −24.

4.C3 **Different signs give a negative result.**
Divide the absolute values, then attach a negative sign.

4.C4 **The products (−2) × 3, (−2) × 2, (−2) × 1, (−2) × 0 rise by 2, so the next is +2.**
The pattern forces the sign.

4.C5 **The second student is right: 7.**
Same signs give a positive product, and multiplying by −1 gives the opposite.

4.C6 **No moves are made, so the position does not change.**
Zero times any number is zero.

4.C7 **It is the same as for multiplication.**
Division is multiplication by the reciprocal, and the reciprocal keeps the sign.

4.C8 **(−30) ÷ 5 = −6 m per stage**
Different signs give a negative quotient, and each stage is a move of −6 m.

4.C9 **−8, and the change is in the negative direction**
An odd power keeps the negative sign.

4.C10 **An even number of negative factors gives a positive product.**
Pairs of negatives cancel, and an odd power leaves one unpaired.

4.C11 **Even powers are positive, odd powers negative, so the answer is −1.**
Three negatives give a negative product.

4.C12 **(−12) ÷ 4 = −3 dollars each**
Different signs give a negative quotient, so each share is a debt of 3 dollars.

CHAPTER 4 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

4.D1 **−50 dollars**
250 − 180 − 120 = −50.

4.D2 **−13 m**
−20 − 15 + 22 = −13.

4.D3 **−11 °C**
−8 − 3 = −11.

4.D4 **1 above par**
+4 + (−3) = +1.

4.D5 **−3 000 dollars**
12 000 − 15 000 = −3 000.

4.D6 **It means the point is below the reference level.**
The sign gives direction from the reference.

4.D7 **Floor 0**
−3 + 8 − 5 = 0.

4.D8 **−12 °C**
−5 + (−7) = −12.

4.D9 **+15 dollars**
−40 + 55 = +15.

4.D10 **Only the total change matters.**
Addition is associative.

4.D11 **2 000 m**
1 200 − (−800) = 2 000.

4.D12 **+4 tonnes**
3 − 1 + 4 − 2 = 4.

CHAPTER 5 · CRITERION A · KNOWING AND UNDERSTANDING

5.A1 **n + 7**
'More than' means add, so the output is the cycles plus 7.

5.A2 **14 dollars**
3(4) + 2 = 14, where 3 dollars is the variable cost and 2 dollars the fixed cost.

5.A3 **Variable y, the number of parts; constant 8, a fixed discount.**
Letters stand for quantities that change; numbers are fixed.

5.A4 **2x − 5**
'Decreased by' means subtract, and 'twice' means multiply by 2.

5.A5 **20 mm**
4(5) = 20; four layers of (base + 2 mm).

5.A6 **The first doubles the parts then adds three; the second adds three parts then doubles.**
Brackets decide the order of operations.

5.A7 **p − 2**
'Less than' means subtract.

5.A8 **4 litres**
10 − 3(2) = 4.

5.A9 **3n + 5**
Multiply per-item cost by the number, then add the fixed cost.

5.A10 **A letter standing for a number that can change.**
It lets one rule cover every possible design.

5.A11 **15**
9 counts the panel and 6 counts the frame, so 9 + 6 = 15.

5.A12 **4x + 1**
Product means multiply.

CHAPTER 5 · CRITERION B · INVESTIGATING PATTERNS

5.B1 **9x**
Collect like terms: 7x + 2x = 9x.

5.B2 **3a + 11**
Collect the a terms, then the constants.

5.B3 **3x + 15**
Multiply each term inside by 3.

5.B4 **5x + 5**
2x + 8 + 3x − 3 = 5x + 5.

5.B5 **No — 10 against 7**
Substitution reveals the difference.

5.B6 **−2m + 2**
4m − 6m = −2m, and the setup time is separate.

5.B7 **Yes, all equivalent**
Expanding shows they match.

5.B8 **−x + 4**
The minus sign changes every sign inside.

5.B9 **2x + 4**
3x + 6 − x − 2 = 2x + 4; the brackets hold the same expression.

5.B10 **They are not like terms — one has the variable, the other does not.**
Only like terms combine.

5.B11 **10x − 15**
Multiply each term by 5; 10x counts the variable part and 15 the fixed part.

CHAPTER 5 · CRITERION C · COMMUNICATING

5.C1 **x = 6**
Subtract 9 from both sides, because the two parts must add to 15.

5.C2 **x = 7**
Divide both sides by 4; both sides change by the same factor, so the equality holds.

5.C3 **x = 3**
Subtract the fixed charge, then divide by the cost per part.

5.C4 **To keep the equation true.**
The balance stays level only if both sides change equally.

5.C5 **x = 6**
Divide by 5 first, then add 2; the reverse order also works if you expand first.

5.C6 **x = 9**
Subtract 2x and add 4 so that the costs are collected on each side.

5.C7 **Substitute: 4 × 7 = 28, so the solution is correct.**
A solution must satisfy the original statement.

5.C8 **x = 12**
Subtract 1, then multiply by 3; x is the running time in hours.

5.C9 **x = 4**
Adding x and subtracting 3 gives x = 4; substituting −4 gives 11, not 3.

5.C10 **An expression has no equals sign; an equation states equality.**
Equations can be solved; expressions can only be evaluated.

5.C11 **x = 5**
2x + 2 = 3x − 3, so x = 5; substituting gives 12 = 12.

5.C12 **x = 7**
3x = 21, so the designs cost the same when 7 parts are used.

CHAPTER 5 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

5.D1 **5 cm by 8 cm**
$2(w + w + 3) = 26$ gives $w = 5$.

5.D2 **23, 24, 25**
$n + (n+1) + (n+2) = 72$ gives $n = 23$.

5.D3 **5 hours**
$30h = 150$.

5.D4 **12**
$2x - 6 = 18$.

5.D5 **9 km**
$2d = 18$.

5.D6 **$x \leq 6$, so at most 6 parts**
$3x \leq 18$, and a part cannot be fractional.

5.D7 **Father 36, son 12**
$3s + s = 48$.

5.D8 **5 cm**
$2w^2 = 50$, $w^2 = 25$.

5.D9 **150 minutes**
$0.10m \leq 15$.

5.D10 **A negative age or a fractional number of people cannot occur.**
The context restricts the solution set.

5.D11 **14 and 21**
$n + (n + 7) = 35$.

5.D12 **8 minutes; check $0.25 \times 8 = 2$**
$0.25t = 2$, and substituting the answer back verifies it.

CHAPTER 6 · CRITERION A · KNOWING AND UNDERSTANDING

6.A1 **14**
The common difference is 3.

6.A2 **Start at 7 and add 3 each time**
The difference is constant.

6.A3 **13**
Add 3 three more times.

6.A4 **No — the differences are 3, 6, 12, so they are not constant.**
A linear pattern has a constant difference between terms.

6.A5 **First term 20, difference −3**
Each term falls by 3.

6.A6 **The value at position n in the sequence**
n is the position number.

6.A7 **16 and 22**
The differences grow by 1 each time.

6.A8 **25**
$5 + 5 \times 4 = 25$.

6.A9 **A position is where the term sits; a term is the value there.**
Position n maps to term value.

6.A10 **55**
$100 - 9 \times 5 = 55$.

6.A11 **Yes — the differences grow, not stay constant**
A geometric pattern outpaces a linear one.

6.A12 **1**
Divide by 3 each time.

CHAPTER 6 · CRITERION B · INVESTIGATING PATTERNS

6.B1 **$5n + 1$**
Difference 5 and first term 6.

6.B2 **$4n - 1$**
$4(1) - 1 = 3$ and the difference is 4.

6.B3 **43**
$2(20) + 3 = 43$.

6.B4 **Yes**
Substituting $n = 1, 2, 3, 4$ gives 2, 6, 10, 14.

6.B5 **$6n + 2$**
$6(1) + 2 = 8$.

6.B6 **n^3**
The terms are perfect cubes.

6.B7 **Check the structure, not only several values.**
A pattern of values is evidence; the structure is proof.

6.B8 **40**
$3(15) - 5 = 40$.

6.B9 **$53 - 3n$**
First term 50 with difference -3.

6.B10 **It gives any term directly without listing the earlier ones.**
A rule is a generalisation.

6.B11 **$5n - 5$**
$5(1) - 5 = 0$.

CHAPTER 6 · CRITERION C · COMMUNICATING

6.C1 **Five points on a line of slope 1**
A linear rule gives collinear points.

6.C2 **The differences between terms grow, so the points do not lie on a line.**
Constant second differences give a curve.

6.C3 **2n**
Its slope is 2 against 1.

6.C4 **Slope 4**
The slope equals the common difference.

6.C5 **The steeper line or higher curve grows faster.**
Steepness is the rate of growth.

6.C6 **3n + 1**
The difference is 3, and 3(3) + 1 = 10.

6.C7 **It is the position where the two sequences have equal value.**
The y-values match at the crossing.

6.C8 **The points are (1,1), (2,4), (3,9), (4,16); the second difference is 2.**
A constant second difference signals a quadratic rule, which is why the design curves.

6.C9 **(5, 25)**
The rule is n^2.

6.C10 **Constant differences produce equally spaced points on a line.**
The slope is the common difference.

6.C11 **n^2 becomes larger from n = 3**
2n grows by 2 each step while n^2 grows by 2n − 1, so n^2 overtakes it at n = 3.

6.C12 **Each row of the table is a point on the graph.**
Table and graph are two notations.

CHAPTER 6 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

6.D1 **78**
$12 \times 13 \div 2 = 78$.

6.D2 **32**
$2n + 2$ with $n = 15$.

6.D3 **6.25 cm**
The rule is $100 \times (1/2)^n$, so $100 \div 16 = 6.25$ cm.

6.D4 **650 dollars**
$200 + 15 \times 30 = 650$.

6.D5 **101**
$5n + 1$ with $n = 20$.

6.D6 **$n = 4$**
$3(4) + 1 = 13$ and $4^2 = 16$, and n^2 stays larger afterwards.

6.D7 **0, 2, 6, 12, 20 and 90**
Substitute $n = 1$ to 5 and $n = 10$.

6.D8 **100×0.92^n; 66% after 5 hours**
$100 \times 0.92^5 \approx 66$.

6.D9 **Both sequences have the same value at position 7.**
The y-values match at the crossing.

6.D10 **101**
$4n + 1$ with $n = 25$.

6.D11 **665.5**
$500 \times 1.1^3 = 665.5$.

6.D12 **The exponential curve overtakes the line and keeps rising above it.**
Exponential growth outpaces linear growth for large n.

CHAPTER 7 · CRITERION A · KNOWING AND UNDERSTANDING

7.A1 **180°**
A half turn is 180°.

7.A2 **55°**
90 − 35 = 55.

7.A3 **72°**
Vertically opposite angles are equal.

7.A4 **50°**
180 − 130 = 50.

7.A5 **360°**
A full turn is 360°.

7.A6 **85°**
360 − 275 = 85.

7.A7 **Alternate angles**
They lie on opposite sides of the transversal.

7.A8 **70°**
180 − 110 = 70.

7.A9 **The three corners can be placed along a straight line.**
A straight angle is 180°.

7.A10 **180°**
They form a C shape between parallel lines.

7.A11 **540°**
(5 − 2) × 180 = 540.

7.A12 **70°**
(180 − 40) ÷ 2 = 70.

CHAPTER 7 · CRITERION B · INVESTIGATING PATTERNS

7.B1 **26 cm**
2(9 + 4) = 26.

7.B2 **8 cm**
32 ÷ 4 = 8.

7.B3 **about 44 m**
C = 2πr = 2 × π × 7 ≈ 43.98 m.

7.B4 **4 cm by 8 cm**
2(w + 2w) = 24 gives w = 4.

7.B5 **Perimeter is for polygons; circumference is the perimeter of a circle.**
Circumference is a special case of perimeter.

7.B6 **16.5 cm**
3 × 5.5 = 16.5.

7.B7 **4, 8, 12, 16**
Perimeter = 4 × side.

7.B8 **about 62.8 m**
C = πd = π × 20 ≈ 62.8 m.

7.B9 **Length × width is the area, not the boundary.**
Perimeter adds side lengths; area multiplies two.

7.B10 **25 cm**
Add all the sides.

7.B11 **100 m**
400 ÷ 4 = 100.

CHAPTER 7 · CRITERION C · COMMUNICATING

7.C1 **66 cm²**
11 × 6 = 66.

7.C2 **20 cm²**
½ × 8 × 5 = 20.

7.C3 **50.3 cm²**
π × 16 ≈ 50.3.

7.C4 **Two identical triangles make a rectangle of the same base and height.**
The triangle is half of the rectangle.

7.C5 **36 cm²**
9 × 4 = 36.

7.C6 **A rectangle, so the area is base × height.**
Cutting along the height moves a triangle to make a rectangle of the same area.

7.C7 **32 cm²**
½(6 + 10) × 4 = 32.

7.C8 **It counts how many unit squares cover the surface.**
Area is a two-dimensional measure.

7.C9 **30 cm²**
Add the parts.

7.C10 **56.25 cm²**
$7.5^2 = 56.25$.

7.C11 **Area = πr^2, so it grows with the square of the radius.**
The radius is squared in the formula.

7.C12 **8 cm**
48 ÷ 6 = 8.

CHAPTER 7 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

7.D1 **70 cm²**
2(w + w + 3) = 34 gives w = 7 and l = 10.

7.D2 **The square (81 cm² against 80 cm²)**
Square side 9; $9^2 = 81$.

7.D3 **78.5 cm²**
r = 31.4 ÷ 2π = 5; π × 25 ≈ 78.5.

7.D4 **42 m²**
½ × 12 × 7 = 42.

7.D5 **240 dollars**
20 m² × 12 = 240.

7.D6 **Area = πr², so doubling r multiplies the area by 4.**
The radius is squared.

7.D7 **32 cm²**
Use the base matching the height: 8 × 4 = 32.

7.D8 **184 cm²**
Outer 600 − inner (26 × 16 = 416) = 184 cm².

7.D9 **40 cm²**
½ × 10 × 8 = 40.

7.D10 **Split it into two rectangles and add their areas.**
Compound areas are sums of simple areas.

7.D11 **21.99 m²**
Outer radius 4, inner 3: π(16 − 9) ≈ 21.99.

7.D12 **The same boundary can enclose different surfaces, e.g. a square against a thin rectangle.**
Perimeter fixes the boundary, not the surface.

CHAPTER 8 · CRITERION A · KNOWING AND UNDERSTANDING

8.A1 **6 faces, 12 edges, 8 vertices**
These counts are fixed for every cuboid.

8.A2 **A cylinder**
Two circles joined by a curved surface.

8.A3 **A flat layout of every face of a solid.**
The net shows the exact shapes and areas of cardboard needed, so waste is reduced.

8.A4 **5 faces**
Two triangular ends and three rectangular sides.

8.A5 **A prism has two identical ends joined by rectangles; a pyramid has one base and triangular faces meeting at a point.**
The shape of the ends and the number of bases decide the type.

8.A6 **The shape seen from above.**
The plan gives the footprint, which decides how many containers fit on a shelf.

8.A7 **3**
Three faces meet at each corner.

8.A8 **A cube**
Six equal square faces define a cube.

8.A9 **It lays out every face so each area can be found and added.**
Surface area is the sum of the face areas.

8.A10 **5 faces**
One square base and four triangular faces.

8.A11 **The shape seen from the front or the side.**
The elevation gives the heights and widths needed to cut each face.

8.A12 **6 vertices; 3 edges meet at each**
Two triangular ends, each with three corners joined by three edges.

CHAPTER 8 · CRITERION B · INVESTIGATING PATTERNS

8.B1 **84 cm³**
$7 \times 4 \times 3 = 84$.

8.B2 **125 cm³**
$5^3 = 125$.

8.B3 **113.1 cm³**
$\pi \times 4 \times 9 \approx 113.1$.

8.B4 **120 cubes**
$6 \times 5 \times 4 = 120$.

8.B5 **It counts how many unit cubes fill the solid.**
Volume is a three-dimensional measure.

8.B6 **4 cm**
$96 \div 24 = 4$.

8.B7 **1, 8, 27, 64**
Volume scales with the cube of the side.

8.B8 **2.5 litres**
1 litre = 1 000 cm³, so 2 500 ÷ 1 000 = 2.5 litres.

8.B9 **Volume = area of cross-section × length.**
The rule works for every prism.

8.B10 **120 cm³**
$\tfrac{1}{2} \times 6 \times 4 \times 10 = 120$.

8.B11 **600 litres**
× 1 000.

CHAPTER 8 · CRITERION C · COMMUNICATING

8.C1 **94 cm²**
$2(20 + 15 + 12) = 94$.

8.C2 **216 cm²**
$6 \times 36 = 216$.

8.C3 **87.96 cm²**
$2\pi(4) + 2\pi(2)(5) \approx 87.96$.

8.C4 **Two circles plus a rectangle that wraps around.**
The curved surface unrolls into a rectangle.

8.C5 **5 cm**
$6s^2 = 150$.

8.C6 **The 3 × 3 × 4 cuboid**
It is closest to a cube.

8.C7 **It shows each triangular face separately so the areas can be added.**
Surface area is the sum of the face areas.

8.C8 **280 cm²**
$2(100 + 20 + 20) = 280$.

8.C9 **Volume is the space inside; surface area is the material covering the outside.**
One is cubic, the other square.

8.C10 **5.96 cm**
$2\pi(4) + 2\pi(2)h = 100$ gives $h \approx 5.96$.

8.C11 **Its dimensions are equal, so no face is stretched.**
Equal dimensions minimise the total face area.

8.C12 **72 cm**
$4(8 + 6 + 4) = 72$.

CHAPTER 8 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

8.D1 **9 000 cm^3 and 2 700 cm^2**
$30 \times 20 \times 15$; $2(600 + 450 + 300)$.

8.D2 **942 litres**
$\pi \times 0.25 \times 1.2 \approx 0.942\ m^3$.

8.D3 **7.8 g/cm^3 — steel**
$7\,800 \div 1\,000 = 7.8$.

8.D4 **512 cm^3**
$6s^2 = 376$ gives $s \approx 7.92$; volume $\approx 496\ cm^3$.

8.D5 **500 m^3 and 140 m^2**
$25 \times 10 \times 2$; $2(25 \times 2) + 2(10 \times 2) = 140$.

8.D6 **Surface area depends on shape, not only on volume.**
A flatter box has a larger surface area.

8.D7 **324.3 cm^2**
$2\pi(12.25) + \pi(7)(11) \approx 324.3$.

8.D8 **1 000 cubes and 6 000 cm^2**
$10^3 = 1\,000$; each has 6 cm^2.

8.D9 **90 m^3 and 96 m^2**
$6 \times 5 \times 3$; $2(6{\times}3) + 2(5{\times}3) + 30 = 96$.

8.D10 **Fill it with water and measure the volume displaced.**
Displacement gives the internal volume.

8.D11 **120 cm^3**
$15 \times 8 = 120$.

8.D12 **Its large faces increase the total surface area for the same volume.**
Shape, not volume alone, decides surface area.

CHAPTER 9 · CRITERION A · KNOWING AND UNDERSTANDING

9.A1 **A translation by the vector (4, −3)**
A translation is described by a vector.

9.A2 **(−1, 6)**
Add the vector to the coordinates.

9.A3 **(−4, 3)**
The x-coordinate changes sign.

9.A4 **Lengths and angles**
A translation is an isometry.

9.A5 **(5, −2)**
The y-coordinate changes sign.

9.A6 **A translation slides; a reflection flips in a mirror line.**
One preserves orientation, the other reverses it.

9.A7 **(4, 3)**
The coordinates are swapped.

9.A8 **(3, 4)**
Subtract the start from the finish.

9.A9 **Reflection preserves all lengths and angles.**
Congruence follows from preserved measures.

9.A10 **(5, −2)**
Add the vector.

9.A11 **It sends the point to an equal distance on the other side of the line.**
The image is the same distance from the mirror line as the original.

9.A12 **(2, 7)**
Change the sign of x.

CHAPTER 9 · CRITERION B · INVESTIGATING PATTERNS

9.B1 **(0, 3)**
$(x, y) \to (-y, x)$.

9.B2 **(−2, −4)**
Both coordinates change sign.

9.B3 **(6, 9)**
Multiply both coordinates by 3.

9.B4 **It is unchanged**
Multiplying by 1 leaves every coordinate the same.

9.B5 **It enlarges and turns the motif upside down.**
The size changes by the absolute value and the sign gives the rotation of 180°.

9.B6 **(2, −1)**
$(x, y) \to (y, -x)$.

9.B7 **1, 4, 9 times the original**
Area scales by k^2.

9.B8 **Centre, angle and direction**
Without all three the rotation is not fixed.

9.B9 **(1, 1), (2, 1), (1.5, 2.5)**
Multiply every coordinate by 0.5.

9.B10 **Congruent shapes are the same size; similar shapes have the same shape but different sizes.**
Enlargement gives similarity, not congruence.

9.B11 **(0, −5)**
Both coordinates change sign.

CHAPTER 9 · CRITERION C · COMMUNICATING

9.C1	**4** Two through opposite vertices, two through midpoints.
9.C2	**3** It fits onto itself every 120°.
9.C3	**Line symmetry is a mirror fold; rotational symmetry is a turn onto itself.** They are tested differently.
9.C4	**5** One through each vertex and the opposite side.
9.C5	**Generally no — only a rectangle or rhombus does.** Test each parallelogram by folding.
9.C6	**2** It fits onto itself after 180°.
9.C7	**Any line through the centre is a mirror line.** Every diameter divides the circle into matching halves.
9.C8	**90°** 360 ÷ 4 = 90.
9.C9	**3, 4, 5, 6** A regular n-gon has n lines of symmetry.
9.C10	**A transformation that preserves the symmetry is consistent with the shape.** Broken symmetry signals an error.
9.C11	**1** Through the apex and the midpoint of the base.
9.C12	**No rotational symmetry** Only a full turn maps the shape onto itself.

CHAPTER 9 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

9.D1 **(0, 0), (−4, 0), (−2, 4)**
Reflect to (−1,2), (−5,2), (−3,6), then add (1, −2).

9.D2 **6 cm²**
Area scales by 9, so 54 ÷ 9 = 6.

9.D3 **45°**
360 ÷ 8 = 45.

9.D4 **A translation by (3, 4)**
Add the vector.

9.D5 **100 cm²**
Area scales by 6.25; 16 × 6.25 = 100.

9.D6 **Both dimensions double, so the area is 2 × 2 times the original.**
Area scales with k^2.

9.D7 **A rotation of 180° about the origin**
Both coordinates change sign.

9.D8 **(−2, −2)**
Multiply by −2.

9.D9 **A regular pentagon**
Regular n-gons have n of each.

9.D10 **Image (4,1), (6,1), (4,4); area ratio 1:1**
Translation preserves area.

9.D11 **3**
9 ÷ 3 = 3.

9.D12 **No line divides it into matching halves.**
All sides and angles differ.

CHAPTER 10 · CRITERION A · KNOWING AND UNDERSTANDING

10.A1 **0 to 1**
0 is impossible and 1 is certain.

10.A2 **3/10**
Favourable outcomes divided by all outcomes.

10.A3 **1/2**
Two equally likely outcomes.

10.A4 **0.7**
Complementary probabilities add to 1.

10.A5 **Theoretical comes from counting outcomes; experimental comes from trials.**
One is exact, the other varies.

10.A6 **1/6**
One favourable outcome out of six.

10.A7 **1/4, or 25%**
0.25 = 25/100, so the percentage is often the clearest for a forecast.

10.A8 **The event is certain**
It happens in every trial.

10.A9 **1/5**
Five equally likely outcomes.

10.A10 **3/5**
18 out of 30.

10.A11 **There cannot be more favourable outcomes than total outcomes.**
The fraction is at most 1.

10.A12 **5/10 = 1/2**
There are 10 cards and 5 are not red, so half the draws are not red.

CHAPTER 10 · CRITERION B · INVESTIGATING PATTERNS

10.B1 **HH, HT, TH, TT**
Each coin has two outcomes, so $2 \times 2 = 4$.

10.B2 **36**
$6 \times 6 = 36$.

10.B3 **1/6**
Six doubles out of 36 outcomes.

10.B4 **No — the coins are distinct**
Order matters when the objects are distinguishable.

10.B5 **100 codes**
10 choices for each digit gives $10 \times 10 = 100$, so a random guess succeeds once in 100.

10.B6 **An outcome is one result; an event is a set of outcomes.**
Events group outcomes.

10.B7 **A, B, C, D**
One outcome per sector.

10.B8 **2, 4, 8**
Each extra coin doubles the sample space.

10.B9 **12**
$6 \times 2 = 12$.

10.B10 **It shows every outcome, so none is missed.**
The denominator must count all outcomes.

10.B11 **1/36**
Only (1, 1) gives a total of 2.

CHAPTER 10 · CRITERION C · COMMUNICATING

10.C1 **2/3**
3 even outcomes plus 5 gives 4 out of 6.

10.C2 **4/13**
13 hearts + 4 kings − 1 king of hearts = 16; 16/52 = 4/13.

10.C3 **1/4**
1/2 × 1/2 = 1/4.

10.C4 **Add for 'or' with mutually exclusive events; multiply for 'and' with independent events.**
The wording of the question decides the rule.

10.C5 **4/9**
The first draw is 5/10 and the second 4/9, so multiply: 20/90 = 4/9.

10.C6 **Every outcome is either even or odd, so the events cover the sample space.**
Mutually exclusive events that cover every outcome have probabilities that sum to 1.

10.C7 **5/36**
1/6 × 5/6 = 5/36; the first roll does not change the second.

10.C8 **The events cannot both happen at the same time.**
Their outcome sets do not overlap.

10.C9 **1/4**
1/2 × 1/2 = 1/4.

10.C10 **Events where the outcome of one does not change the probability of the other.**
For independent events, P(A and B) = P(A) × P(B).

10.C11 **1**
The two events cover all outcomes.

10.C12 **1/12**
Three outcomes out of 36: (5,6), (6,5), (6,6).

CHAPTER 10 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

10.D1 **0 dollars**
1/5 × 10 = 2, so the game breaks even.

10.D2 **Odd digits are heads; count digits until the first odd one.**
The rule must give P(head) = 1/2.

10.D3 **Digits 00–02 are faulty**
3 favourable outcomes out of 100.

10.D4 **0 dollars**
1/100 × 100 = 1.

10.D5 **Few trials give results that vary widely from the theoretical value.**
Experimental probability settles as trials increase.

10.D6 **1 dollar**
1/2 × 4 = 2; 2 − 1 = 1.

10.D7 **Digits 0–6 score; 7–9 miss**
7 favourable digits out of 10.

10.D8 **The average wait over many repetitions is 4 minutes.**
A single wait may differ; the average settles.

10.D9 **Each player has the same chance of the higher roll.**
Symmetry of the sample space.

10.D10 **0 dollars**
1/6 × 6 = 1, so the game breaks even.

10.D11 **Digits 0 and 1 are favourable out of 10, but 00–24 out of 100 gives exactly 25%.**
The rule must match the target probability.

10.D12 **3/200 and 0.75 dollars**
3/200 × 50 = 0.75.

CHAPTER 11 · CRITERION A · KNOWING AND UNDERSTANDING

11.A1 **A bar chart**
Categories are compared by bar height.

11.A2 **20**
Add the frequencies.

11.A3 **Labelled axes, a scale and units**
Without units the display cannot be interpreted.

11.A4 **A histogram**
Continuous data are grouped into intervals.

11.A5 **A bar chart compares categories; a histogram shows the distribution of grouped values.**
One has gaps between bars; the other does not.

11.A6 **The question is leading and biased.**
It suggests the expected answer.

11.A7 **The most frequent category**
The mode is the most common value.

11.A8 **Parts of a whole**
A pie chart shows proportions.

11.A9 **Five responses were not tallied.**
The frequencies must add to the sample size.

11.A10 **Without units the numbers cannot be interpreted.**
Units give the values meaning.

11.A11 **A pie chart**
A pie chart divides a circle into sectors.

11.A12 **A smaller group chosen to represent a larger one**
The sample stands for the population.

CHAPTER 11 · CRITERION B · INVESTIGATING PATTERNS

11.B1 **9**
36 ÷ 4 = 9.

11.B2 **9**
The middle of five ordered values.

11.B3 **6**
The most frequent value.

11.B4 **Mean 24, median 6**
The outlier pulls the mean far above the median.

11.B5 **9.5**
The mean of the two middle values.

11.B6 **When the data contain an extreme value.**
The median ignores the extremes.

11.B7 **14**
70 ÷ 5 = 14.

11.B8 **3, 4, 5**
Each set's mean is its middle value.

11.B9 **The mode**
Only the mode works without numerical order.

11.B10 **3.33**
10 ÷ 3 ≈ 3.33.

11.B11 **The mean adds all values; the median uses only the middle ones.**
This is why the mean is sensitive to outliers.

CHAPTER 11 · CRITERION C · COMMUNICATING

11.C1 **18**
30 − 12 = 18.

11.C2 **The spread of the middle half of the data.**
IQR = Q3 − Q1.

11.C3 **Q1 = 7, median 9, Q3 = 11**
Split the ordered data into quarters.

11.C4 **It shows five summary values in one picture.**
The display makes comparison quick.

11.C5 **One is more spread out than the other.**
Range measures spread, not centre.

11.C6 **IQR = 8**
Q1 = 3 and Q3 = 11, so the middle half of households differ by 8 people.

11.C7 **It uses only the two extreme values.**
A single outlier changes the range a lot.

11.C8 **The middle half of the data, from Q1 to Q3**
The box spans the interquartile range.

11.C9 **Range 15, IQR 3**
Q1 = 5.5 and Q3 = 8.5.

11.C10 **Two sets can share a centre but differ greatly in spread.**
One measure alone is incomplete.

11.C11 **20**
40 − 20 = 20.

11.C12 **There are values far from the bulk of the data.**
Long whiskers indicate possible outliers.

CHAPTER 11 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

11.D1 **The median, 50 dollars**
The mean of 138 dollars is distorted by the outlier.

11.D2 **Class 2 scores higher on average but is far less consistent.**
Both centre and spread must be reported.

11.D3 **The sample is too small and may be biased.**
A small sample need not represent the population.

11.D4 **A few very high wages pull the mean up.**
The median would be lower and more typical.

11.D5 **That patient has a 0.7 chance of responding; it does not guarantee the outcome.**
A population percentage becomes a probability for one individual, not a certainty.

11.D6 **Same centre, but B is much more spread out.**
The IQR compares the spread of the middle half.

11.D7 **The spread, the sample size and whether there are outliers.**
A mean alone is an incomplete summary.

11.D8 **Scores rose on average but became much less consistent.**
Centre and spread moved in different ways.

11.D9 **80, 70 and 50 students**
40% of 200 is 80, and so on.

11.D10 **The sample does not represent the population, so the statistic does not apply.**
Bias breaks the link between sample and population.

11.D11 **The values are widely spread around a central value.**
A large range with matching centre indicates spread.

11.D12 **Which average was used, and what happened to the spread?**
The choice of average changes the claim.

Formula bank (1 of 4)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

RATIO, PROPORTION AND RATES

Ratio

a : b compares two quantities in the same unit; a/b is the unit rate.

Equivalent ratios

a : b = (a×k) : (b×k) for any non-zero k.

Sharing in a ratio

share = (part ÷ total parts) × amount.

Proportional relationship

y = kx, and the graph is a straight line through the origin.

Speed

speed = distance ÷ time; distance = speed × time.

Density

density = mass ÷ volume.

Percentage change

new = original × (1 + p/100) for an increase; × (1 − p/100) for a decrease.

Formula bank (2 of 4)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

INTEGERS AND ALGEBRA

Absolute value

$|x|$ is the distance of x from 0 on a number line.

Sign rules

$(-a)(-b) = ab$; $(-a)(b) = -ab$; $a - b = a + (-b)$.

Like terms

only terms with the same letters and powers combine.

Expanding brackets

$a(b + c) = ab + ac$.

Linear equation

$ax + b = c \longrightarrow x = (c - b)/a$, applying the inverse operation to both sides.

nth term of a linear sequence

term $= mn + c$, where m is the constant difference.

Formula bank (3 of 4)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

GEOMETRY AND SPACE

Angle sum of a triangle

180°.

Angles on a straight line

sum to 180°.

Angles at a point

sum to 360°.

Interior angle sum of a polygon

$(n - 2) \times 180°$.

Exterior angle sum of a polygon

360°.

Rectangle

area = $l \times w$; perimeter = $2(l + w)$.

Triangle

area = $\frac{1}{2} \times \text{base} \times \text{height}$.

Parallelogram

area = base × height.

Trapezium

area = $\frac{1}{2}(a + b) \times \text{height}$.

Circle

circumference = $2\pi r = \pi d$; area = πr^2.

Prism

volume = area of cross-section × length.

Cylinder

volume = $\pi r^2 h$; curved surface area = $2\pi rh$.

Sphere

volume = $\frac{4}{3}\pi r^3$.

Formula bank (4 of 4)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

TRANSFORMATIONS, PROBABILITY AND DATA

Reflection in y = x

$(x, y) \rightarrow (y, x)$.

Rotation 90° clockwise about the origin

$(x, y) \rightarrow (y, -x)$.

Enlargement, scale factor k, centre the origin

$(x, y) \rightarrow (kx, ky)$.

Probability of an event

P(event) = favourable outcomes ÷ total equally likely outcomes.

Complement

P(not A) = 1 − P(A).

Mean

sum of values ÷ number of values.

Median / quartiles

the middle value; quartiles split the ordered data into four parts.

Range and IQR

range = max − min; IQR = Q3 − Q1.

Glossary (1 of 3)

('Absolute value', 'The distance of a number from zero on a number line, written |x|.', 4) ch. 4
The distance of a number from zero on a number line, written |x|.

('Bar model', 'A strip divided into equal parts used to represent fractions, ratios and percentages.', 1) ch. 1
A strip divided into equal parts used to represent fractions, ratios and percentages.

('Best buy', 'The option with the lowest unit price for the same quantity.', 3) ch. 3
The option with the lowest unit price for the same quantity.

('Bias', 'A systematic error that makes a sample unrepresentative of the population.', 11) ch. 11
A systematic error that makes a sample unrepresentative of the population.

('Box plot', 'A display of the minimum, lower quartile, median, upper quartile and maximum.', 11) ch. 11
A display of the minimum, lower quartile, median, upper quartile and maximum.

('Categorical data', 'Data that name groups rather than measure amounts.', 11) ch. 11
Data that name groups rather than measure amounts.

('Complement (probability)', 'The set of outcomes that are not in the event; P(not A) = 1 − P(A).', 10) ch. 10
The set of outcomes that are not in the event; P(not A) = 1 − P(A).

('Composite solid', 'A solid made by joining two or more simpler solids.', 8) ch. 8
A solid made by joining two or more simpler solids.

('Continuous data', 'Numerical data that can take any value in an interval.', 11) ch. 11
Numerical data that can take any value in an interval.

('Cross-section', 'The shape exposed when a solid is cut parallel to a face.', 8) ch. 8
The shape exposed when a solid is cut parallel to a face.

('Density', 'Mass per unit volume, in g/cm³ or kg/m³.', 3) ch. 3
Mass per unit volume, in g/cm^3 or kg/m^3.

('Diameter', 'A chord of a circle that passes through the centre; twice the radius.', 7) ch. 7
A chord of a circle that passes through the centre; twice the radius.

('Discrete data', 'Numerical data that counts separate values.', 11) ch. 11
Numerical data that counts separate values.

('Double number line', 'Two aligned number lines used to compare two quantities in a ratio.', 1) ch. 1
Two aligned number lines used to compare two quantities in a ratio.

('Enlargement', 'A transformation that multiplies every distance from a centre by the scale factor.', 9) ch. 9
A transformation that multiplies every distance from a centre by the scale factor.

('Equivalent expressions', 'Expressions that have the same value for every value of the variable.', 5) ch. 5
Expressions that have the same value for every value of the variable.

('Equivalent ratios', 'Ratios that describe the same comparison, for example 2:3 and 6:9.', 1) ch. 1
Ratios that describe the same comparison, for example 2:3 and 6:9.

('Exterior angle', 'The angle between a side of a polygon and the extension of the next side.', 7) ch. 7
The angle between a side of a polygon and the extension of the next side.

('Frequency table', 'A table that records how often each value or group occurs.', 11) ch. 11
A table that records how often each value or group occurs.

('Function machine', 'A diagram that applies a rule to an input to produce an output.', 6) ch. 6
A diagram that applies a rule to an input to produce an output.

('Inequality', 'A statement that one quantity is greater than or less than another.', 4) ch. 4
A statement that one quantity is greater than or less than another.

('Integer', 'A whole number that may be positive, negative or zero.', 4) ch. 4
A whole number that may be positive, negative or zero.

('Interquartile range', 'Q3 − Q1, the spread of the middle half of the data.', 11) ch. 11
Q3 − Q1, the spread of the middle half of the data.

('Inverse operation', 'The operation that undoes another, such as subtraction undoing addition.', 5) ch. 5
The operation that undoes another, such as subtraction undoing addition.

Glossary (2 of 3)

('Line of best fit', 'A straight line that follows the trend of a scatter of points.', 11) ch. 11
A straight line that follows the trend of a scatter of points.

('Linear equation', 'An equation in which the variable appears to the first power only.', 5) ch. 5
An equation in which the variable appears to the first power only.

('Linear sequence', 'A sequence with a constant difference between consecutive terms.', 6) ch. 6
A sequence with a constant difference between consecutive terms.

('Mean', 'The total of the values divided by the number of values.', 11) ch. 11
The total of the values divided by the number of values.

('Median', 'The middle value when the data are ordered.', 11) ch. 11
The middle value when the data are ordered.

('Mode', 'The most frequent value in a data set.', 11) ch. 11
The most frequent value in a data set.

('nth term', 'A rule that gives the value of any term from its position n.', 6) ch. 6
A rule that gives the value of any term from its position n.

('Outcome', 'A possible result of a probability experiment.', 10) ch. 10
A possible result of a probability experiment.

('Part-part and part-whole ratio', 'Comparing two parts, or one part with the whole.', 1) ch. 1
Comparing two parts, or one part with the whole.

('Population', 'The whole group a statistical question is about.', 11) ch. 11
The whole group a statistical question is about.

('Probability', 'A number from 0 to 1 describing how likely an outcome is.', 10) ch. 10
A number from 0 to 1 describing how likely an outcome is.

('Proportion', 'Two quantities linked by a constant multiplier, so their ratio is fixed.', 2) ch. 2
Two quantities linked by a constant multiplier, so their ratio is fixed.

('Quartile', 'A value that splits ordered data into four equal parts.', 11) ch. 11
A value that splits ordered data into four equal parts.

('Radius', 'The distance from the centre of a circle to any point on it.', 7) ch. 7
The distance from the centre of a circle to any point on it.

('Range', 'The largest value minus the smallest value.', 11) ch. 11
The largest value minus the smallest value.

('Rate', 'A comparison of two quantities in different units.', 3) ch. 3
A comparison of two quantities in different units.

('Ratio', 'A comparison of two quantities of the same kind, written a : b.', 1) ch. 1
A comparison of two quantities of the same kind, written a : b.

('Reflection', 'A transformation that flips a shape in a mirror line.', 9) ch. 9
A transformation that flips a shape in a mirror line.

('Rotation', 'A transformation that turns a shape about a centre through a given angle.', 9) ch. 9
A transformation that turns a shape about a centre through a given angle.

('Sample', 'The part of a population that is actually measured.', 11) ch. 11
The part of a population that is actually measured.

('Sample space', 'The set of all possible outcomes of an experiment.', 10) ch. 10
The set of all possible outcomes of an experiment.

('Scale drawing', 'A drawing in which every length is multiplied by the same scale factor.', 2) ch. 2
A drawing in which every length is multiplied by the same scale factor.

('Scale factor', 'The multiplier from an original length to the image length.', 9) ch. 9
The multiplier from an original length to the image length.

('Similar figures', 'Figures with the same shape and corresponding angles equal.', 2) ch. 2
Figures with the same shape and corresponding angles equal.

Glossary (3 of 3)

('Simplest form', 'A ratio whose parts have no common factor greater than 1.', 1) ch. 1
A ratio whose parts have no common factor greater than 1.

('Simulation', 'A model of a real experiment using random numbers or objects.', 10) ch. 10
A model of a real experiment using random numbers or objects.

('Speed', 'Distance travelled per unit of time.', 3) ch. 3
Distance travelled per unit of time.

('Stem-and-leaf diagram', 'A display that keeps the data values while grouping them by place value.', 11) ch. 11
A display that keeps the data values while grouping them by place value.

('Surface area', 'The total area of the outside faces of a solid.', 8) ch. 8
The total area of the outside faces of a solid.

('Term-to-term rule', 'A rule that gets from one term of a sequence to the next.', 6) ch. 6
A rule that gets from one term of a sequence to the next.

('Theoretical probability', 'Probability calculated from the number of equally likely outcomes.', 10) ch. 10
Probability calculated from the number of equally likely outcomes.

('Transformation', 'A change to a shape: translation, reflection, rotation or enlargement.', 9) ch. 9
A change to a shape: translation, reflection, rotation or enlargement.

('Translation', 'A transformation that slides a shape without turning or stretching it.', 9) ch. 9
A transformation that slides a shape without turning or stretching it.

('Unit price', 'The price of one unit of a quantity, used to compare offers.', 3) ch. 3
The price of one unit of a quantity, used to compare offers.

('Unit rate', 'The value of one part when a ratio is scaled so that part is 1.', 1) ch. 1
The value of one part when a ratio is scaled so that part is 1.

('Variable', 'A letter that stands for a number that can change.', 5) ch. 5
A letter that stands for a number that can change.

('Venn diagram', 'A diagram using overlapping regions to show set relationships.', 10) ch. 10
A diagram using overlapping regions to show set relationships.

REFERENCES

References and acknowledgements

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- IB MYP mathematics subject page — https://www.ibo.org/programmes/middle-years-programme/curriculum/mathematics/ Used for: the four branches of MYP mathematics, standard and extended mathematics, and the inquiry and application focus.
- IB MYP projects — https://www.ibo.org/programmes/middle-years-programme/curriculum/myp-projects/ (page states last updated 19 February 2026). Used for: the Community Project completed by students finishing MYP in Year 3 or Year 4, the Personal Project completed by Year 5 students, and the student-centred nature of both.
- IB MYP general regulations (2026 assessment procedures) — https://www.ibo.org/globalassets/new-structure/become-an-ib-school/pdfs/myp-2026-general-regulations.pdf Used for: the required use of MYP objectives and assessment criteria, minimum teaching hours, and Year 5 eAssessment eligibility.

Principal content reference

- First-edition MYP Mathematics 2 Student Coursebook (repository archive copy). Used as the principal content reference for chapter scope, teaching order, worked examples and exercise coverage. Preserved unchanged.

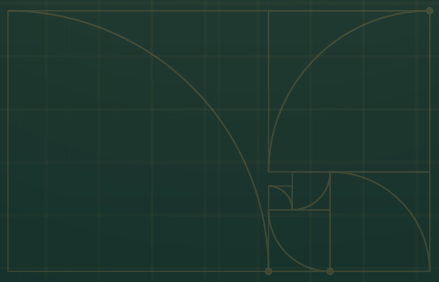
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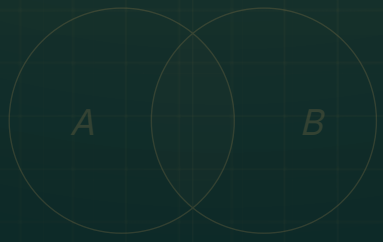
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Eleven chapters that take ratio, proportion, integers, algebra, pattern, geometry, transformations, probability and data from first ideas to confident use. Each chapter opens with a task: you investigate, notice a pattern, state a mathematical conclusion, and only then meet the formal explanation — with worked examples that model a complete solution and practice separated into Criterion A, B, C and D sets so you can see which skill each task trains.



Every chapter closes with a Service as Action opportunity, a Community Project connection and a Personal Project extension for MYP 5. The back of the book carries complete student-facing answers with reasoning, a formula bank, a glossary and references.



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